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REPORT ON

**HYDROGEOLOGIC ASSESSMENT AND WATER
QUALITY MONITORING OF THE
WOOD WASTE LANDFILL SITE
JORDAN RIVER, BRITISH COLUMBIA**

Submitted to:

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1.0 INTRODUCTION

Golder Associates Ltd. (Golder) was retained by Western Forest Products Inc. (WFP) to conduct a hydrogeologic and water quality monitoring program for the former wood waste landfill area at the Jordan River Operations in Jordan River, BC (the "Site"; see Figure 1). The British Columbia Ministry of Environment (MOE) sent a letter to WFP, titled "Abandonment of Permit PR-06965 Issued under the Authority of the Environmental Management Act", outlining the requirements to be taken by WFP for the abandonment of the waste discharge permit. The purpose of the groundwater and surface water assessment is to provide the MOE with information for acting upon Section 20(5)(b) of the *Environmental Management Act* for the abandonment of a waste discharge permit.

A hydrogeologic investigation including a quarterly groundwater and surface water monitoring program was initiated in September 2006. The second and third quarterly water quality monitoring events were carried out in December 2006 and March 2007, respectively. The results, discussion and recommendations of the hydrogeologic investigation and quarterly water quality monitoring program carried out to date are presented in this report.

This report should be interpreted and used in accordance with the limitations and considerations set out in the Golder Associates Ltd. *Limitations and Use of this Report*, which appears in Section 10.0.

2.0 BACKGROUND AND PROJECT UNDERSTANDING

Golder understands that WFP operated a permitted wood waste landfill in the Jordan River area between approximately 1982 and 1999 and wishes to abandon waste discharge permit PR-06965 for the Site. Golder further understands that as part of the discharge permit administration, a limited surface water quality monitoring program was conducted by the MOE at two stations. Based on communication with MOE personnel, the Abandoned Spillway (Site #1) surface water sample location was established approximately 30 m upstream from the culvert area located on the south bank of the spillway (Figure 2). MOE indicated at the time of their sampling program, groundwater seepage was observed on the north bank of the spillway, but the sampling was conducted from the south bank due to more favorable access. The MOE sampled the south bank location in 1995, 1997 and 1999. The Groundwater Seep sample station (Site #2) is located west of the landfill, on the east bank of the Jordan River, and was sampled by the MOE in 2000 and 2001. The available laboratory analytical report containing the water quality results from the June 2000 monitoring by the MOE have been compiled along with the analytical data from the three quarterly monitoring events conducted in 2006 and 2007 by Golder as part of this program.

The water quality analytical reports for the 1995, 1997, 1999 and 2001 monitoring events conducted by the MOE were not available, which prevented verification of the quality and reliability of then data. As a result, the water quality data provided by the MOE for these events was used for comparative purposes only, and was not relied upon in developing conclusions and recommendations for the current assessment.

In 2001, Golder was requested by WFP to conduct a visual assessment of the Site and to review laboratory analytical data from the Groundwater Seep (Site #2) sample location from the MOE's June 14, 2000 monitoring program. The results of the assessment and review were presented in the Golder report titled "*Field Reconnaissance of Wood Bark Landfill Seepage Jordan River*", dated April 12, 2001 (file ref. 012-1033, E/01/78).

WFP sent a letter to the MOE, dated May 21, 2002, stating their intention to abandon the waste discharge permit (PR-06965) for the wood waste landfill. The MOE responded with a letter to WFP, dated July 24, 2006, outlining the requirements that should be undertaken by WFP in order to comply with Section 20 5(b) of the *Environmental Management Act* for the abandonment of the waste discharge permit PR-06965. A summary of these requirements is presented below:

- Conduct a hydrogeological survey to assess the extent of the leachate plume and the direction of groundwater flow within and from an area which includes the former landfill site;
- Carry out a quarterly monitoring program for water quality parameters as follows; chemical oxygen demand (COD), tannin and lignin, pH, phenolics, chloride, specific conductance, total metals, sulphide, sulphate and 96hr LC50 toxicity testing to assess groundwater leachate characteristics at Site #1 (abandoned spillway) and Site #2 (seepage on the east bank of the Jordan River);
- Maintain a database of the analyses and submit an interpretative report with conclusions and recommendations to the Regional Manager, Environmental Protection; and,
- Based on the findings and recommendations of the hydrogeological survey, the results of the monitoring program and other information obtained in connection with leachate from the former landfill site, carry out additional requirements respecting restoration of the environment or the control and monitoring of waste discharged, as specified by the Director.

WFP retained Golder to conduct three quarterly hydrogeological and water quality assessments (September and December 2006 and March 2007) as per the requirements outlined in the MOE letter to WFP dated July 24, 2006.

3.0 SITE DESCRIPTION

The WFP wood waste landfill is located on the east side of the Jordan River and north of Westcoast Road (Highway 14) in Jordan River, BC (see Figures 1 and 2). According to WFP personnel, the landfill received wood bark and chippings from approximately 1982 to 1999. Prior to the land filling of wood waste, the area was reported to have been a sand and gravel borrow pit. Currently, the landfill site is covered with a layer of sand and gravel fill that supports immature (less than approximately 15 years old) vegetation such as alder, broom and grasses. The lateral extent of the landfill is inferred to be approximately 220 m long by 150 m wide based on visual inferences from the presence of mature conifer forested areas located to the south and east of the landfill and moderate to steep slopes along the north margin of the landfill.

A former BC Hydro hydroelectric power generation building is located east of the landfill area. A drainage that flows through the power generation building site and passes along the south side of the wood waste landfill is known as the "Abandoned Spillway". The Abandoned Spillway drainage flows into the Jordan River near the southwest corner of the landfill and appears to be tidally influenced.

4.0 REGULATORY CONSIDERATIONS

The British Columbia Ministry of Environment (MOE) Approved and Working Water Quality Guidelines (BCWQG)^{1,2} were used as benchmarks to screen the surface water (groundwater seep and abandoned spillway) data. The groundwater seep data were compared to the fresh water aquatic life criteria of the BCWQG. The results of the surface water quality analyses for the abandoned spillway were compared to the marine water aquatic life criteria of the BCWQG due to tidal influences and calculated salinity (*i.e.*, > 1.5 ‰).

Environmental matters pertaining to contaminated sites or provincial lands generally fall under the jurisdiction of the MOE pursuant to the *Environmental Management Act* (EMA). The key regulation under the EMA relating to the assessment and remediation of contaminated sites is the Contaminated Sites Regulation (CSR)³. The standards

¹ BC Ministry of Environment, "British Columbia Approved Water Quality Guidelines 2006 edition", updated August 2006.

² BC Ministry of Environment, "A Compendium of Working Water Quality Guidelines for British Columbia", updated August 2006.

³ BC Contaminated Sites Regulation (CSR) [en.BC Reg. 324/2004, s.68; am. BC Regs.405/2004; 76/2005].

presented in the CSR provide numerical concentrations for the evaluation of groundwater quality and are divided into different categories of land use. The water standards are available for specific uses (e.g., aquatic life, irrigation, livestock watering and drinking water). As the site groundwater is not used for extractive purposes (agriculture, drinking water, etc.), the aquatic life standards have been used.

The Site is located adjacent to the Jordan River, near the Strait of Juan de Fuca and the Jordan River is tidally influenced. Available information indicates there are no registered domestic water wells downgradient of the landfill area. Surficial groundwater in the area of the existing landfill is expected to flow southward from the upland area to the surface water receiving environment of either the Jordan River or the Abandoned Spillway and/or into the marine foreshore of the Strait of Juan de Fuca (Figure 2). The water quality standards adopted for the groundwater component of the hydrogeologic assessment, therefore, are for groundwater flow discharging to surface water used by marine and fresh water aquatic life, which are represented by the CSR freshwater and marine aquatic life standards, as appropriate. The use of either the freshwater or the marine standards was determined based on salinity with an approximate salinity cutoff of 1.5 ‰ used to differentiate between marine and freshwater.

The Site is also subject to the provisions of the federal *Fisheries Act*. Section 36(3) of the *Fisheries Act* prohibits the discharge of a “deleterious substance” into fish-frequented waters and applies to the point of discharge. Specifically what constitutes a “deleterious substance” is a matter of expert opinion. Golder was retained by WFP to address the issues identified by the MOE as noted above; however, Golder notes that additional regulatory requirements may apply. The study design follows Ministry of Environment prescriptions and may not be adequate to provide a comprehensive perspective vis-à-vis statutory requirements.

5.0 METHODS

5.1 Review of Existing Information

Existing information on the general characteristics of the Site was compiled from selected geologic maps, unpublished reports and data from publicly accessible websites such as the British Columbia MOE Water Resources Atlas⁴. The Water Resources Atlas provides information on local water wells, general subsurface geologic conditions and aquifer status. The 2001 Golder report was also reviewed as part of the current hydrogeologic assessment.

⁴ British Columbia Ministry of Environment Water Resources Atlas: <http://srmapps.gov.bc.ca/apps/wrbc/>

5.2 Drilling Investigation and Groundwater Monitoring Well Installation

Subsurface geologic and hydrogeologic conditions were investigated at the Site by conducting a borehole drilling and monitoring well installation program. The drilling program was carried out between September 20 and 21, 2006 with full time inspection by Golder field staff. The program included drilling and installing five monitoring wells (MW06-01 to MW06-04 and MW06-06) and drilling one exploratory borehole (BH06-05). The drilling was conducted using a Bombardier Muskeg HY track mounted auger drill rig supplied and operated by Downrite Drilling Ltd. of Chilliwack, BC. Soil conditions were visually logged in the field and samples collected at regular intervals during the drilling.

Five of the six boreholes were completed as groundwater monitoring wells. Groundwater sampling and laboratory testing, hydraulic testing and groundwater level monitoring were carried out following completion of the subsurface investigation and monitoring well installation program. The locations of the five monitoring wells, including one borehole, from the Site are indicated on the Site plan, Figure 2.

The locations of the monitoring wells were determined based on available site access to areas where potential shallow groundwater contamination near the landfill could be investigated. Monitoring well MW06-01 was installed within the landfill to characterize groundwater conditions in the area of inferred direct contact between wood waste and groundwater (*i.e.*, the source of potential leachate generation). Monitoring wells MW06-02, MW06-03 and MW06-04 were installed outside (southwest) of the landfill area to intercept potential leachate migration from the Site. Due to limited upgradient site access, monitoring well MW06-06 was installed in an area inferred to be cross gradient (southeast) of the landfill and for background groundwater monitoring purposes. Borehole BH06-05 was put down for exploration purposes near MW06-06 to assess the deeper soil stratigraphy.

The monitoring wells were constructed of 51 mm (2 inch) diameter PVC (Schedule 40) pipe with No. 10 well screens. Clean silica sand (10/20) was placed around the well screens, and bentonite chips were placed above the sand pack to establish hydraulic seals. The remaining portion of the boreholes was backfilled using either soil cuttings from the drilling or bentonite chips to near the ground surface. Surficial concrete seals and above-ground locking steel covers were used to complete each monitoring well installation.

The locations and relative elevations of the monitoring wells and borehole were surveyed by Loach Construction Surveying Ltd. of Victoria, BC. Monitoring well construction details and borehole logs for each installation are provided in Appendix I.

5.3 Groundwater Level Monitoring

Groundwater levels in the monitoring wells were measured during the field program for the three monitoring events. The static water levels and approximate groundwater elevations are summarized in Tables II-1 and II-2 and Graphs II-1 and II-2, respectively in Appendix II.

5.4 Surface Water Monitoring Locations

5.4.1 Abandoned Spillway Sample Location (Site #1)

During the September 2006 water quality monitoring event, Golder initially sampled a surface water sampling site referred to as the "Lower Abandoned Spillway". This sampling site is located approximately 5 m upstream from an existing culvert that crosses under access road to the landfill area. Following the water sampling program, Golder was informed by the MOE that their original Abandoned Spillway (Site #1) sampling site was located approximately 30 m upstream from the culvert and accessed on the south bank of the spillway.

During the December 2006 water quality monitoring event, Golder established the sample location on the north side of the spillway across from the original MOE Abandoned Spillway (Site #1) sample location. The location and relative elevation of the Lower Abandoned Spillway sample station was surveyed by Loach Construction Surveying Ltd. and is indicated on the Site plan, Figure 2. The approximate position of the new Abandoned Spillway sample station was measured in the field by Golder using a long tape.

During the three monitoring events, it was observed that the lower Jordan River system and lower section of the Abandoned Spillway that was sampled appear to be tidally-influenced based on observed flow reversal. The Abandoned Spillway (Site #1) was sampled during spillway outflow at low tide during the three monitoring events.

5.4.2 Groundwater Seep Sample Location (Site #2)

The Groundwater Seep (Site #2) location, sampled by Golder in 2001 and during the present series of sampling events, and historically by the MOE, is located west of the landfill on the east bank of the Jordan River. The Groundwater Seep area along the Jordan River appears to be periodically submerged at high tide; it was sampled during low tide conditions during the three recent monitoring events. The flow rate of the groundwater seep was estimated during the three monitoring events at the time of sampling. The location and relative elevation of the Groundwater Seep sample location was surveyed by Loach Construction Surveying Ltd. and is indicated on the Site plan, Figure 2.

5.5 Surface and Groundwater Sampling and Laboratory Analysis

Following the drilling program, the five monitoring wells were manually developed using dedicated Waterra™ tubing. The well development removes sediment and water introduced during the drilling process and allows for the collection of groundwater samples that are representative of formation water. Prior to sampling, the monitoring wells were purged to remove approximately three well volumes of groundwater within the well for the purpose of obtaining representative formation water. The monitoring wells were purged and sampled during the three monitoring events on September 27 and December 19, 2006 and March 22, 2007.

The surface water and groundwater samples were collected in clean laboratory bottles, packaged with ice packs and shipped for analysis to ALS Laboratory Group, Environmental Division (ALS) located in Vancouver, BC. Field parameters including pH, temperature, specific conductivity and oxidation-reduction potential (ORP or “redox potential”) were measured on aliquots from the surface water and groundwater samples submitted for laboratory analysis. Groundwater samples submitted for dissolved metals analysis were filtered in the field with a disposable 45 µm membrane filter cartridge and subsequently preserved with nitric acid prior to shipment to the laboratory. The surface water and groundwater chemistry data for this investigation are compiled in Tables III-1 and III-2, respectively, Appendix III. The laboratory chemical reports for this assessment are provided in Appendix V.

ALS analyzed the surface and groundwater samples collected during the field investigation by Golder. ALS is certified by the Canadian Association for Environmental Analytical laboratories (CAEAL) for all analytical methods (chemistry) used in this field program.

Surface water and groundwater samples collected from the field investigation were submitted for analysis of the following parameters:

- Total and dissolved metals, including hardness;
- Chemical oxygen demand (COD);
- Anions (bromide, chloride, fluoride, sulphide, sulphate, nitrate and nitrite);
- Tannin and lignin; and,
- Total phenolics.

In addition to the physical and chemical parameters analyzed, 96-hour LC₅₀ acute toxicity testing was conducted on the Groundwater Seep and Lower Abandoned Spillway surface water samples at Golder’s North Vancouver Aquatic Toxicity Laboratory. The acute toxicity testing reports for this assessment are provided in Appendix V. The testing was

conducted according to the Environmental Canada protocol for conducting acute toxicity tests using rainbow trout (EPS 1/RM/13, Second Edition, 2000). Golder's laboratory is accredited by the Standards Council of Canada (SCC), in cooperation with the Canadian Association for Environmental Analytical Laboratories (CAEAL) and the Washington Department of Ecology (WDOE) Environmental Laboratory Accreditation Program.

5.6 Single-Well Response Testing

Following the September 2006 groundwater sampling program, single-well response testing was conducted on the five monitoring wells (MW06-01 to MW06-04 and MW06-06) on October 5, 2006. This method was used to estimate the hydraulic conductivity of the local surficial aquifer.

The single-well response testing was conducted using the rising head method. In this test, a measured volume of water was removed from the well using a disposable bailer. The bailer is inserted and the water level is allowed to equilibrate. The bailer is then removed and water levels measured during recovery to static conditions.

At the time of testing, water levels in the monitoring wells were recorded using Solinst® Levellogger pressure transducers (model 3001 LT F15/M5). The data from the pressure transducers was downloaded and analyzed using the AQTESOLV® software program to calculate the hydraulic conductivity of each well. The hydraulic conductivity analysis reports for this investigation are compiled and provided in Appendix IV.

Prior to the single-well response testing on October 5, 2006, limited continuous water level monitoring was conducted in monitoring wells MW06-2 and MW06-4 to determine whether or not these wells were tidally-influenced. The water levels in the monitoring wells were recorded using Solinst® Levellogger pressure transducers (model 3001 LT F15/M5 and LT F60/M20) over a two-hour period. The water levels from the monitoring wells were compared with relative sea level or tidal changes from the nearest tidal monitoring station (#7020) located in Sooke, BC. The website: <http://www.waterlevels.gc.ca/english/Canada.shtml>, was accessed for tidal information in the area.

5.7 Soil Sample Testing

Two soil samples that were collected during the drilling program were submitted for sieve analysis to Golder's materials testing laboratory. The sieve analyses were conducted to check the grain size distribution of each sample against visual field estimates conducted during drilling. The grain size distribution results and graphs attached in Appendix V.

5.8 Quality Assurance/Quality Control

5.8.1 Field Program

The following procedures were followed as part of quality assurance and quality control (QA/QC).

5.8.2 Chain-of-Custody Procedures

Chain-of-custody procedures were followed during all sampling events. Chain-of-custody forms accompanied all submissions to the laboratory. Each sample was given a unique sample control number (SCN), which was used to identify samples. Samples were submitted to the laboratory under chain-of-custody protocol using forms that did not identify the sampling locations, expected concentrations, or QA/QC samples, such as field replicate samples.

5.8.3 Field Replicates

The QA/QC target for the field investigation was approximately one replicate sample for every ten samples analyzed for a particular compound (*i.e.*, 10 % replicates) per sampling event. During the three monitoring events three replicate groundwater samples were submitted along with fifteen groundwater samples and six surface water samples.

The relative percent difference (RPD - the absolute difference between the two values, divided by their mean) of replicate analyses was used to evaluate the sample result variability. A RPD value of less than 20 percent for groundwater is generally considered an indication of acceptable analytical precision. Higher RPD values are evaluated on a case-by-case basis and may reflect compositional variability inherent to the sample, or variability related to sample collection, handling or analysis. Where the concentration of a given parameter is less than five times the laboratory method detection limit, the RPD is not calculated. For parameters with concentrations below five times the laboratory detection limit, the acceptable analytical precision is determined by calculating the absolute difference between replicate samples. The absolute difference should then be less than two times the laboratory detection limit (USEPA, 1994)⁵.

⁵ USEPA, 1994, "USEPA Contract Laboratory Program National Functional Guidelines for Inorganic Data Review", EPA-540/R-94-013

6.0 RESULTS

6.1 Existing Geology and Hydrogeology Review

6.1.1 Previous Studies

The results of Golder's 2001 assessment indicated that shallow groundwater quality at the Jordan River groundwater seep has been affected by leachate that is apparently being generated within WFP's wood waste landfill. The available analytical results report concentrations of chemical oxygen demand of 150 mg/L, pH of 5.5, phenols of 0.002 mg/L, chloride of 240 mg/L, tannin and lignin of 16.1 mg/L and 96h LC₅₀ acute toxicity test results >100 (%v/v). A low seepage flow rate was also reported. The report recommended that information on groundwater conditions and quality in the area of the landfill be gathered to better understand and address the potential impacts that leachate from the landfill may be having on groundwater quality in the area adjacent to the river.

6.1.2 Local Geology and Topography

The local bedrock geology as reported by the BC MOE Water Resources Atlas consists of sedimentary rocks consisting of conglomerate, sandstone, siltstone, shale of the Tertiary Carmanah Group.

Two different types of surficial geologic deposits occur in the area. The first unit appears to be a wave cut platform composed of gravel, cobbles and boulders inferred to have been deposited in a high-energy glacial regime. This unit has been observed to outcrop on the bank of the Jordan River north of the groundwater seep area and was also observed at the base of the slope located north of the landfill. The second surficial soils unit is composed of fluvial silt, sand and gravel, interpreted to be part of the Salish Sediments, which overlie the coarse grained sediments. These finer grained materials are interpreted to have been generated in a lower energy environment and laid down as a deltaic deposit by the Jordan River system. This unit extends laterally west and east from the mouth of the river.

An online review of the available registered water well records of the general area was conducted on the BC MOE Water Resources Atlas. The website reports twelve water wells in the region. Eleven wells are located greater than 400 m west of Jordan River and one well is located greater than 400 m northeast of the Site. The drill logs of these wells generally report granular surficial soil (silt, sand and gravel) generally less than 16 m in thickness overlying sedimentary bedrock.

6.1.3 Local Hydrogeology

Information from the BC MOE Water Resources Atlas was reviewed to identify the local aquifers at the Site. The local bedrock aquifer underlying the Site and the general area east of the Jordan River is Aquifer 606 IIIA (12). At this time, the Water Resources Atlas does provide aquifer classification for the general area west of the Jordan River to Port Renfrew, BC.

Aquifer 606 is designated as a bedrock aquifer reported to have low demands and productivity and high vulnerability (due to the absence of a low-permeability overlying confining layer). One domestic supply well (WTN 81862) located upgradient and approximately 500 m northeast of the site appears to be constructed within this bedrock aquifer. The surficial soils (approximately 0 to 15 m thick) reported at this well location are granular (silt, sand and gravel) in nature and represent an unconfined aquifer condition, hence, the high vulnerability classification for the aquifer. The surficial soils at the Site overlying the bedrock aquifer do not appear to have a formal MOE aquifer designation and classification.

6.2 Subsurface Investigation Results

6.2.1 Geologic Conditions

The sub-surface geologic conditions at the site were assessed during the borehole drilling program on September 20 and 21, 2006. Detailed descriptions of the soil conditions encountered in the boreholes are provided in the Record of Borehole and Monitoring Wells in Appendix I. The subsurface soil conditions encountered in the landfill area in MW06-01 included loose, sand and gravel fill (approximately 0.6 m thick) overlying loose, wood waste fill (approximately 7.3 m thick). Compact native sandy gravel, approximately 2.1 m thick, was observed underlying the wood waste fill. Approximately 1.3 m of native sand was observed to underlie the sandy gravel.

Wood waste fill was observed in MW06-02 from 0.6 m to 2.1 m below ground surface and appeared to be mixed with sand and gravel fill. The wood waste fill was not observed at the other borehole locations investigated at the Site. Loose granular fill (*i.e.*, sand and gravel, with cobbles and woody debris) was observed throughout the site in monitoring wells MW06-03 and MW06-06 and borehole BH06-05 and ranged in thickness from 0.6 m to 1.2 m.

In general, the fill soils and wood waste are underlain by native sand, sand and gravel, and gravel layers ranging in thickness from approximately 2.1 m in MW06-02 to a maximum proven thickness of 3.4 m in MW06-01. Native silt, containing trace amounts

of sand and gravel, was encountered in MW06-02, MW06-04, BH06-05 and MW06-06. The minimum penetrated thickness of the silt unit was 0.3 m in MW06-02 and ranged in thickness from 0.3 m in MW06-02 to a maximum proven thickness (bottom not seen) of approximately 3.6 m in BH06-05. The vertical extent of the silt layer was not confirmed and silt was not encountered in MW06-01 and MW06-03, however, the silt likely underlies most of the site, given the relatively consistent stratigraphic sequence encountered among the boreholes and monitoring wells in the investigation. The generalized sub-surface geologic conditions inferred from the boreholes and monitoring wells put down in the area of the wood waste landfill area are illustrated on conceptual geologic cross section A-A', in Figure 3.

A single Standard Penetration Test (SPT) was conducted in monitoring well MW06-01, within the lower, saturated sand soil horizon to assess apparent *in situ* density of the native sand unit underlying the wood waste. The resistance value of the SPT split spoon sampler was approximately 5 blows per 305 mm, suggesting the sand at this location is loose. Little resistance was encountered within this sand unit during drilling at the other monitoring well and borehole locations. Slight heaving sand conditions were encountered during drilling at MW06-01 and MW06-03. Drilling resistance encountered within gravel was greater than that within the sand units, suggesting the gravels were in a generally compact to dense state.

Two grain size analyses were conducted to assess the soil gradation for samples collected from MW06-02 and MW06-03 monitoring well borehole locations. The grain size analysis graphs and tables are provided in Appendix V.

The observed subsurface conditions varied between the boreholes. Similar and possibly greater variations in subsurface soil conditions should be expected throughout the Site. Thus, this variation should be considered when interpreting conditions between boreholes.

6.2.2 Hydrogeologic Conditions

Groundwater levels measured in the monitoring wells and approximate groundwater elevations from the September and December 2006 and March 2007 monitoring events are summarized in Tables II-1 and II-2 and Graphs II-1 and II-2, respectively in Appendix II.

The shallow subsurface geologic conditions encountered during the drilling program indicate the presence of an inferred shallow, unconfined groundwater aquifer that underlies the Site. This shallow aquifer is interpreted to be "perched" above a native silt unit that restricts the downward migration of groundwater. Therefore, the dominant groundwater flow path is inferred to be within the shallow sand and gravel unit that overlies the silt.

Limited continuous water level monitoring was conducted on October 5, 2006 in monitoring wells MW06-02 and MW06-04 for a two hour period to determine whether or not these wells were tidally-influenced. Groundwater elevations (metres above mean sea level or "m amsl") based on the groundwater levels measured from the three monitoring events indicate that minor tidal influence in monitoring well MW06-02 and moderate tidal influence in MW06-04 is inferred.

The water levels from the monitoring wells were compared with relative sea level or tidal changes from the nearest tidal monitoring station (#7020) located in Sooke, BC.

The results of the groundwater level monitoring suggest that variations in groundwater levels in the other monitoring wells installed within and outside the landfill area are relatively minor. Apparent groundwater mounding was not observed in the course of the monitoring program.

6.3 Single-Well Response Testing and Groundwater Seepage Rate Estimate

6.3.1 Single-Well Response Testing

Hydraulic conductivity values estimated from the single-well response tests were calculated using the Bouwer-Rice (1976) solution for unconfined aquifers in the AQTESOLV® software. The estimated hydraulic conductivity values from the single well response tests are summarized in Table 1. The results of the analyses indicate the estimated hydraulic conductivities range from 3×10^{-7} m/s within sand, to 2×10^{-4} m/s within sand and gravel. It should be noted that recent research indicates that single well response tests underestimate soil hydraulic conductivities and that actual values can be between 2 to 10 times higher than those reported by this method.⁶

TABLE 1: Estimates of Hydraulic Conductivity from Single-Well Response Testing

Monitoring Well	Test Type	Stratigraphic Unit	Hydraulic Conductivity K (m/s)
MW06-01	Rising Head	Sand and Gravel	2×10^{-4}
MW06-02	Rising Head	Sandy Gravel	2×10^{-4}
MW06-03	Rising Head	Sand	3×10^{-7}
MW06-04	Rising Head	Gravel	2×10^{-2}
MW06-06	Rising Head	Sand and Gravel	4×10^{-5}

⁶ Schulze-Makuch D., D. A. Carlson, D. S. Cherkauer, and P. Malik. 1999. Scale dependency of hydraulic conductivity in heterogeneous media. *Ground Water* 37, no. 6: 904-919

Although monitoring well MW06-04 appears to be tidally influenced, the effects of tidal influence during the single well response testing are considered negligible given the short duration of the test.

6.3.2 Groundwater Seep Flow Rate Estimate

The rate of groundwater seepage was estimated during the three monitoring events at the groundwater seep. The groundwater seep area along the Jordan River appears to be periodically submerged during high tides and was estimated and sampled during low tide conditions for the three monitoring events. The estimated flow rate from the groundwater seep at the time of sampling was approximately 1.8 L/min in September 2006 and 17 L/min in December 2006. The groundwater seep flow rate was not estimated in March 2007 due to the lateral extent and sheet flow nature (approximately 3.5 m wide) of the seep in the area. Groundwater seepage was not observed to be present in other river bank locations at the Site during the three Golder monitoring events.

6.4 Estimated Groundwater Gradient and Flow Direction

Groundwater levels measured in the monitoring wells for the three monitoring events are summarized in Tables II-1 and II-2 and Graphs II-1 and II-2 in Appendix II.

The apparent horizontal groundwater hydraulic gradient, based on available static water levels measured for the three monitoring events ranges from approximately 0.006 m/m in October 2006 to 0.011 m/m in March 2007 or approximately 0.6 % to 1.1 % and towards the southwest to northwest (approximately azimuth of 250° to 305°, respectively). The gradient and direction calculations were estimated based on relative groundwater elevations in monitoring wells MW06-01 and MW06-03 and the groundwater seep elevation. Additional calculations were carried out to assess the apparent horizontal groundwater gradient among monitoring wells MW06-01, MW06-02 and MW06-03. These results indicate that the horizontal groundwater hydraulic gradient ranges from approximately 0.003 m/m in December 2006 to 0.011 m/m in October 2006 or approximately 0.3 % to 1.1 % towards the south and west (approximately azimuth of 200° to 290°). The more likely flow path is thus inferred to be towards the southwest to northwest direction (approximately azimuth of 250° to 305°) due to a slightly steeper hydraulic gradient.

Monitoring well MW06-06 is inferred to be hydraulically connected cross-gradient and located outside the groundwater migration pathway of wood waste landfill.

6.5 Groundwater Flow Direction and Velocity

The estimated direction of dominant groundwater flow is generally anticipated to be towards groundwater seep located southwest to northwest (approximately 250° to 305°) of the landfill area. Using the highest value for the hydraulic conductivity value of 2×10^{-4} m/s estimated for native gravel and sand in monitoring well MW06-01, an inferred seasonal groundwater gradient range from of 0.006 to 0.011 metres per metre, and an effective porosity of 0.3 (assumed value, typical of sand and gravel), the average linear groundwater velocity was calculated as follows:

- $V = Ki/n$

Where:

- V = average linear groundwater velocity (m/s),
- K = saturated hydraulic conductivity (m/s),
- i = hydraulic gradient (dimensionless), and,
- n = the average estimated porosity of the soil (dimensionless).

Under these conditions, the estimated average linear groundwater velocity range within the shallow the aquifer is 0.35 m/d to 0.63 m/day. The estimated distance to the abandoned spillway (Site #1) and groundwater seep (Site #2) location from the inferred edge of the wood waste landfill is approximately 50 m and 20 m, respectively. Thus, the estimated travel time to the abandoned spillway and groundwater seep locations along the groundwater flow path from the edge of the landfill area would be approximately 79 to 143 days and 32 to 57 days, respectively.

6.6 Water Quality Monitoring Results

6.6.1 Quality Assurance and Quality Control

Results of Field Replicates in Groundwater

Fifteen groundwater samples, three groundwater seep samples and three surface water samples were collected from the Site over the three monitoring events in 2006 and early 2007. One groundwater field replicate was collected from each monitoring event and was submitted for laboratory analysis. The RPD results for the three monitoring events are provided in Table III-3 in Appendix III.

The RPD results, for the groundwater field replicate sample for MW06-1, ranged from 0 % to 14.3 % for the dissolved and total metals and other chemical parameters to which the RPD was applicable. The results were within the accepted RPD threshold of 20 %.

The RPD calculation was not applied to parameters that were below laboratory method detection limits or parameters that were less than five times the laboratory method detection limit as discussed in Section 5.8.3. Replicate sample parameters that were excluded from the RPD calculation had absolute difference values that were less than two times the laboratory detection limit and therefore were within the acceptable limit except for total barium ([0.079]) and total cobalt ([0.083]).

The RPD results, for the groundwater field replicate sample for MW06-04, ranged from 0 % to 45.1 % for the dissolved and total metals and other chemical parameters to which the RPD was applicable. Generally, most results were within the accepted RPD threshold of 20 % except for bromide (45.1 %), total copper (29 %), phosphorus (23 %) and uranium (22 %). Replicate sample parameters that were excluded from the RPD calculation had absolute difference values that were less than two times the laboratory detection limit and therefore were within the acceptable limit.

The RPD results, for the groundwater field replicate sample for MW06-06, ranged from 0.6 % to 64 % for the dissolved and total metals and other chemical parameters to which the RPD was applicable. Generally, most results were within the accepted RPD threshold of 20 % except for COD (64 %), total aluminum (61 %), chromium (58 %), copper (56 %), iron (44 %), magnesium (35 %), manganese (36 %), silicon (36 %), strontium (20.3 %), titanium (57 %) and zinc (55 %). Replicate sample parameters that were excluded from the RPD calculation had absolute difference values that were less than two times the laboratory detection limit and therefore were within the acceptable limit except for total barium ([0.047]).

The RPD calculation was applied to both dissolved and total metals whereas typically the calculations would be applied to dissolved metals for the groundwater samples. The analytical results of groundwater samples that were analyzed for dissolved metals indicate that the samples were within the accepted RPD threshold of 20 %. The groundwater samples collected and analyzed for total metals were not field filtered and were persevered with nitric acid prior to analysis. As the field observations and analytical results indicate, the RPD calculations are highly variable. Thus, the RPD is expected to be highly variable due to sediment entrainment, sequential sampling and no blended prior to laboratory analysis.

Chemistry Laboratory QA/QC Program

ALS analyzed the groundwater and surface water samples for the physical and chemical parameters collected during the three quarterly monitoring events (September and December 2006 and March 2007). A review of the ALS laboratory analytical data determined that ALS appeared to have met their own standards and internal RPD target of 20 % for all parameters tested. The laboratory RPD results are provided in Tables III-4 to III-6 in Appendix III.

QA/QC Measures during Toxicity Testing

QA/QC measures incorporated into the acute rainbow trout toxicity tests include the use of standardized test procedures, negative (clean) controls tested with every sample, reference toxicant tests performed on each batch of fish used for testing, and independent review of all test data prior to report release. For the samples in question, test acceptability criteria were met, negative control results were acceptable, water quality variables were within acceptable ranges, and reference toxicant test results were within expected ranges based on historical performance. There was one anomaly associated with the samples tested under Work Order 0600589. For each of the two samples tested, one test treatment at an intermediate test concentration had poor survival whereas all the other treatments had 100 % survival. This poor survival was subsequently traced to accidentally using two test containers that had previously been used for conducting tests with zinc and were therefore likely contaminated. Therefore, the results for those two test concentrations were excluded from calculation of the final test results.

Ministry of Environment Samples

Sample results provided by the Ministry of Environment were not provided with information sufficient to enable Golder's independent review of QA/QC. The data were therefore relied upon on a good faith basis.

6.6.2 Surface Water Quality

Surface water samples were collected from the two surface water monitoring locations, Abandoned Spillway (Site #1) and Groundwater Seep (Site #2), for the three monitoring events. Surface water chemistry results from the monitoring program are presented in Appendix III (Table III-1).

The abandoned spillway and groundwater seep locations include previous monitoring by the MOE and current monitoring program by Golder. Analytical parameters related to wood waste leachate included: pH, COD, total phenols, tannin and lignin and sulphide. The applicable BCWQG were compared with the total metals analytical results, as dissolved metals were not analyzed as part of this sampling and analytical program. Laboratory detection limits for some of the groundwater quality parameters measured in this monitoring program were higher than the applicable BCWQG due to matrix interferences encountered during laboratory analysis. It is inferred that the matrix interferences encountered were due to salt water influences in the samples.

Abandoned Spillway (Site #1)

Monitoring was previously conducted by the MOE with toxicity testing conducted in June 1995 and toxicity, pH, chloride, COD, phenols and tannin and lignin analyzed in April 1997 and March 1999. Total metal analysis was not conducted during the three MOE monitoring events in June 1995, April 1997 and March 1999.

The analytical results for the abandoned spillway surface water monitoring location were compared with the BCWQG for marine aquatic life due to tidal influence conditions in that area. Field conductivity ranged from 302 $\mu\text{S}/\text{cm}$ in March 2007 to 19,900 $\mu\text{S}/\text{cm}$ in September 2006. Salinity was calculated using the field conductivity measurements. The calculated salinity for the abandoned spillway range from 0.16 ‰ in March 2007 to 12.7 ‰ in September 2006 indicating freshwater (<1.5 ‰) to brackish water conditions. Total metals were analyzed for the abandoned spillway surface water samples for the three monitoring events. Most total metals analyzed were below the BCWQG for marine life except for boron and copper in September 2006 and March 2007, respectively. Certain elemental concentrations (boron, calcium, magnesium, potassium, sodium and strontium) as well as hardness reported for the lower abandoned spillway monitoring location during the September 2006 monitoring event were significantly higher than in the abandoned spillway location monitored during the December 2006 and March 2007 monitoring events. These elements are associated with sea water and appeared to influence the lower part of the spillway during the September 2006 monitoring event. The lower concentrations of these elements reported from the December 2006 and March 2007 monitoring events suggest that limited sea water influence may have been present in the spillway during these events.

Field-measured pH ranged from 6.1 (March 2007) to 7.0 (September 2006), COD concentrations ranged from 3 mg/L (March 1999) to 981 mg/L (September 2006), total phenol concentrations ranged from <0.0010 mg/L (April 1997, March 1999, September 2006 and March 2007) to 0.0100 mg/L (December 2006) and tannin and lignin concentrations ranged from 0.4 mg/L (April 1997) to 1.6 mg/L (December 2006). Although sulphide concentrations were less than laboratory detection (<0.02 mg/L) for the three sampling events, the detection limit was set higher than the BCWQG (0.002 mg/L). Field oxidation/reduction potential ranged from -3 mV in September 2006 to 76 mV in December 2006 suggesting a range of slightly reducing to slightly oxidizing conditions.

Surface water collected from the abandoned spillway monitoring locations in 2006 and 2007 was observed to be visually clear with no odours or apparent sheen.

Groundwater Seep (Site #2)

Monitoring was conducted by the MOE where toxicity testing, pH, chloride, COD, phenols and tannin and lignin were analyzed in June 2000 and April 2001. Total metal analysis was not conducted from the two monitoring events carried out by the MOE.

The analytical results for the groundwater seep surface water monitoring location were compared with the BCWQG for freshwater aquatic life. Total metal analytical results for the groundwater seep samples in 2006 and 2007 were generally below the applicable BCWQG for freshwater aquatic life, except for copper, iron, manganese and phosphorus. Hardness concentrations reported are considered low and ranged from 63.2 mg/L to 119 mg/L for the three Golder monitoring events. Salinity was also calculated using the field conductivity measurements for this location. Field conductivity measured ranged from 194 $\mu\text{S}/\text{cm}$ in December 2006 to 350 $\mu\text{S}/\text{cm}$ in September 2006. The calculated salinity for the groundwater seep range from 0.09 ‰ in December 2006 to 0.18 ‰ in September 2006 indicating freshwater (< 1.5 ‰) conditions.

Field measured pH was mildly acidic and ranged from 5.5 (June 2000) to 6.0 in March 2007. The April 2001 monitoring indicated a pH value of 7.0. Chloride concentration (240 mg/L) for June 2000 and sulphide concentrations of 0.092 mg/L and 0.191 mg/L for December 2006 and March 2007, respectively, exceeded the BCWQG. COD concentrations ranged from 42 mg/L (April 2001) to 150 mg/L (June 2000), total phenol concentrations ranged from <0.0010 mg/L (April 2001, September 2006 and March 2007) to 0.043 mg/L (December 2006) and tannin and lignin concentrations ranged from 0.63 mg/L (April 2001) to 16.1 mg/L (June 2000). Field oxidation/reduction potential measured ranged from -20 mV in December 2006 to -48 mV in March 2007 suggesting slightly reducing conditions.

Surface water collected from the groundwater seep monitoring location generally has a slight cloudy brown appearance with apparent iron staining observed on the surface of the water. Slight hydrogen sulphide ("rotten egg") odours were present during the three Golder monitoring events.

6.6.3 Groundwater Quality

Groundwater samples were collected from the five monitoring well locations (MW06-01 to MW06-04 and MW06-06) for the three monitoring events in 2006 and 2007. Analytical parameters related to wood waste leachate include: pH, COD, total phenols, tannin and lignin and sulphide.

Laboratory results for groundwater were compared with both the CSR freshwater and marine aquatic life standards. Although dissolved and total metal analytical results were compared to the CSR water quality standards, the CSR standards specify that groundwater is to be assessed using dissolved metal concentrations and surface water be assessed using total metal concentrations. Laboratory detection limits for some of the groundwater quality parameters that were assessed in this monitoring program were higher than the applicable CSR standard due to matrix interferences encountered during laboratory analysis.

Field-measured groundwater pH for the three events ranged from 5.3 (MW06-02 and MW06-03) to 6.0 (MW06-04) in the monitoring wells located within and near the landfill. The field measured pH for the cross-gradient background monitoring well (MW06-06) was 5.7 to 5.9 for the three events.

COD concentrations ranged from 173 mg/L (MW06-03) to 1,840 mg/L (MW06-02) for the three events. COD concentrations in the cross-gradient background monitoring well (MW06-01) ranged from 95 mg/L to 330 mg/L for the three events. Total phenol concentrations exceeded the CSR aquatic life freshwater and marine standard of 0.01 mg/L in monitoring well MW06-03 in September 2006 and in wells MW06-01 to MW06-04 in December 2006. Total phenols were detected in MW06-06 during the December 2006 monitoring event and below laboratory detection for all other monitoring events. Tannin and lignin concentrations ranged from 2.13 mg/L in MW06-04 to 29.4 mg/L in MW06-03 for the well located within or near the landfill. Currently, there are no regulatory standards or guidelines for COD and tannin and lignin nor are there reliable, established concentration-effect relationships for toxicity to fish from these substances. Sulphide concentrations range from less than laboratory detection (<0.02 mg/L) in MW06-06 (September 2006) to 0.13 mg/L in MW06-02 and 0.10 mg/L in MW06-06 (December 2006). Sulphide exceedances were reported in MW06-04 for the three monitoring events and may be due to sea water (high in sulphate) intrusion with sulphate reduction taking place. Exceedances of CSR sulphide standards were reported in the five monitoring wells in at least one event.

Generally, dissolved metal concentrations were low or below laboratory detection for dissolved metals analyzed in monitoring wells MW06-01, MW06-02, MW06-04 and MW06-06. The analytical results from monitoring well MW06-03 for September 2006 indicate dissolved cadmium (September and December 2006) and selenium (September 2006) concentrations of 0.00033 mg/L and 0.0105 mg/L, respectively, exceeded the freshwater aquatic life standard but not the marine life standard. The dissolved copper concentration reported for September and December 2006 exceeded both the CSR freshwater and marine aquatic life standards with concentrations of 0.0525 mg/L and 0.0201 mg/L, respectively. Dissolved lead concentration exceeded the CSR marine

aquatic life standard with a reported concentration of 0.0333 mg/L for September 2006. The results for December 2006 for MW06-03 report cadmium (0.00064 mg/L) and copper (0.0201 mg/L) concentrations that exceed the freshwater and marine standards, respectively.

The analytical results for monitoring well MW06-04 were elevated in comparison to the other groundwater monitoring wells due to the inferred sea water influence in the area of the well. The range of analytical results reported from the three monitoring events for MW06-04 include hardness (460 to 2,510 mg CaCO₃/L equivalent), chloride (1,880 mg/L to 6,540 mg/L), sulphate (268 mg/L to 870 mg/L), dissolved magnesium (92 mg/L to 492 mg/L), dissolved potassium (48 mg/L to 134 mg/L) and dissolved sodium (1,170 mg/L to 3,220 mg/L) which indicates sea water influence.

Total metals were also analyzed from the groundwater samples collected from the monitoring wells for comparative purposes only. As expected, the total metal concentrations were generally higher than the dissolved metal concentrations in the wells that were tested and exceeded a number of CSR standards. The presence of the total metals exceedances are in part, attributed to the sediment content of the samples. Purged water from the five monitoring wells generally had a silty brown appearance and mild H₂S “rotten egg” odour was detected in MW06-02 and MW06-04 during the December 2006 and in MW06-03 and MW06-04 the during March 2007 monitoring events. Groundwater chemistry results from the monitoring program are presented in Appendix III (Table III-2).

6.6.4 Toxicity Testing Results

Rainbow trout acute toxicity testing was conducted on the abandoned spillway and groundwater seep water samples collected from the MOE and Golder monitoring events. Results of previous MOE toxicity testing of the groundwater seep and abandoned spillway locations did not specify what fish species was used for the testing (rainbow trout presumed). The historic MOE laboratory analytical reports were not available. The acute toxicity testing reports for the current assessment (2006 and 2007) are provided in Appendix V. The analytical results of the toxicity testing on the abandoned spillway and groundwater seep sample locations are presented in Table 2.

TABLE 2: Summary of Rainbow Trout Toxicity Testing Results

Sampling Date	96h LC ₅₀ (Percent Sample, v/v)	
	Abandoned Spillway	Groundwater Seep
June 1995 (MoE)	100 (NR)	-
April 1997 (MoE)	100 (NR)	-
March 1999 (MoE)	100 (NR)	-
June 14, 2000 (MoE)	-	>100 (NR)
April 2001 (MoE)	-	100 (NR)
September 27, 2006 (Golder)	100 (50)	>100 (40)
December 19, 2006 (Golder)	>100 (0)	>100 (0)
March 22, 2007 (Golder)	>100 (0)	>100 (0)

Note: () indicate the % fish mortality at 100 % toxicity test concentration where known. (NR) indicates the information was not reported. - indicates that toxicity testing was not conducted at the monitoring location.

Toxicity testing laboratory results for the abandoned spillway indicate that some fish mortality was present in the September 2006 monitoring event where five out of ten fish survived the test at 100 % concentration. No fish mortality at 100 % concentration was recorded for the December 2006 and March 2007 monitoring events. Laboratory information from the MOE monitoring was not available to confirm mortality rates.

The analytical results of the toxicity testing on the groundwater seep sample location are 100 % for April 2001 and greater than 100 % for the June 2000, September 2006, December 2006 and March 2007. Toxicity testing laboratory results for the groundwater seep indicate that some fish mortality was present in the September 2006 monitoring event where six out of ten survived the testing at 100 % concentration. No fish mortality at 100 % concentration was recorded for the December 2006 and March 2007 monitoring events. Laboratory information from the MOE monitoring was not available to confirm mortality rates for June 2000 and April 2001 monitoring events.

7.0 DISCUSSION

7.1 Hydrogeologic Conditions

Groundwater monitoring wells installed in the area of the existing wood waste landfill indicate that wood waste has a thickness of approximately 7 m (MW06-01) in the centre of the landfill. A wood waste mixed with sand and gravel layer of approximately 1.5 m thick was encountered in MW06-02 with no wood waste encountered within the other borehole locations. The lateral extent of the landfill is inferred to be bounded by the mature conifer forested areas located to the south and east of the landfill and the moderate to steep slopes along the north margin of the landfill. Some sand and gravel was deposited as layers and lenses within the wood waste. Native soil conditions in the area are dominated by sand and gravel with localized layers of silty sand, silt and organic silt (e.g., MW06-03, BH06-05, and MW06-06).

Groundwater within the native soils is apparently “perched” on a unit of grey silt, containing traces of gravel, sand and clay, which underlies the sand and gravel in the upper portion of the native soil stratigraphy. The silt is the lowest stratigraphic unit identified during Golder’s subsurface investigations at the Site. The vertical extent of the silt is unknown. Groundwater levels in the wells are located at depths of about 1 m to 3 m below existing ground surface (based on the three monitoring events) except for monitoring well location MW06-01, in the interior of the landfill. The groundwater within the wood waste landfill is at a depth of about 8 m in MW06-01, where the wood waste and capping materials have been deposited in a mound that rises approximately 10 m or more above the surrounding natural grade.

Based on the three monitoring events, groundwater flow direction from the landfill area is inferred to range from the south to the northwest. Based on monitoring wells MW06-01 and MW06-03 and the groundwater seep location, the direction of dominant groundwater flow ranges from southwest to northwest (between approximately 250° to 305°) with a relatively shallow gradient (about 0.6 % to 1.1 %). Estimated horizontal groundwater travel times are likely in the order of less than 1 m/day to perhaps several metres per day based on inferred groundwater flow velocities and native soil permeability estimates. Groundwater mounding in the area of the landfill did not appear to be significant (approximately 0.7 m difference between September 2006 and March 2007) based on groundwater level observations made during the three monitoring events.

The Groundwater seep location appears to be localized and relatively low flow with measured flow rates of 1.8 L/min in September 2006 to 17 L/min in December 2006. The seepage flow rate in March 2007 may be slightly higher than the December seepage flow estimate based on the lateral extent of the seep.

7.2 Water Quality Monitoring

7.2.1 Groundwater Quality Monitoring

Groundwater quality in monitoring wells installed inside, peripheral to and outside the existing wood waste landfill demonstrate that leachate from buried wood waste has had some localized, minor effects on groundwater quality at the Site. Potential impacts to groundwater quality were evaluated by assessing the levels of several chemical parameters associated with wood waste leachate in groundwater monitoring wells in the area of the existing wood waste landfill.

Monitoring well MW06-06, which is located approximately 100 m east of the inferred limits of the landfill (Figure 2), was used as a cross-gradient “background” groundwater quality monitor based on its position outside of the inferred zone of influence of the wood waste landfill. MW06-06 is also considered to be outside the area of potential influence of seawater associated with the tidally influenced Jordan River. Table 2 provides a summary of the wood waste leachate-related parameters measured with respect to groundwater sampled and analyzed from the various monitoring wells.

TABLE 3: Summary of Groundwater Quality Parameters of Interest

Parameter (mg/L)	Monitoring Well Location				
	MW06-01	MW06-02	MW06-03	MW06-04	MW06-06
	Middle of landfill	Edge of landfill	In groundwater flowpath down-gradient of landfill	River bank and sea water influenced area	“Background” groundwater quality
pH	5.6 to 5.8	5.3 to 5.6	5.3 to 5.8	5.7 to 6.0	5.7 to 6.0
COD	310 to 705	177 to 1,840	150 to 661	220 to 618	95 to 330
Phenols	<0.0010 to 0.0320	<0.0010 to 0.0320	<0.0010 to 0.0510	<0.0010 to 0.0220	<0.0010 to 0.0050
Tannin and lignin	9 to 24.5	7 to 13	8 to 29.4	2.13 to 6	1 to 4.09
Dissolved Iron	19.3 to 21.9	6.38 to 13.6	5.04 to 16.2	2.44 to 4.96	2.63 to 5.60

Based on the above table, the following observations are made:

- Monitoring wells, MW06-01, MW06-02 and MW06-03 that may be influenced by wood waste leachate from the landfill has slightly more acidic groundwater based on lower pH measurements than monitoring well MW06-06 which is not inferred to be influenced by wood waste leachate and is outside of the landfill area.
- Monitoring well MW06-01, located in the middle of the landfill, had slightly higher COD concentrations and comparable phenol concentrations to the cross-gradient background monitoring well MW06-06.
- Monitoring well MW06-02, located at the edge of the landfill near the river bank groundwater seep, reported the highest COD concentration (December 2006).
- Monitoring well MW06-03, located in the groundwater flowpath down gradient of the landfill, had comparable COD and tannin and lignin concentrations with MW06-01.
- Monitoring well MW06-04, interpreted to be strongly influenced by saltwater from the tidally influenced Jordan River, had tannin and lignin and COD concentration ranges of 2.1 mg/L to 6 mg/L and 220 mg/L to 618 mg/L, respectively. Sulphide exceedances were also reported in the well for the three monitoring events of 2006 and 2007 and may indicate sulphate reduction taking place from sea water influences.
- Phenols were above the laboratory detection limit in the monitoring wells during the December 2006 monitoring and below detection for the September 2006 and March 2007 monitoring events, except for MW06-03, where phenols were detected in September 2006.
- Dissolved iron concentrations in groundwater may reflect mobilization of iron from sediments and/or wood waste contributions. Highest dissolved iron concentrations were in MW06-01 (centre of landfill). Second highest dissolved iron concentrations were in MW06-02 and MW06-03, (directly downgradient of landfill). Lowest dissolved iron concentrations were in MW06-04 (tidally influenced) and MW06-06 (background).

The analytical results from the monitoring well groundwater samples report concentrations of dissolved anions that are generally below than the applicable CSR standards except for sulphide which exceeded the CSR standards in all well locations tested including background monitoring well MW06-06. Phenol concentrations exceeded the CSR standards in monitoring well locations MW06-01 to MW06-04 and not in

background monitoring location MW06-06. Other wood waste leachate parameters, COD and tannin and lignin, are detectable from groundwater sampled in the landfill area including the background groundwater monitoring location. These parameters are likely associated with the wood waste landfill but may occur naturally as suggested by the analytical results in the background location (MW06-06).

The analytical results from monitoring wells MW06-01, MW06-02, MW06-04 and MW06-06 report concentrations of dissolved metals that are lower than the applicable CSR standards. Exceedances of CSR freshwater and marine aquatic life standards for dissolved metals were indicated by the laboratory analytical results for MW06-03 for cadmium and copper for September and December 2006, lead and selenium for September 2006 and no exceedances reported in March 2007.

7.2.2 Surface Water Quality Monitoring

Surface water quality monitoring was previously conducted by the MOE at the Abandoned Spillway (Site #1) and Groundwater Seep (Site #2) locations. The parameters tested were pH, COD, phenols and tannin and lignin and toxicity. The laboratory analytical results for the three monitoring events conducted in 2006 and early 2007 for the abandoned spillway location, groundwater seep and monitoring well MW06-04 indicate the water quality at these locations is affected by some mixing with seawater from the tidally influenced Jordan River system. A summary of wood waste leachate-related parameters in the surface water sampled and analyzed from the previous MOE monitoring and from the current monitoring program is provided in Table 4.

TABLE 4: Summary of Surface Water Quality Parameters of Interest

Parameter (mg/L)	Surface Water Location	
	Abandoned Spillway	Groundwater Seep
pH	6.1 to 7.0	5.5 to 7.0
COD	3 to 55. One anomaly of 981	42 to 150
Phenols	<0.0010 to 0.0100	<0.0010 to 0.0430
Tannin and lignin	0.4 to 2	9 to 16.1. One anomaly of 0.63
96hr LC ₅₀ (%)	100 and >100	100 and >100

Based on the above table, the following observations are made:

- The lower pH range of the groundwater seep appears to be slightly more acidic than the lower range from the abandoned spillway; however, mildly acidic conditions are not uncommon in coastal BC areas. Both locations report neutral pH conditions during one monitoring event.
- The COD range of the abandoned spillway appears to be lower than the groundwater seep. The COD concentration of 981 mg/L in the abandoned spillway location in September 2006 is considerably higher than previous sampling events and may be related to the apparent influx of sea water.
- Phenols were generally below laboratory detection in the two locations except the June 2000 MOE monitoring event for the groundwater seep and the December 2006 monitoring event for both locations.
- Tannin and lignin concentrations were generally lower in the abandoned spillway compared with the groundwater seep. The tannin and lignin concentration of 0.63 mg/L in the groundwater seep occurred in April 2001 and may be the result of winter dilution.

The analytical results of the toxicity testing on the abandoned spillway locations were 100 % for four monitoring events and greater than 100 % for the two most recent monitoring events (December 2006 and March 2007). The analytical results of the toxicity testing on the groundwater seep locations were greater than 100 % for four monitoring events and 100 % for the MOE monitoring event in 2001.

The acute toxicity testing results indicate that water collected from the two monitoring locations in September 2006 caused some fish mortality as indicated in Table 2. The greater than 100 % 96h LC₅₀ acute toxicity testing results from the December 2006 and March 2007 monitoring events indicate that water collected from these two monitoring locations resulted in no mortality to the rainbow trout fish species used for the testing.

8.0 SUMMARY AND RECOMMENDATIONS

The following conclusions are formed based on the hydrogeologic and water quality monitoring program for the WFP Jordan River Operations wood waste landfill site:

- The analytical results for the groundwater seep and abandoned spillway locations indicate that the water analyzed exceeds some total metals and pH when compared to the BCWQG for freshwater and marine aquatic life.
- The analytical results for the groundwater seep location indicate that the water analyzed exceeds BCWQG for freshwater life for sulphide (2 events) and chloride (1 event).
- The abandoned spillway and MW06-04 monitoring locations are tidally influenced.
- The analytical results for the groundwater monitoring well locations indicate that the groundwater analyzed exceeds some dissolved metals in MW06-03 when compared to the CSR for freshwater and marine aquatic life.
- Groundwater analytical results from each monitoring well, including the background monitoring location, indicated exceedances of the CSR standards for sulphide and phenols for at least one monitoring event.
- Discharge of wood waste to the landfill area has not been carried out by WFP since approximately 1999.

Golder Associates have carried out this scope of work on the basis of information requirements provided by the BC Ministry of Environment. There are a number of instances where ambient water quality guidelines have been exceeded. These guidelines provide useful, though conservative, screening benchmarks. The exceedances should not be interpreted as a finding of an environmental problem; however, they indicate that further evaluation is warranted to determine if these exceedances are significant. In some cases, for example, the exceedance is the result of a detection limit issue or may be the result of analytical method interference by salt water. Further evaluation is recommended.

The toxicity test results have found that the 96h LC_{50} is $\geq 100\%$ (v/v). In colloquial references to toxicity test results, such findings are often termed “non-toxic”. However, Golder notes that fish mortality did result in some of the tests. The cause of this mortality is presently unknown although it could simply be the result of marine water influx into this intertidal area. While rainbow trout are capable of adaptation to marine water (which

naturally occurs over time in an estuary), a sudden salt-water challenge can result in mortality due to an inability of the fish to regulate electrolyte levels.

The selection of the 96h LC₅₀ toxicity test was carried out on the basis of the BC Ministry of Environment's letter dated May 21, 2002. The rainbow trout acute lethality test is a common test that is applied to freshwater effluents. Its application to surface waters as a tool to ascertain whether or not "pollution" has been caused (*Environmental Management Act*) or whether a substance is a "deleterious substance" (*Fisheries Act*) is limited because the effects measured (mortality) would not adequately describe whether the water is "polluted".

Notwithstanding the administrative disposition of the existing permit, Golder recommends that a more robust and appropriate monitoring program be designed to provide data that will allow WFP to identify if broader statutory implications exist at the site. The process of designing such a program should take into account the ecological uses of the areas being sampled, the characteristics of the receiving environment and the use of appropriate biological and chemical analyses to reflect those uses. For example, the Abandoned Spillway is clearly connected to the estuary of the Jordan River in a manner such that fish movement in and out of the Abandoned Spillway may be taking place. It should be determined if this is fish habitat and, if so, the monitoring program should be structured on that basis.

For reasons of cost efficiency and beneficial information yield, it is further recommended that following the preparation of a study plan, it be provided to the BC Ministry of Environment, Environment Canada, and Fisheries and Oceans Canada for their input, prior to carrying out the plan.

9.0 CLOSURE

We trust that the information contained herein meets your current needs. Please do not hesitate to contact the undersigned directly with any questions you may have with respect to this report.

Yours very truly,

GOLDER ASSOCIATES LTD.

Dwayne Smith, B.Sc., GIT
Project Geologist

Lee Nikl, M.Sc., R.P.Bio.
Associate/Senior Environmental Scientist

Reviewed by:

Jeffrey Phillipone, Ph.D., P.Geo.
Associate, Project Manager

DS/LN/JF/knb

Attachments: Figures, Appendices

N:\FINAL\2007\1414\07-1414-0039 WFP MONITORING JORDAN R\FINAL REPORT\RPRT 12-12-07 HYDROGEOLOGICAL ASSESS JORDAN RIVER.DOC

10.0 LIMITATIONS AND USE OF THIS REPORT

Standard of Care: Golder Associates Ltd. (Golder) has prepared this report in a manner consistent with that level of care and skill ordinarily exercised by members of the engineering and science professions currently practicing in British Columbia, subject to the time limits and physical constraints applicable to this report. No other warranty, express or implied is made.

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Unless otherwise stated, the suggestions, recommendations and opinions given in this report are intended only for the guidance of the Client in the design of the specific project. The extent and detail of investigations, including the number of test holes, necessary to determine all of the relevant conditions which may affect construction costs, techniques and equipment choice, scheduling and sequence of operations would normally be greater than has been carried out for design purposes. Contractors bidding on, or undertaking the work, should rely on their own investigations, as well as their own interpretations of the factual data presented in the report, as to how subsurface conditions may affect their work.

Soil, Rock and Groundwater Conditions: Classification and identification of soils, rocks, and geologic units have been based on commonly accepted methods employed in the practice of geotechnical engineering and related disciplines. Classification and identification of the type and condition of these materials or units involves judgement, and boundaries between different soil, rock or geologic types or units may be transitional rather than abrupt. Accordingly, Golder does not warrant or guarantee the exactness of the descriptions.

Special risks occur whenever engineering or related disciplines are applied to identify subsurface conditions and even a comprehensive investigation, sampling and testing program may fail to detect certain conditions. The environmental, geologic, geotechnical, geochemical and hydrogeologic conditions that Golder interprets to exist between sampling points may differ from those that actually exist.

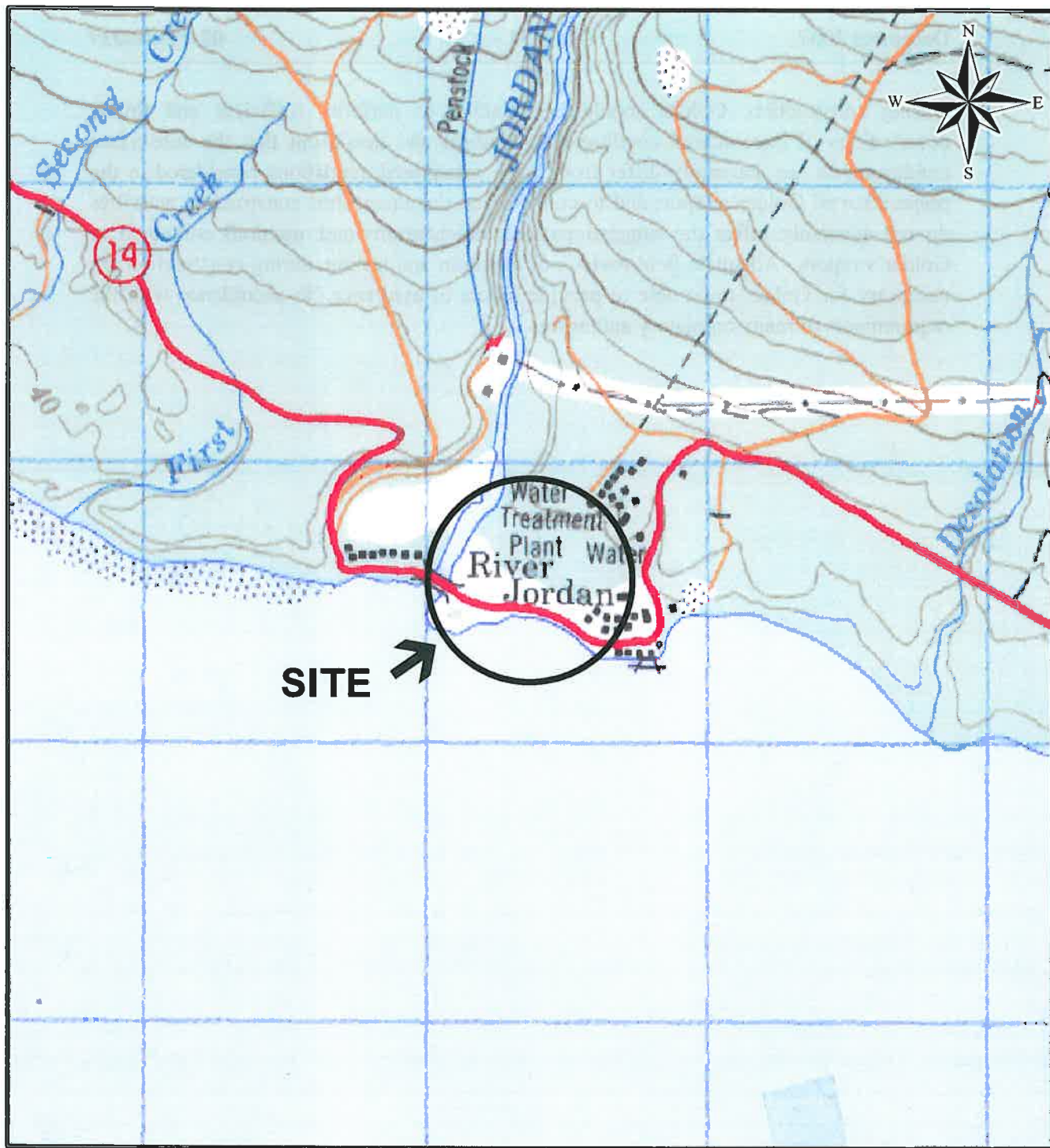
Groundwater conditions shown in the factual data and described in the report are the observed conditions at the time of their measurement. Groundwater conditions may vary between reported locations and can be affected by annual, seasonal and special meteorological conditions or tidal fluctuations. Groundwater conditions may also be altered by construction activity on or in the vicinity of the project site.

Sample Disposal: All contaminated samples and materials shall remain the property and responsibility of the Client for proper disposal. Golder will dispose of all uncontaminated soil and/or rock samples 90 days following issue of this report or, upon written request of the Client, will store uncontaminated samples and materials at the Client's expense.

Follow-Up and Construction Services: All details of the design and proposed construction may not be known at the time of submission of Golder's report. Golder should be retained to review the final design, project plans and documents prior to construction, to confirm that they are consistent with the intent of Golder's report.

During construction, Golder should be retained to perform sufficient and timely observations of encountered conditions to confirm and document that the subsurface conditions do not materially differ from those interpreted conditions considered in the preparation of Golder's report and to confirm and document that construction activities do not adversely affect the suggestions, recommendations and opinions contained in Golder's report. Adequate field review, observation and testing during construction are necessary for Golder to be able to provide letters of assurance, in accordance with the requirements of many regulatory authorities.

Project N:\Active\2007\1414-07-1414-0039 WFP Monitorm Jordan R\GIS\mapshp\mshp\Figure1_sayplan.mxd - D:\N\Active\2007\1414-07-1414-0039 WFP Monitorm Jordan R\GIS\mapshp\mshp\Figure1_sayplan.mxd



LEGEND

 SITE

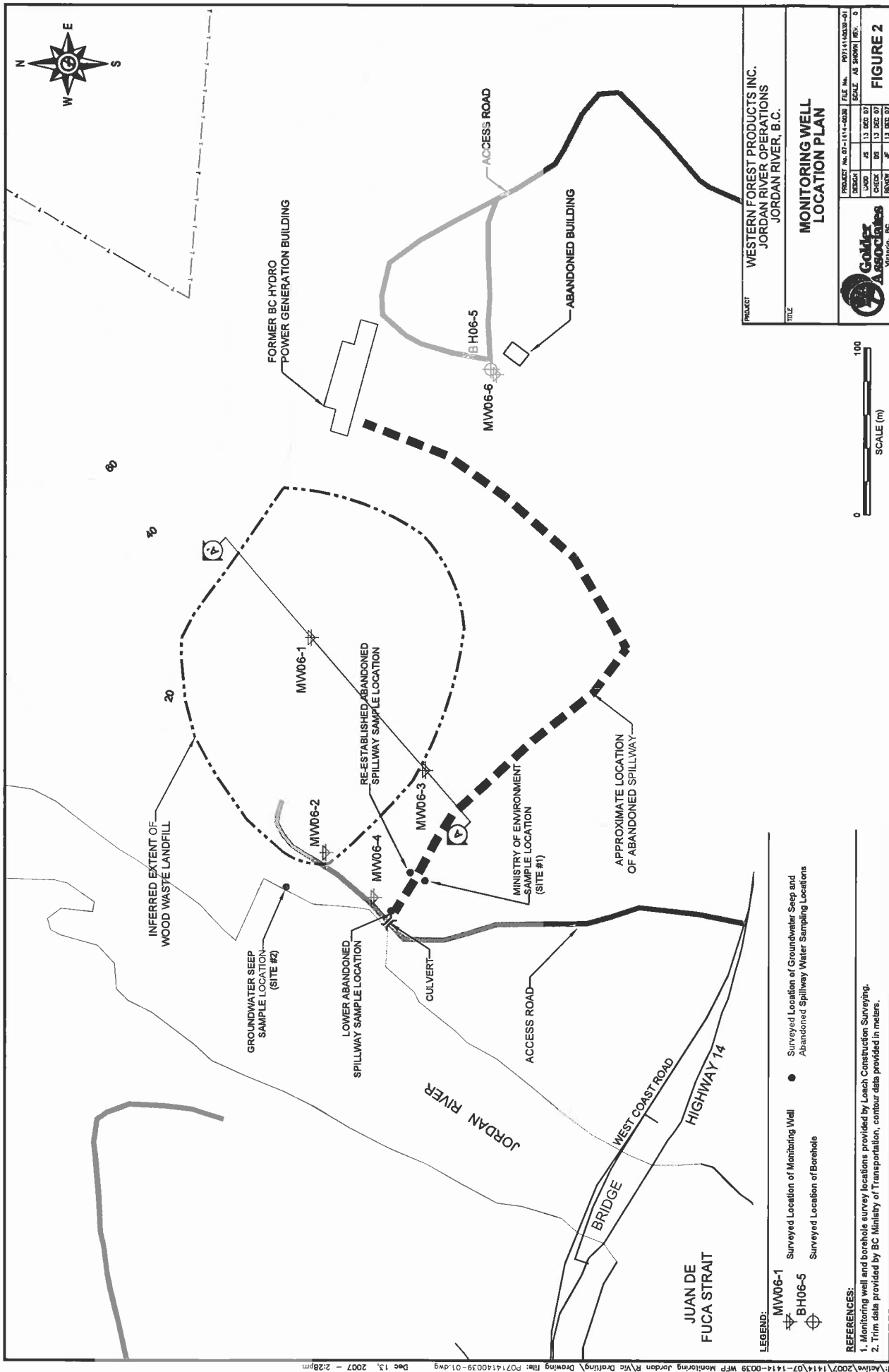
REFERENCE

Digital data, National Topographic Series (1:50,000)

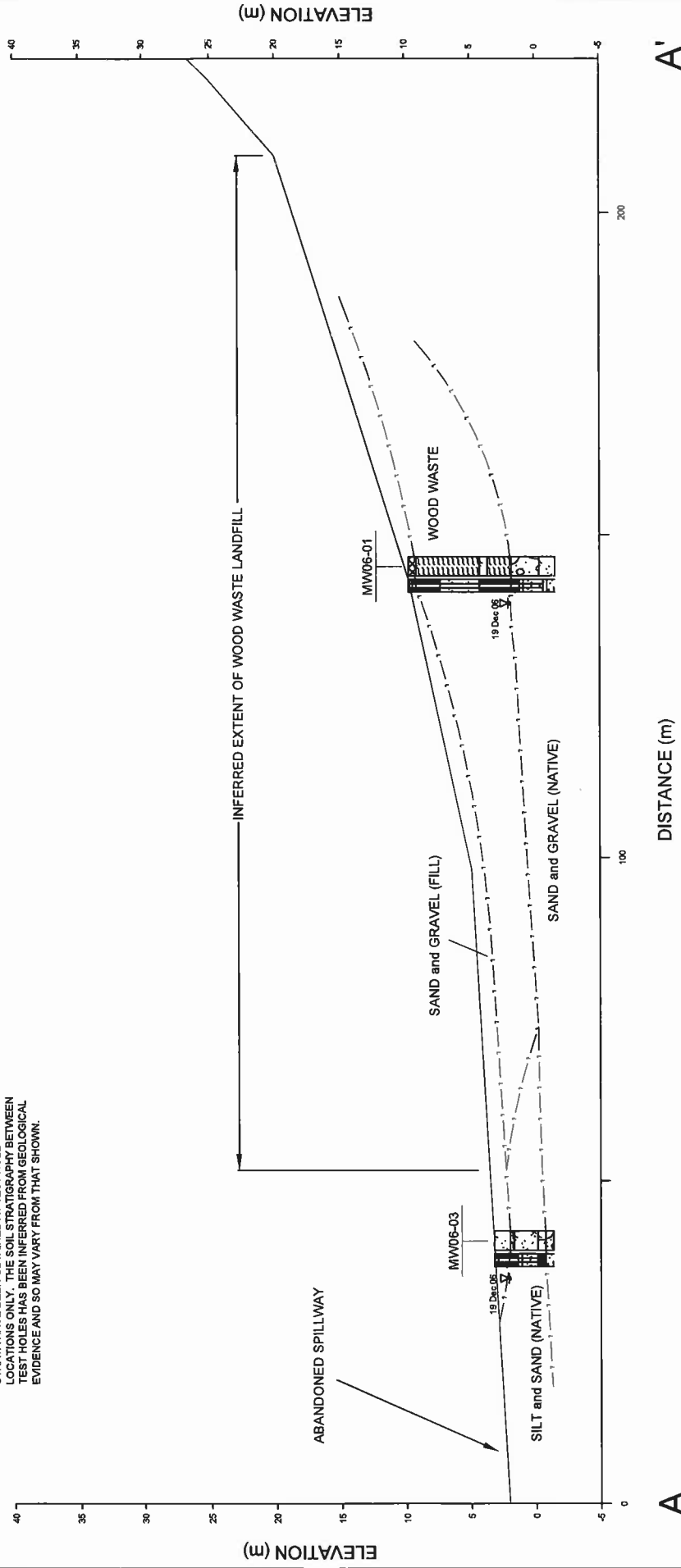
Projection: Transverse Mercator Datum: NAD 83 Coordinate System: UTM Zone 9

500 0 500
SCALE 1:20,000 METRES

PROJECT		WESTERN FOREST PRODUCTS INC. JORDAN RIVER OPERATIONS JORDAN RIVER, B.C.		
TITLE		KEY PLAN		
 Golder Associates Victoria, BC		PROJECT No. 07-1414-0039		SCALE AS SHOWN
		DESIGN	DS	10 Dec. 2007
		GIS	RD	13 Dec. 2007
		CHECK	DS	13 Dec. 2007
		REVIEW	JF	13 Dec. 2007
		FIGURE 1		



SPECIAL NOTE: DATA CONCERNING THE VARIOUS STRATA HAVE BEEN OBTAINED AT TEST HOLE LOCATIONS ONLY. THE SOIL STRATIGRAPHY BETWEEN TEST HOLES HAS BEEN INFERRED FROM GEOLOGICAL EVIDENCE AND SO MAY VARY FROM THAT SHOWN.



- LEGEND:**
- Denotes Approximate Testhole Location
 - Denotes Standpipe Piezometer Installation
 - Observed Soil Stratigraphy at hole location (for detailed descriptions refer to testhole logs)
 - Inferred Soil Stratigraphy
 - Groundwater Level Observed 5 October 2006

NOTE:
1. Conceptual cross section only, not to be used for design or construction purposes.

REFERENCES:
1. Monitoring well and borehole survey locations provided by Leach Construction Surveying.
2. Trim data provided BC Ministry of Transportation, contour data provided in meters.

PROJECT: WESTERN FOREST PRODUCTS INC.
JORDAN RIVER OPERATIONS
JORDAN RIVER, B.C.

TITLE: CONCEPTUAL GEOLOGIC CROSS SECTION A - A'

FIGURE 3

PROJECT No. 07-1414-0038	FILE No. 3071414038-01	SCALE: AS SHOWN	REV. 0
DESIGN	13 DEC 07		
CHECK	13 DEC 07		
REVIEW	13 DEC 07		

Golden Associates
VICTORIA, BC



APPENDIX I
RECORD OF BOREHOLE AND MONITORING WELLS

PROJECT No.: 07-1414-0039

RECORD OF MONITORING WELL: MW06-01

SHEET 1 OF 2

LOCATION: Centre of Landfill

BORING DATE: 20 Sept 06

DATUM: Local

SAMPLER HAMMER, kg: DROP, mm

PENETRATION TEST HAMMER, Variable

DEPTH SCALE METRES	BORING METHOD	SOIL PROFILE		SAMPLES		BECKER PENETRATION RESISTANCE, BLOWS/0.3m				BOUNCE CHAMBER PRESSURE (kPa)				ADDITIONAL LAB. TESTING	PIEZOMETER OR STANDPIPE INSTALLATION		
		DESCRIPTION	STRATA PLOT	ELEV. DEPTH (m)	NUMBER	TYPE	BLOWS/0.3m	SHEAR STRENGTH				WATER CONTENT PERCENT					
								Cu, kPa	nat V. rem V.	+ ⊕ U -	Q - O	Wp	W			Wi	
																	10
								20	40	60	80						
0	Down Hole Drilling Ltd. Bombardier Muskeg Hy Tracked Auger Rig	Ground Surface		0.00											Concrete and Monument		
		Loose, wet, brown, SAND, trace to some gravel, trace silt, with cobbles and boulders (fill).		0.61												Bentonite with Cuttings	
1																	
2																	
3																	
4		Soft, wet, brown to dark brown, woodwaste, trace sand, trace gravel (fill).			1	AS										Sand Fill	
5																	
6																	
		Loose, wet, brown, SAND, some gravel, trace silt, with woody debris (fill).		5.49	2	AS											
7		Soft, wet, brown to dark brown, woodwaste, trace sand, trace gravel (fill).		6.10													Bentonite
8																	
9																	
	Compact, free water, brown, sandy GRAVEL, trace silt, with cobbles and woody debris (Native).		7.92												Sand		
10																	
		CONTINUED NEXT PAGE															

BECKER CHAMBER (KPa) 07-1414-0039.GPJ GLDR CAN GDT 7/30/07

DEPTH SCALE

1 : 50



LOGGED: DS

CHECKED:

PROJECT No.: 07-1414-0039

RECORD OF MONITORING WELL: MW06-01

SHEET 2 OF 2

LOCATION: Centre of Landfill

BORING DATE: 20 Sept 06

DATUM: Local

SAMPLER HAMMER, kg; DROP, mm

PENETRATION TEST HAMMER, Variable

DEPTH SCALE METRES	BORING METHOD	SOIL PROFILE		SAMPLES		BECKER PENETRATION RESISTANCE, BLOWS/0.3m				BOUNCE CHAMBER PRESSURE (kPa)				ADDITIONAL LAB. TESTING	PIEZOMETER OR STANDPIPE INSTALLATION
		DESCRIPTION	STRATA PLOT	ELEV. DEPTH (m)	NUMBER	TYPE	BLOWS/0.3m	SHEAR STRENGTH		WATER CONTENT PERCENT					
								Cu, kPa	nat V. + rem V. ⊕	U - O -	Wp	Wl			
10	Down Hole Drilling Ltd.	Loose, free water, brown, SAND, trace gravel, trace silt (Native).		10.06										Slotted PVC Pipe Sand Slough	
11				3	DO	5									
12		End of MONITORING WELL.		11.26											
13															
14															
15															
16															
17															
18															
19															
20															

DEPTH SCALE

1 : 50



LOGGED: DS

CHECKED:

BECKER CHAMBER (KPA) 07-1414-0039.GPJ GLDR CAN.GDT 7/30/07

PROJECT No.: 07-1414-0039

RECORD OF MONITORING WELL: MW06-02

SHEET 1 OF 1

LOCATION: Southwest of Landfill

BORING DATE: 20 Sept 06

DATUM: Local

SAMPLER HAMMER, kg; DROP, mm

PENETRATION TEST HAMMER, Variable

DEPTH SCALE METRES	BORING METHOD	SOIL PROFILE		SAMPLES		BECKER PENETRATION RESISTANCE, BLOWS/0.3m				BOUNCE CHAMBER PRESSURE (kPa)				ADDITIONAL LAB. TESTING	PIEZOMETER OR STANDPIPE INSTALLATION
		DESCRIPTION	STRATA PLOT	ELEV. DEPTH (m)	NUMBER	TYPE	BLOWS/0.3m	SHEAR STRENGTH Cu, kPa		WATER CONTENT PERCENT					
								10	20	30	40	10	20		
								nat V. + Q - ● rem V. ⑥ U - ○							
								20	40	60	80	Wp	W	Wi	
0		Ground Surface		0.00											
	Down Hole Drilling Ltd. Bombardier Mucking Hy Tracked Auger Rig	Loose, moist to wet, brown, SAND and GRAVEL, trace silt (fill).		0.61											
1		Loose, wet, brown, SAND and GRAVEL, trace silt, with woodwaste (fill).			1	AS									
2					2	AS									
3		Loose, wet to free water, SAND, trace silt, trace gravel, with rooting (Native).			2.13										
4					3.05	3	AS								
		Loose, to compact, free water, grey-brown, sandy GRAVEL, trace silt (Native).		4.27	4	AS									
		Loose, wet to free water, SILT, trace sand, trace clay, with rootlets (Native).		4.57											
5		End of MONITORING WELL.													
6															
7															
8															
9															
10															

DEPTH SCALE

1 : 50



LOGGED: DS

CHECKED:

BECKER CHAMBER (KPa) 07-1414-0039.GPJ GLDR CAN.GDT 7/30/07

PROJECT No.: 07-1414-0039

RECORD OF MONITORING WELL: MW06-03

SHEET 1 OF 1

LOCATION: South of Landfill

BORING DATE: 21 Sept 06

DATUM: Local

SAMPLER HAMMER, kg; DROP, mm

PENETRATION TEST HAMMER, Variable

DEPTH SCALE METRES	BORING METHOD	SOIL PROFILE		SAMPLES		BECKER PENETRATION RESISTANCE, BLOWS/0.3m				BOUNCE CHAMBER PRESSURE (kPa)				ADDITIONAL LAB. TESTING	PIEZOMETER OR STANDPIPE INSTALLATION		
		DESCRIPTION	STRATA PLOT	ELEV. DEPTH (m)	NUMBER	TYPE	BLOWS/0.3m	SHEAR STRENGTH				WATER CONTENT PERCENT					
								Cu, kPa		nat V. rem V. + -		+ -				Wp	
								10	20	30	40	10	20	30	40		
								20	40	60	80						
0	Down Hole Drilling Ltd. Bombardier Muskeg Hy Tracked Auger Rig	Ground Surface		0.00													Concrete and Monument
		Loose, moist to wet, brown, SAND, some gravel, trace to some silt, with cobbles and rooting (fill).															Bentonite
1		Loose, wet, dark brown, organic SILT, trace sand, with rooting (Native).		1.22													
				1.52													
2		Loose, wet to free water, brown, SAND, trace gravel, trace silt (Native).			1	AS											
3		Loose, free water, brown-grey, silty SAND, trace gravel with woody debris (Native).		3.35	2	AS											Slotted PVC Pipe Sand
		Loose, free water, brown-grey, sandy GRAVEL, trace silt, with woody debris (Native).		3.96	3	AS											Bentonite
4																	Slough
5		End of MONITORING WELL.		4.57													
6																	
7																	
8																	
9																	
10																	

DEPTH SCALE

1 : 50



LOGGED: DS

CHECKED:

BECKER CHAMBER (KPA) 07-1414-0039.GPJ GLDR CAN.GDT 7/30/07

PROJECT No.: 07-1414-0039

RECORD OF MONITORING WELL: MW06-04

SHEET 1 OF 1

LOCATION: Southwest of Landfill

BORING DATE: 21 Sept 06

DATUM: Local

SAMPLER HAMMER, kg; DROP, mm

PENETRATION TEST HAMMER, Variable

DEPTH SCALE METRES	BORING METHOD	SOIL PROFILE		SAMPLES		BECKER PENETRATION RESISTANCE, BLOWS/0.3m				BOUNCE CHAMBER PRESSURE (kPa)				ADDITIONAL LAB. TESTING	PIEZOMETER OR STANDPIPE INSTALLATION		
		DESCRIPTION	STRATA PLOT	ELEV. DEPTH (m)	NUMBER	TYPE	BLOWS/0.3m	SHEAR STRENGTH				WATER CONTENT PERCENT					
								Cu, kPa		nat V. + Q - rem V. ⊕ U - ○							
								10	20	30	40	10	20			30	40
								20	40	60	80	10	20	30	40		
0	Down Rita Drilling Ltd. Bombardier Musking Hy Tracked Auger Rig	Ground Surface															
		Loose, wet, brown, SILT and SAND (Native).		0.00	1	AS									Concrete and Monument		
		Compact to dense, wet to free water, gravelly SAND, trace silt (Native).		0.61	2	AS									Bentonite		
			Dense, free water, GRAVEL, trace to some sand, trace silt (Native).		1.52	3	AS								Slotted PVC Pipe Sand		
		Loose, free water, SILT, trace gravel trace sand, trace clay with rooting (Native).			3.35	4	AS								Bentonite		
4														Slough			
5		End of MONITORING WELL.															
6																	
7																	
8																	
9																	
10																	

DEPTH SCALE

1 : 50



LOGGED: DS

CHECKED:

BECKER CHAMBER (KPa) 07-1414-0039 GPJ GLDR CAN GDT 7/30/07

PROJECT No.: 07-1414-0039

RECORD OF BOREHOLE: BH06-05

SHEET 1 OF 1

LOCATION: East of Landfill

BORING DATE: 21 Sept 06

DATUM: Local

SAMPLER HAMMER, kg; DROP, mm

PENETRATION TEST HAMMER, Variable

DEPTH SCALE METRES	BORING METHOD	SOIL PROFILE		SAMPLES		BECKER PENETRATION RESISTANCE, BLOWS/0.3m				BOUNCE CHAMBER PRESSURE (kPa)				ADDITIONAL LAB. TESTING	PIEZOMETER OR STANDPIPE INSTALLATION		
		DESCRIPTION	STRATA PLOT	ELEV. DEPTH (m)	NUMBER	TYPE	BLOWS/0.3m	SHEAR STRENGTH				WATER CONTENT PERCENT					
								Cu, kPa		nat V. + Q - ● rem V. ⊕ U - ○		Wp ——— W ——— WI					
								10	20	30	40	10	20			30	40
								20	40	60	80	10	20	30	40		
0		Ground Surface		0.00													
	Down Hole Drilling Ltd. Bombardier Mustang Hy Tracked Auger Rig	Loose, wet, brown, SAND, some gravel, trace silt (fill).		0.81													
1		Loose, wet, brown to reddish brown, organic SILT, trace sand with woody debris (Native).		1.83	1	AS											
2		Compact, wet, brown, SAND, some silt, trace to some gravel (Native).		2.44													
		Compact to dense, free water, brown-grey, gravelly SAND, trace silt (Native).		3.66	2	AS											
3		Loose, free water, grey, silty SAND, trace gravel (Native).		3.96													
4					3	AS											
5																	
6		Loose, wet, grey, SILT, trace to some sand, trace gravel, trace clay (Native).			4	AS											
7																	
						5	AS										
8		End of BOREHOLE.		7.62													
9																	
10																	

DEPTH SCALE

1 : 50



LOGGED: DS

CHECKED:

BECKER CHAMBER (KPa) 07-1414-0039.GPJ GLDR CAN.GDT 7/30/07

PROJECT No.: 07-1414-0039

RECORD OF MONITORING WELL: MW06-06

SHEET 1 OF 1

LOCATION: East of Landfill

BORING DATE: 21 Sept 06

DATUM: Local

SAMPLER HAMMER, kg; DROP, mm

PENETRATION TEST HAMMER, Variable

DEPTH SCALE METRES	BORING METHOD	SOIL PROFILE		SAMPLES		BECKER PENETRATION RESISTANCE, BLOWS/0.3m				BOUNCE CHAMBER PRESSURE (kPa)				ADDITIONAL LAB. TESTING	PIEZOMETER OR STANDPIPE INSTALLATION
		DESCRIPTION	STRATA PLOT	ELEV. DEPTH (m)	NUMBER	TYPE	SHEAR STRENGTH Cu, kPa		WATER CONTENT PERCENT		WATER CONTENT PERCENT				
							10	20	30	40	10	20	30		
0	Down Rio Drilling Ltd. Bombardier Mustang Hy Tracked Auger Rig	Ground Surface		0.00											Concrete and Monument Bentonite Slotted PVC Pipe Sand Slough
		Loose, wet, brown, gravelly SAND, trace silt (fill).													
1		Loose, wet, dark brown, organic SILT, trace sand, with woody debris (Native).		0.61											
		Loose to compact, wet, brown, gravelly SAND, trace silt (Native).		1.07	1	AS									
2		Compact, free water, brown-grey, SAND and GRAVEL, trace silt (Native).		1.52											
3					2	AS									
4		Loose, free water, grey, SILT, trace sand, trace clay (Native).		3.81											
4		End of MONITORING WELL.													
5															
6															
7															
8															
9															
10															

DEPTH SCALE

1 : 50



LOGGED: DS

CHECKED:

BECKER CHAMBER (KPA) 07-1414-0039.GPJ GLDR CAN.GDT 7/30/07

APPENDIX II

**COMPILATION OF GROUNDWATER LEVELS AND ELEVATIONS
DATA**

**Groundwater Levels in Monitoring Wells
WFP - Jordan River Operation Landfill**

Well ID	Ground Elevation (m)	Top of Pipe Elevation (m)	Stick Up (m)	20-Sep-06	21-Sep-06	22-Sep-06	27-Sep-06	05-Oct-06	19-Dec-06	22-Mar-07
MW06-01	9.73	10.54	0.81	8.30	8.29	8.32	8.32	8.34	7.69	7.63
MW06-02	3.34	4.09	0.75	3.17	3.11	3.14	3.09	3.19	2.45	2.75
MW06-03	3.21	3.95	0.74	-	2.26	2.33	2.33	2.40	1.41	1.42
MW06-04	1.28	2.06	0.78	-	1.42	1.26	1.46	1.84	0.68	1.73
BH06-05	2.55	-	-	-	-	-	-	-	-	-
MW06-06	2.57	3.41	0.84	-	1.13	1.15	1.20	1.23	0.69	0.78
Groundwater Seep	0.61	-	-	-	-	-	-	-	-	-
Lower Abandoned Spillway	0	-	-	-	-	-	-	-	-	-

Notes:

1. The site survey conducted presents elevations that are referenced to an assumed datum (i.e. 500 m elevation) as a geodetic benchmark in the area was not available.
2. The groundwater levels presented are measured from below ground surface.
3. The elevation data presented in the table designates the Lower Abandoned Spillway location as the geodetic 0 m elevation and the other survey locations are reference relative to that location.

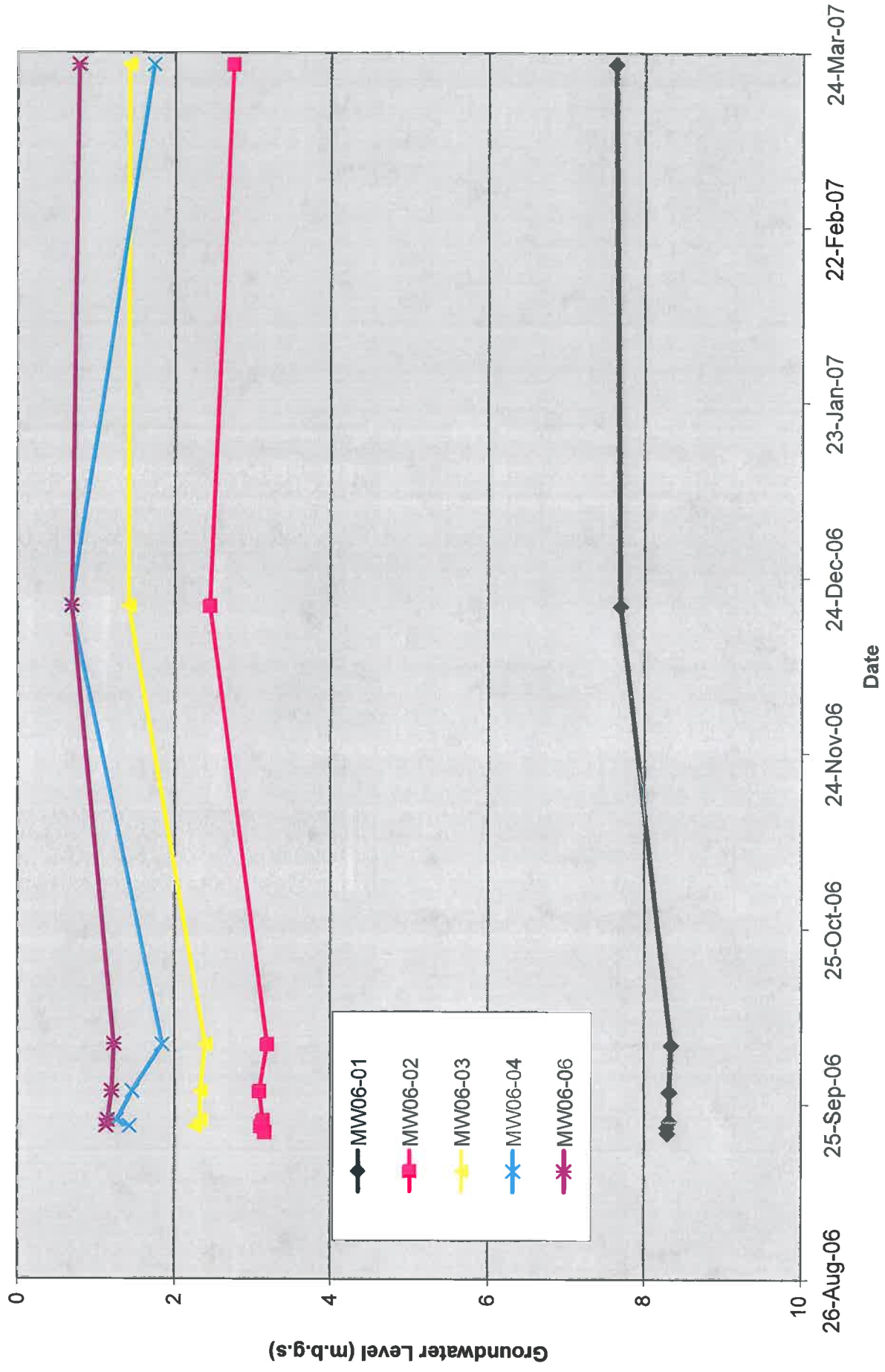
**Groundwater Elevations in Monitoring Wells
WFP - Jordan River Operations Landfill**

Well ID	Ground Elevation (m)	Top of Pipe Elevation (m)	Stick Up (m)	20-Sep-06	21-Sep-06	22-Sep-06	27-Sep-06	05-Oct-06	19-Dec-06	22-Mar-07
MW06-01	9.73	10.54	0.81	1.44	1.44	1.41	1.41	1.39	2.04	2.11
MW06-02	3.34	4.09	0.75	0.18	0.23	0.20	0.25	0.15	0.89	0.59
MW06-03	3.21	3.95	0.74	-	0.95	0.88	0.88	0.81	1.80	1.79
MW06-04	1.28	2.06	0.78	-	-0.14	0.02	-0.18	-0.56	0.60	-0.45
BH06-05	2.55	-	-	-	-	-	-	-	-	-
MW06-06	2.57	3.41	0.84	-	1.44	1.42	1.37	1.34	1.88	1.79
Groundwater Seep	0.61	-	-	-	-	-	-	-	-	-
Lower Abandoned Spillway	0	-	-	-	-	-	-	-	-	-

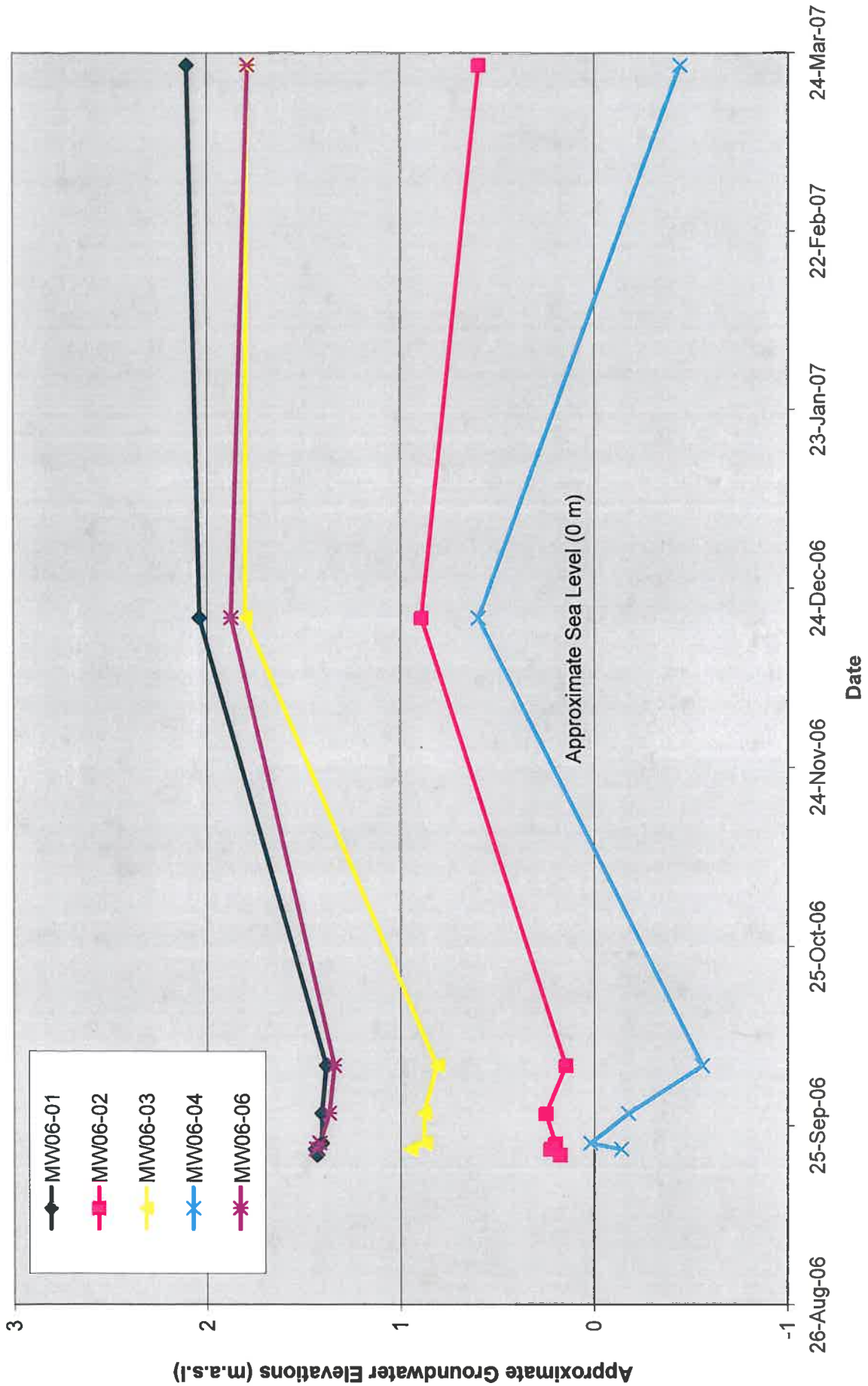
Notes:

1. The site survey conducted presents elevations that are referenced to an assumed datum (i.e. 500 m elevation) as a geodetic benchmark in the area was not available.
2. The groundwater levels presented are measured from below ground surface.
3. The elevation data presented in the table designates the Lower Abandoned Spillway location as the geodetic 0 m elevation and the other survey locations are reference relative to that location.

Graph II-1: Groundwater Levels - WFP Jordan River Operations Landfill



Graph II-2: Approximate Groundwater Elevations - WFP Jordan River Operations Landfill



APPENDIX III

**COMPILATION OF GROUNDWATER AND SURFACE WATER
CHEMISTRY DATA**

Table III-1
Surface Water Analytical Results, WFP Jordan River Operations Landfill, Jordan River, BC

Location Sample Control Number Date Sampled	BC Water Quality Guidelines (Criteria) for Freshwater Aquatic Life (Approved and Working) ^{1,2}	BC Water Quality Guidelines (Criteria) for Marine Aquatic Life (Approved and Working) ^{1,2}	Abundant Spillway (Site #1) MoE Apr-97	Abundant Spillway (Site #1) MoE Mar-99	Lower Abundant Spillway (Site #1) MoE 27-Sep-06	Abundant Spillway 0557-08 (9-Dec-06)	Abundant Spillway 0887-03 (22-Jul-07)	Groundwater Sep (Site #2) MoE 14-Jun-00	Groundwater Sep (Site #2) MoE 27-Sep-06	Groundwater Sep (Site #2) MoE 19-Dec-06	Groundwater Sep (Site #2) MoE 0887-05 22-Jul-07
pH	6.5 to 9 (unrestricted change permitted)	7.0 to 8.7 (unrestricted change permitted)	6.3	6.3	6.97	6.34	6.12	5.5	5.70	5.78	6.02
Temperature (°C)	15°C (maximum daily)	+2°C from natural ambient	-	-	14.2	14.2	11.7	-	15.2	11.7	11.3
Conductivity (µS/cm)	150 µS/cm (maximum daily)	150 µS/cm (maximum daily)	-	-	1900	1900	1900	-	1900	1900	1900
Dissolved Oxygen (mg/L)	1.5 g/L of NaCl maximum	1.5 g/L of NaCl maximum	-	-	12.7	12.7	12.7	-	0.18	0.09	0.11
Calcium Salinity (mg/L)	1.5 g/L of NaCl maximum	1.5 g/L of NaCl maximum	-	-	3320	205	33.3	-	119	84.9	43.2
Hardness (CaCO₃)	150 mg/L (average)	150 mg/L (average)	-	-	100	100	100	-	100	100	100
Triclinic Testing	0.2 to 0.3 mg/L	0.2 to 0.3 mg/L	-	-	8.3	8.3	8.3	-	3.0	7.4	3.0
96 hr LC50 (%)	1.5 g/L of NaCl maximum	1.5 g/L of NaCl maximum	-	-	14	5	11500	-	32	10.9	7.4
Laboratory Dissolved Oxygen	100 mg/L (average)	100 mg/L (average)	-	-	1600	1600	1600	-	179	114	24
Bioassay	0.002 mg/L	0.002 mg/L	-	-	1.47	0.343	0.138	-	0.55	0.095	0.191
Chloride	40 mg/L (average)	40 mg/L (average)	-	-	1.47	0.343	0.138	-	0.55	0.095	0.191
Fluoride	0.2 to 0.2 mg/L (average)	0.2 to 0.2 mg/L (average)	-	-	981	55	30	-	127	115	90
Sulphate	0.05 mg/L	0.05 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Ammonia	0.05 mg/L	0.05 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Phenols	0.05 mg/L	0.05 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Heavy Metals	0.05 mg/L	0.05 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Trace Metals	0.05 mg/L	0.05 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Aluminum	0.02 mg/L	0.02 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Antimony	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Barium	1 mg/L (average)	1 mg/L (average)	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Beryllium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Bismuth	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Boron	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Cadmium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Chromium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Copper	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Iron	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Lead	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Lithium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Magnesium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Manganese	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Molybdenum	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Nickel	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Phosphorus	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Potassium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Selenium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Silver	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Sodium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Strontium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Thallium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Tin	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Titanium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Vanadium	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Zinc	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Speciated Metals	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Chromium (III) T-Cr	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001
Chromium (VI) T-Cr	0.005 mg/L	0.005 mg/L	-	-	0.94	1.66	2	-	0.002	0.040	0.001

Notes:
1. BC Ministry of Environment, "British Columbia Approved Water Quality Guidelines 2004 edition", updated August 2006.
2. BC Ministry of Environment, "A Compendium of Working Water Quality Guidelines for British Columbia", updated August 2006.

All concentrations in milligrams per litre (mg/L), unless otherwise noted.

CI = standard is chloride dependent

V = Standard is volume dependent (CRVY/Guid)

Phenols refer to non-chlorinated total phenols

Indices = laboratory detection limits are greater than standard or guideline used.

BCWQG for total iron is currently under review in draft. BCWQG for total iron is currently under review in draft. BCWQG for total iron is currently under review in draft.

Phenols = Phenols are currently under review in draft. BCWQG for total iron is currently under review in draft. BCWQG for total iron is currently under review in draft.

Salinity = Salinity is currently under review in draft. BCWQG for total iron is currently under review in draft. BCWQG for total iron is currently under review in draft.

Salinity = Salinity is currently under review in draft. BCWQG for total iron is currently under review in draft. BCWQG for total iron is currently under review in draft.

Golden Associates

Table III-1

Reference:
1. British Columbia Contaminated Sites Regulation (CSR) [en. BC Reg. 324/2004, s 68; am. BC Regs. 405/2004, 76/2005].

Notes:
All concentrations in milligrams per litre (mg/L), unless otherwise noted.
Standards from the Contaminated Sites Regulation (CSR), its associated Schedules and Technical Guidance Documents.
Aquatic life standards assume a 10:1 dilution is available.
Standards for surface water: metals are total metals; standards for groundwater: metals are dissolved metals
Recommended that surface water samples for metals be analysed for dissolved concentrations.
Cl⁻ = standard in chloride dependent
V²⁺ = standard in valence dependent (GRVIV)Cr(III)
Pyrrolid cys in non-chlorinated and open.

December 2007

Table III-4
September 2006 Laboratory RPD Results
WFP Jordan River Operations Landfill
Jordan River, BC

07-1414-0039

Project 06-1414-092-4000 Water Analysis
Report to Golder Associates Ltd.
ALS File No. Z3248r
Date Received 29/09/2006
Date: 23/10/2006

Duplicate Results

Sample ID	0557-03	0557-03	RPD %
Date Sampled	27/09/2006	QC# 525177	
Time Sampled			
ALS Sample	3		
Nature	Water		

Physical Tests

Hardness	58.3	54.8	6.19
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Dissolved Anions

Bromide	I <1.0	<1.0	0
Chloride	C 13	14	7.41
Fluoride	F <0.40	<0.40	0
Sulphate	S <10	<10	0

Nutrients

Nitrate Nitrogen	0.16	0.18	11.8
Nitrite Nitrogen	<0.020	<0.020	0

Inorganic Parameters

Sulphide	S <0.20	<0.20	0
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Organic Parameters

Chemical Oxygen Demand	177	183	3.33
Tannin and Lignin	13.0	13.0	0

Table III-5
December 2006 Laboratory RPD Results
WFP Jordan River Operations Landfill
Jordan River, BC

Project 06-1414-092-6000 Water Analysis
 Report to Golder Associates Ltd.
 ALS File No. Z8280
 Date Received 21/12/2006
 Date: 05/01/2007

Duplicate Results

Sample ID	0569-02	0568-02	RPD %
Date Sampled	19/12/2006	QC# 535374	
Time Sampled			
ALS Sample ID	2		
Nature	Water		

Physical Tests

Hardness	CaCO ₃ 107	107	0
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Total Metals

Aluminum	T-Al	24.7	24.4	1.22
Antimony	T-Sb	<0.0025	<0.0025	0
Arsenic	T-As	0.0111	0.0111	0
Barium	T-Ba	<0.10	<0.10	0
Beryllium	T-Be	<0.025	<0.025	0
Boron	T-B	<0.50	<0.50	0
Cadmium	T-Cd	0.00057	0.00061	8.78
Calcium	T-Ca	34.2	34.7	1.45
Chromium	T-Cr	0.0548	0.0545	0.183
Cobalt	T-Co	0.0134	0.0133	0.749
Copper	T-Cu	0.237	0.233	1.7
Iron	T-Fe	43.8	44.5	2.04
Lead	T-Pb	0.0152	0.0154	1.31
Lithium	T-Li	<0.25	<0.25	0
Magnesium	T-Mg	19.0	19.2	1.05
Manganese	T-Mn	1.81	1.83	1.23
Mercury	T-Hg	<0.00020	<0.00020	0
Molybdenum	T-Mo	<0.0050	<0.0050	0
Nickel	T-Ni	0.038	0.038	0
Selenium	T-Se	<0.0050	<0.0050	0
Silver	T-Ag	<0.00025	<0.00025	0
Sodium	T-Na	17	17	0
Thallium	T-Tl	<0.0010	<0.0010	0
Titanium	T-Ti	1.02	1.03	0.876
Uranium	T-U	0.0060	0.0063	4.68
Vanadium	T-V	<0.15	<0.15	0
Zinc	T-Zn	0.087	0.083	4.71

Dissolved Metals

Aluminum	D-Al	0.538	0.528	1.68
Antimony	D-Sb	<0.0025	<0.0025	0
Arsenic	D-As	<0.0050	<0.0050	0
Barium	D-Ba	<0.10	<0.10	0
Beryllium	D-Be	<0.025	<0.025	0
Bismuth	D-Bi	<1.0	<1.0	0
Boron	D-B	<0.50	<0.50	0
Cadmium	D-Cd	<0.00025	<0.00025	0
Calcium	D-Ca	25.0	25.2	0.787
Chromium	D-Cr	<0.0025	<0.0025	0
Cobalt	D-Co	<0.0025	<0.0025	0
Copper	D-Cu	<0.0050	<0.0050	0
Iron	D-Fe	21.8	22.1	0.809
Lead	D-Pb	<0.0050	<0.0050	0
Lithium	D-Li	<0.25	<0.25	0
Magnesium	D-Mg	10.7	10.8	0.93
Manganese	D-Mn	1.27	1.29	1.56
Mercury	D-Hg	<0.00020	<0.00020	0
Molybdenum	D-Mo	<0.0050	<0.0050	0
Nickel	D-Ni	<0.025	<0.025	0
Phosphorus	D-P	<1.5	<1.5	0
Potassium	D-K	12	12	0
Selenium	D-Se	<0.0050	<0.0050	0
Silicon	D-Si	17.5	17.5	0
Silver	D-Ag	<0.00025	<0.00025	0
Sodium	D-Na	15	15	0
Strontium	D-Sr	0.083	0.085	2.38
Thallium	D-Tl	<0.0010	<0.0010	0
Tin	D-Sn	<0.15	<0.15	0
Titanium	D-Ti	<0.25	<0.25	0
Uranium	D-U	<0.0010	<0.0010	0
Vanadium	D-V	<0.15	<0.15	0
Zinc	D-Zn	<0.025	<0.025	0

Speciated Metals

Chromium (VI)	T-Cr	<0.060	<0.060	0
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Inorganic Parameters

Sulphide	S	<0.20	0.22	9.52
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Organic Parameters

Chemical Oxygen Demand	705	731	3.82
Tannin and Lignin	9.3	9.3	0

Table III-6
March 2007 Laboratory RPD Results
WFP Jordan River Operations Landfill
Jordan River, BC

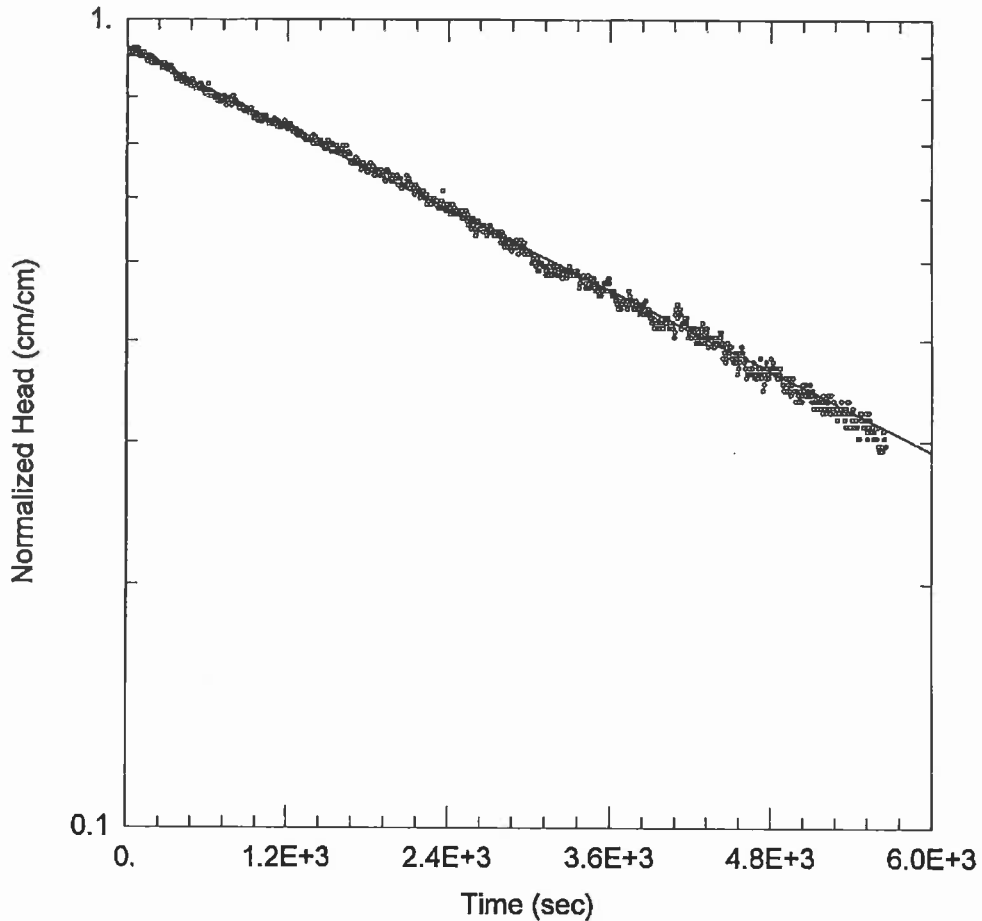
Project 07-1414-0039-1000
 Report To DWAYNE SMITH, GOLDER ASSOCIATES LTD.
 ALS File No. L489415
 Date Received 23-Mar-07
 Date 04-Apr-07

REPLICATE RESULTS

Sample ID	Matrix	ALS ID	Analyte	Replicate 1	Replicate 2	Units	RPD	RPD Limit	Diff	Diff Limit	Qualifier
Dissolved Metals											
L489415-1	Water	WG575331-4	Aluminum (Al)-Dissolved	0.311	0.268	mg/L	-	-	0.042	0.2	J
L489415-1	Water	WG575331-4	Antimony (Sb)-Dissolved	<0.0025	<0.0025	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Arsenic (As)-Dissolved	<0.0050	<0.0050	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Barium (Ba)-Dissolved	<0.020	<0.020	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Beryllium (Be)-Dissolved	<0.0050	<0.0050	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Bismuth (Bi)-Dissolved	<0.20	<0.20	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Boron (B)-Dissolved	<0.10	<0.10	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Cadmium (Cd)-Dissolved	<0.00025	<0.00025	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Calcium (Ca)-Dissolved	13.5	13.6	mg/L	0.19	20	-	-	-
L489415-1	Water	WG575331-4	Chromium (Cr)-Dissolved	<0.0025	<0.0025	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Cobalt (Co)-Dissolved	<0.0025	<0.0025	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Copper (Cu)-Dissolved	<0.0050	<0.0050	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Iron (Fe)-Dissolved	5.33	5.37	mg/L	0.84	20	-	-	-
L489415-1	Water	WG575331-4	Lead (Pb)-Dissolved	<0.0050	<0.0050	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Lithium (Li)-Dissolved	<0.050	<0.050	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Magnesium (Mg)-Dissolved	3.76	3.78	mg/L	0.38	20	-	-	-
L489415-1	Water	WG575331-4	Manganese (Mn)-Dissolved	0.169	0.169	mg/L	0.13	20	-	-	-
L489415-1	Water	WG575331-4	Mercury (Hg)-Dissolved	<0.00020	<0.00020	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Molybdenum (Mo)-Dissolved	<0.0050	<0.0050	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Nickel (Ni)-Dissolved	<0.025	<0.025	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Phosphorus (P)-Dissolved	<0.30	<0.30	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Potassium (K)-Dissolved	<2.0	<2.0	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Selenium (Se)-Dissolved	<0.0050	<0.0050	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Silicon (Si)-Dissolved	6.74	6.76	mg/L	0.34	20	-	-	-
L489415-1	Water	WG575331-4	Silver (Ag)-Dissolved	<0.00025	<0.00025	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Sodium (Na)-Dissolved	4.2	4.2	mg/L	-	-	0.0	8	J
L489415-1	Water	WG575331-4	Strontium (Sr)-Dissolved	0.0684	0.0693	mg/L	1.2	20	-	-	-
L489415-1	Water	WG575331-4	Thallium (Tl)-Dissolved	<0.0010	<0.0010	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Tin (Sn)-Dissolved	<0.030	<0.030	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Titanium (Ti)-Dissolved	<0.050	<0.050	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Uranium (U)-Dissolved	<0.0010	<0.0010	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Vanadium (V)-Dissolved	<0.030	<0.030	mg/L	N/A	20	-	-	RPD-NA
L489415-1	Water	WG575331-4	Zinc (Zn)-Dissolved	<0.0050	<0.0050	mg/L	N/A	20	-	-	RPD-NA
Organic Parameters											
L489415-1	Water	WG577986-2	COD	170	170	mg/L	-	-	0	80	J
L489415-3	Water	WG576537-3	Tannin & lignin	2	2	mg/L	-	-	0	4	J

APPENDIX IV

**COMPILATION OF SINGLE WELL RESPONSE ANALYTICAL
REPORTS**



SINGLE WELL RESPONSE TEST: MW06-3

Data Set: N:\...MW06-3 Bouwer-Rice Nov 20-06.aqt

Date: 12/11/07

Time: 11:57:41

PROJECT INFORMATION

Company: Golder Associates Ltd.

Client: Western Forest Products Inc.

Project: 07-1414-0039

Location: Jordan River, B.C.

Test Well: MW06-3

Test Date: October 5, 2006

AQUIFER DATA

Saturated Thickness: 217. cm

Anisotropy Ratio (K_z/K_r): 1.

WELL DATA (MW06-3)

Initial Displacement: 47.5 cm

Static Water Column Height: 95. cm

Total Well Penetration Depth: 95. cm

Screen Length: 95. cm

Casing Radius: 2.54 cm

Well Radius: 7.5 cm

Gravel Pack Porosity: 0.3

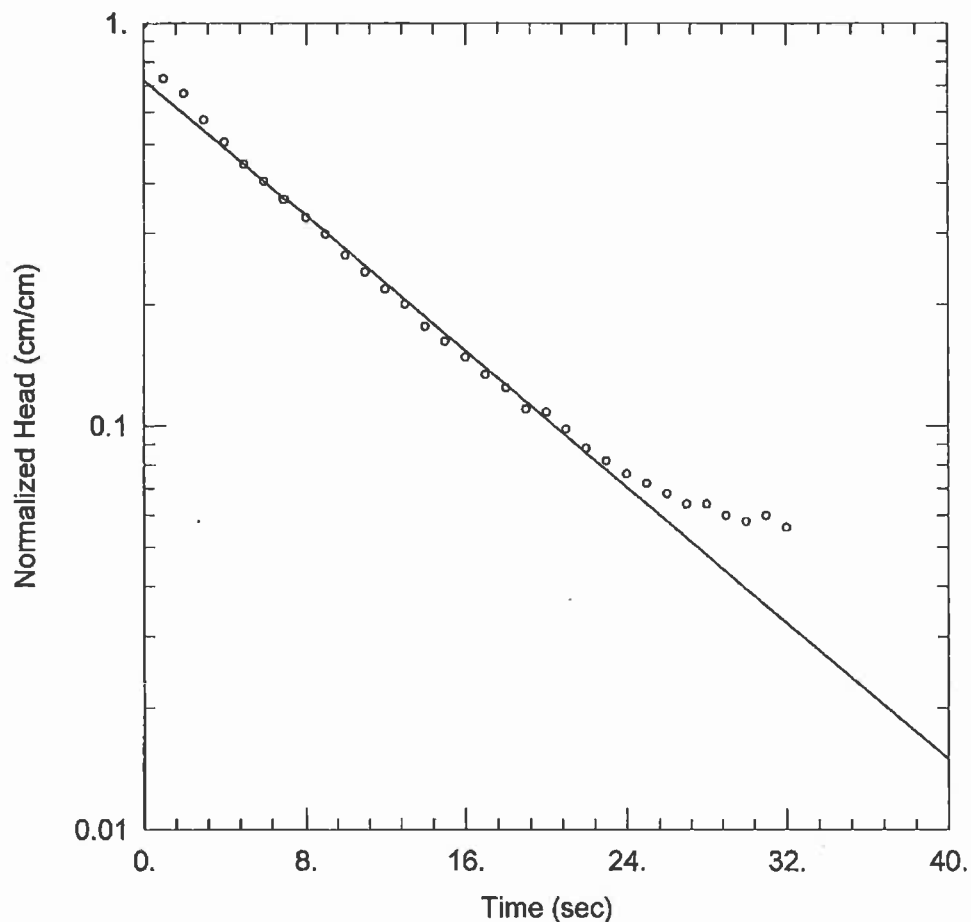
SOLUTION

Aquifer Model: Unconfined

Solution Method: Bouwer-Rice

$K = 3.314E-7$ m/sec

$y_0 = 43.7$ cm



SINGLE WELL RESPONSE TEST: MW06-4

Data Set: N:\...MW06-4 Bouwer-Rice Nov 20-06.aqt

Date: 12/11/07

Time: 11:58:12

PROJECT INFORMATION

Company: Golder Associates Ltd.

Client: Western Forest Products Inc.

Project: 07-1414-0039

Location: Jordan River, B.C.

Test Well: MW06-4

Test Date: October 5, 2006

AQUIFER DATA

Saturated Thickness: 151. cm

Anisotropy Ratio (K_z/K_r): 1.

WELL DATA (MW06-4)

Initial Displacement: 50. cm

Static Water Column Height: 151. cm

Total Well Penetration Depth: 151. cm

Screen Length: 120. cm

Casing Radius: 2.54 cm

Well Radius: 7.5 cm

Gravel Pack Porosity: 0.3

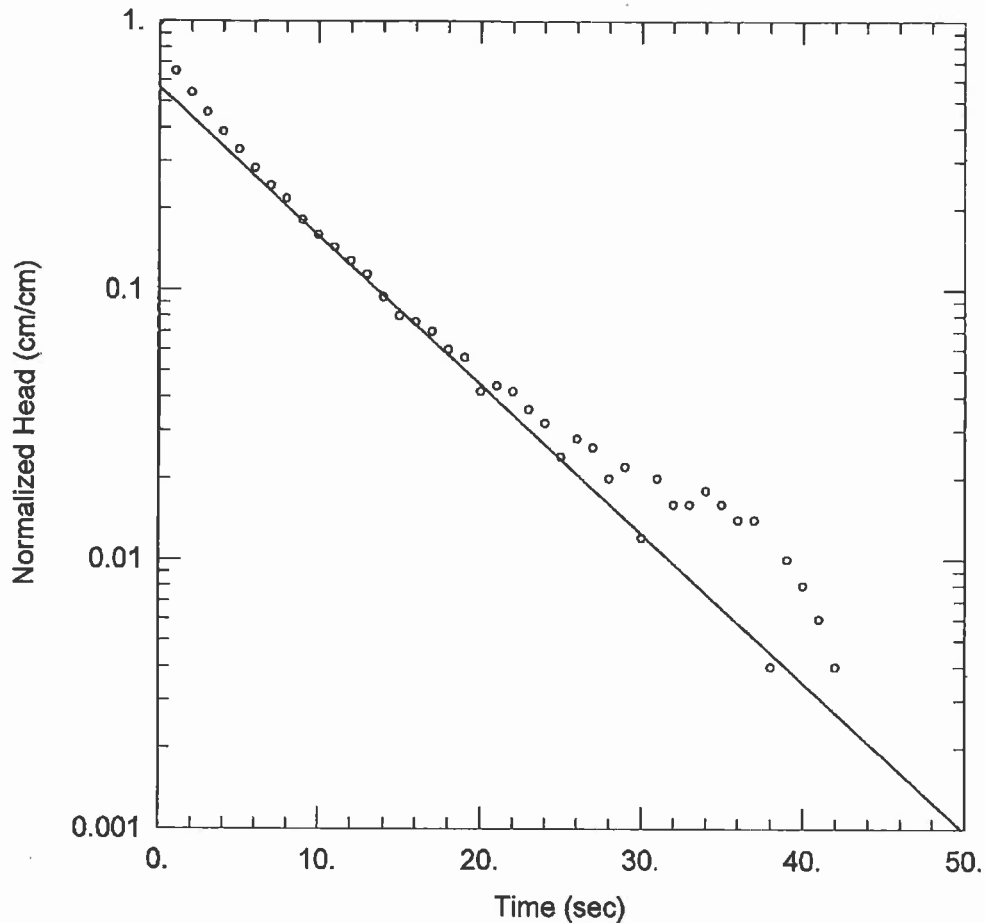
SOLUTION

Aquifer Model: Unconfined

Solution Method: Bouwer-Rice

$K = 0.0001867$ m/sec

$y_0 = 36.09$ cm



SINGLE WELL RESPONSE TEST: MW06-1

Data Set: N:\...MW06-1 Bouwer-Rice Nov 20-06.aqt

Date: 12/11/07

Time: 11:56:22

PROJECT INFORMATION

Company: Golder Associates Ltd.

Client: Western Forest Products Inc.

Project: 07-1414-0039

Location: Jordan River, B.C.

Test Well: MW06-1

Test Date: October 5, 2006

AQUIFER DATA

Saturated Thickness: 294. cm

Anisotropy Ratio (K_z/K_r): 1.

WELL DATA (MW06-1)

Initial Displacement: 50. cm

Static Water Column Height: 206. cm

Total Well Penetration Depth: 206. cm

Screen Length: 152. cm

Casing Radius: 2.54 cm

Well Radius: 7.5 cm

Gravel Pack Porosity: 0.3

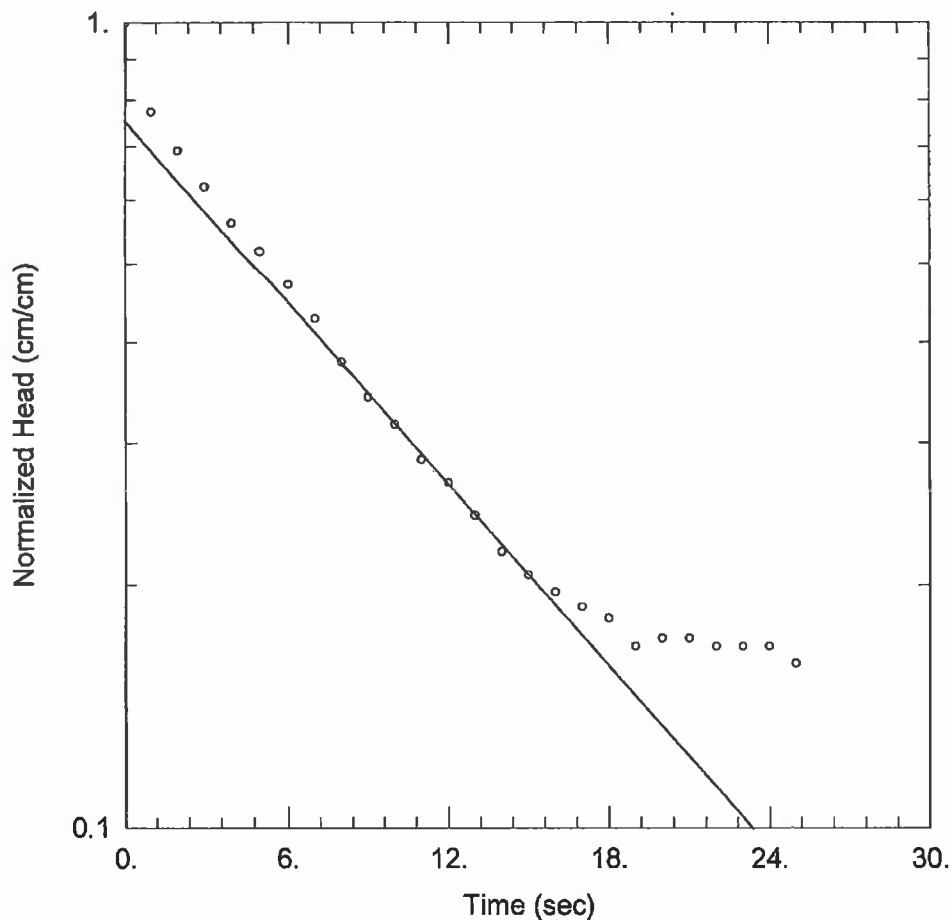
SOLUTION

Aquifer Model: Unconfined

Solution Method: Bouwer-Rice

$K = 0.0001868$ m/sec

$y_0 = 28.32$ cm



SINGLE WELL RESPONSE TEST: MW06-2

Data Set: N:\...MW06-2 Bouwer-Rice Nov 20-06.aqt

Date: 12/11/07

Time: 11:57:03

PROJECT INFORMATION

Company: Golder Associates Ltd.

Client: Western Forest Products Inc.

Project: 07-1414-0039

Location: Jordan River, B.C.

Test Well: MW06-2

Test Date: October 5, 2006

AQUIFER DATA

Saturated Thickness: 138. cm

Anisotropy Ratio (K_z/K_r): 1.

WELL DATA (MW06-2)

Initial Displacement: 50. cm

Static Water Column Height: 108. cm

Total Well Penetration Depth: 108. cm

Screen Length: 108. cm

Casing Radius: 2.54 cm

Well Radius: 7.5 cm

Gravel Pack Porosity: 0.3

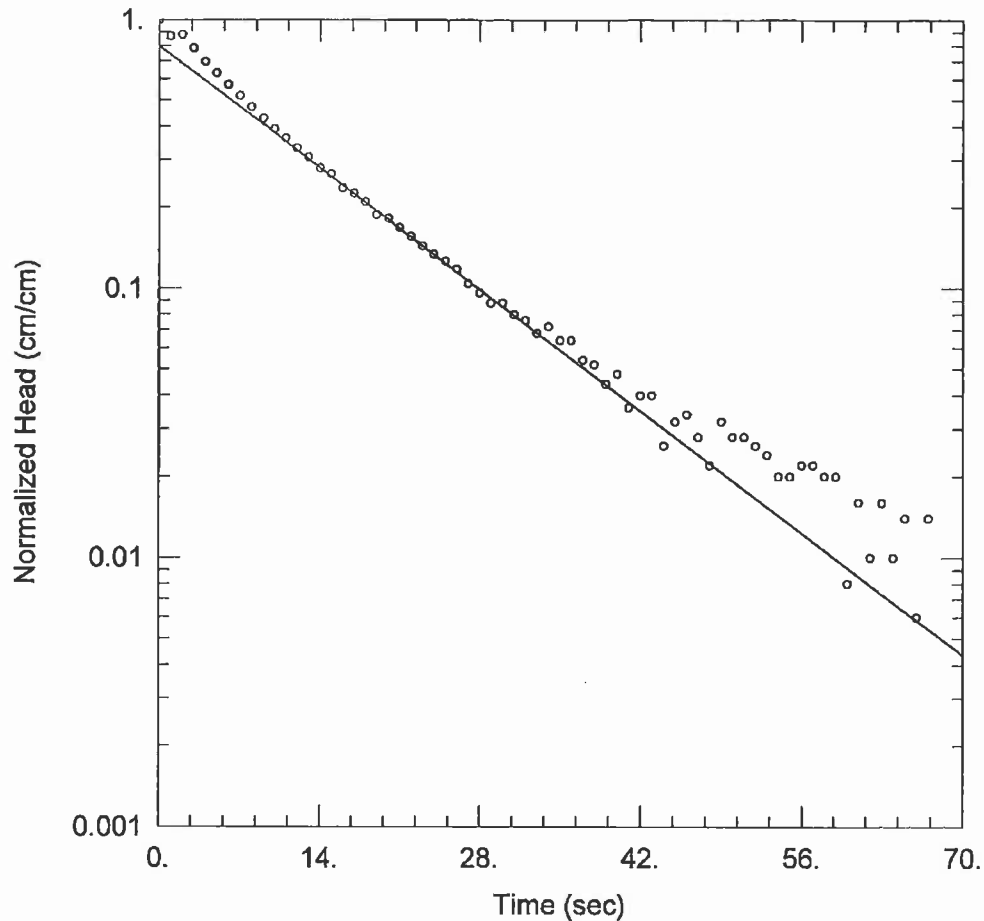
SOLUTION

Aquifer Model: Unconfined

Solution Method: Bouwer-Rice

$K = 0.0001474$ m/sec

$y_0 = 37.61$ cm



SINGLE WELL RESPONSE TEST: MW06-6

Data Set: N:\...MW06-6 Bouwer-Rice Nov 20-06.aqt

Date: 12/11/07

Time: 11:58:41

PROJECT INFORMATION

Company: Golder Associates Ltd.

Client: Western Forest Products Inc.

Project: 07-1414-0039

Location: Jordan River, B.C.

Test Well: MW06-6

Test Date: October 5, 2006

AQUIFER DATA

Saturated Thickness: 242. cm

Anisotropy Ratio (K_z/K_r): 1.

WELL DATA (MW06-6)

Initial Displacement: 50. cm

Static Water Column Height: 242. cm

Total Well Penetration Depth: 242. cm

Screen Length: 152. cm

Casing Radius: 2.54 cm

Well Radius: 7.5 cm

SOLUTION

Aquifer Model: Unconfined

Solution Method: Bouwer-Rice

$K = 3.961E-5$ m/sec

$y_0 = 39.82$ cm

APPENDIX V

**LABORATORY CHEMICAL AND TOXICITY ANALYSIS AND
MATERIALS TESTING REPORTS**

Golder Associates Ltd.

195 Pemberton Avenue
North Vancouver, British Columbia, Canada V7P 2R4
Telephone 604-986-4331
Fax 604-662-8548



E/07/1348
06-1414-092

January 17, 2007

Golder Associates
2640 Douglas Street
Victoria, BC V8T 4M1

Attention: Mr. Dwayne Smith

**RE: TOXICITY TESTING ON THE SAMPLES IDENTIFIED AS 2703-01
AND 2703-02 (COLLECTED ON DECEMBER 19, 2006)
WORK ORDER: 0600589**

Dear Mr. Smith:

We have conducted two 96-h LC50 toxicity tests using rainbow trout on the above samples, received at Golder Associates Ltd. North Vancouver Toxicology Laboratory on December 22, 2006. The tests were performed according to the Environment Canada protocol for conducting acute toxicity tests using rainbow trout (EPS 1/RM/13, EC 2000). During testing, two tanks that are normally reserved for use in zinc reference toxicant tests were inadvertently used for the 25% (v/v) concentration in sample 2703-1 and the 6.25% (v/v) concentration in sample 2703-2 resulting in lower fish survival in these tanks. For this reason, the affected concentrations have been excluded from the test results. The results of these tests are based on the appended data and are presented in Table 1.

If you wish to schedule additional testing or have any questions regarding the data presented in this report, please do not hesitate to contact me by e-mail (rharrison@golder.com) or telephone.

Yours very truly,
GOLDER ASSOCIATES LTD.

Robert Harrison, B.Sc. Hons.
Laboratory Biologist – Fish Team

Verified By:

QA/QC Committee:
Julianna Kalocai, M.Sc. R.P. Bio.
Barri Rudolph, B.Sc.

REH/JGK/pdk
Attachment: Table 1

C:\Documents and Settings\PKersey\Local Settings\Temporary Internet Files\OLK\B1\0600589.doc



TABLE 1: 96-h LC50 Toxicity Test Results

Sample Identification	Sample Collection Date (Time)	96-h LC50 [% vol/vol]
2703-01	December 19, 2006 (1500)	>100
2703-02	December 19, 2006 (1600h)	>100

Toxicity testing was carried out in accordance with applicable test methodologies and/or standards of practice. Our liability is limited solely to the cost of re-testing in the event of non-compliance with such test specifications or standards of practice. Golder accepts no responsibility or liability for the interpretation or use of these testing results by others, nor for any delay, loss, damage or interruptions of testing, collection, preparation, and delivery of samples or test results resulting from events or circumstances beyond our control.

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA SUMMARY**

Client Golder
Lab Project No. 06-1414-092
Lab Work Order No. 0600589

Lab Analysts REH, AXF, MFB
Test Type 96hr LC50
Test Initiation Date Dec 19, 2006
22

SAMPLE

Identification 2703-1
Amount Received 2 x 20L
Date Collected Dec 19, 2006
Date Received Dec 22, 2006
Other -

DILUTION/CONTROL WATER (initial water quality)

Fresh Water (dechlorinated) ✓
Temperature (°C) 14
pH 7.5 to 7.6
Dissolved Oxygen (mg/L) 10.2
Conductivity (µS/cm) 61
Hardness (mg/L as CaCO₃) 12
Alkalinity (mg/L as CaCO₃) 12
Other -

TEST SPECIES INFORMATION

Source San Valley
Collection Date/Batch 11/2/06
Control Fish Size (mean, SD and range measured at end of test)
Date Measured Dec 26, 2006
Fork Length (mm) 37 ± 3 (34* and 41)
Wet Weight (g) 0.42 ± 0.12 (0.32-0.57)
Reference Toxicant SDS 0.19

Current Reference Toxicant Result

Reference Toxicant Test Date Dec 8/06
Duration of Acclimation (days) 17
96-h LC50 (and 95% CL) 32 (27 and 38) mg/L
Reference Toxicant Warning Limits (mean ± 2SD) and CV
28 ± 13 mg/L SDS CV: 23%

TEST CONDITIONS

Dissolved Oxygen Range (mg/L) 8.5 → 10.2
Temperature Range (°C) 8.9 → 10.3
pH Range 14 → 16
Conductivity Range (µS/cm) 6.8 → 7.7 6.3 - 8.1
Aeration Provided? (give rate) 60 → 2340 61 - 238
Photoperiod (L:D h) 6.5 ± 1 mL/min/L
No. Organisms/Volume 16:8
Loading Density (g/L) 10/12L
Acclimation Before Testing (days) 0.835
Mortality In Previous Week of Acclimation (%) 31
Other 0%

TEST RESULTS

The 96hr LC50 for sample 2703-1 is
>100% (V/V)

Data Verified By B. Nish

Date Verified 16 Jan 2007

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA**

WHOLE SAMPLE WATER QUALITY

Temp. (°C)	Initial	pH Adjustment ¹	After 30-min Pre-aeration
	14	/	14
pH	6.1	/	6.3
DO (mg/L)	7.4	/	7.7
Conc: (µS/cm)	233	/	238

1. Document pH adjustment procedure (if used) under "Comments".

Client Golder
 Lab Project No. 06-144-092
 Lab Work Order No. 0600589
 Trout Batch No. and 7-d Acclimation Mortality 12106/02
 No. Fish/Volume 2703-1 10/12L
 Sample ID Dec 14/06
 Date/Time Collected Dec 14/06 @ 1600
 Test Initiation Date/Time

Total Pre-Aeration Time 30 min

Concentration % (v/v)	Number of Survivors (1 to 96 hours)						Dissolved Oxygen (mg/L)						Temperature (°C)						pH						Conductivity (µS/cm)	
	1	2	4	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	96		
0% (v/v)				10	10	10	10	10.2	8.8	8.8	9.4	9.6	14	16	15	14	14.0	7.6	7.5	7.6	7.3	7.1	6.1	6.9		
control				10	10	10	10	10.2	8.5	8.6	9.2	8.8	14	16	15	14	14.0	7.5	7.1	7.8	7.1	7.0	7.4	7.3		
0.025				10	10	10	10	10.1	9.2	9.2	9.6	9.7	14	16	15	14	14.0	7.2	7.3	7.6	7.3	7.0	8.5	8.8		
12.5				10	10	10	10	9.7	9.3	9.1	9.8	9.8	14	16	15	14	14.0	7.0	7.3	7.2	7.5	7.4	10.8	10.8		
25				10	10	10	10	9.3	9.3	9.2	9.9	9.9	14	16	15	14	14.0	6.7	7.6	7.2	7.9	7.6	15.2	15.0		
50				10	10	10	10	7.7	9.4	9.1	9.8	9.9	14	16	15	14	14.0	6.3	7.6	7.4	8.1	8.0	23.8	22.7		
100																										
Technician Initials				DM	AXE	DM	WFS	WFS	WFS	DM	AXE	DM	WFS	WFS	DM	AXE	DM	WFS	WFS	DM	AXE	DM	WFS	WFS		

WQ Instruments Used: Temperature Calibrated thermometer DO II-A-20 Conductivity II-A-5303.86
 Sample Description Dark brown
 Comments 1 Tank used previously used for Zn testing only

Test Set Up By MM/SMS Data Verified By BSM Date Verified 15-Jan-2007

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA SUMMARY**

Client Golder
Lab Project No. 06-1414-092
Lab Work Order No. 0600589

Lab Analysts AxF, REH, MFB
Test Type 96hr LC50
Test Initiation Date Dec 22, 2006

SAMPLE

Identification 2703-2
Amount Received 2 X 20L
Date Collected 19 Dec 2006
Date Received 22 Dec 2006
Other -

DILUTION/CONTROL WATER (initial water quality)

Fresh Water (dechlorinated) ☒
Temperature (°C) 14
pH 6.8-7.5
Dissolved Oxygen (mg/L) 9.4-10.3
Conductivity (µS/cm) 2320-60
Hardness (mg/L as CaCO₃) 12
Alkalinity (mg/L as CaCO₃) 12
Other -

TEST SPECIES INFORMATION

Source San Valley
Collection Date/Batch 112106
Control Fish Size (mean, SD and range measured at end of test)
Date Measured Dec 26, 2006
Fork Length (mm) 36 ± 2 (34-40 and 40)
Wet Weight (g) 0.40 ± 0.07 (0.31 and 0.53)
Reference Toxicant SDS

Current Reference Toxicant Result

Reference Toxicant Test Date Dec 8/06
Duration of Acclimation (days) 17
96-h LC50 (and 95% CL) 32 (27 and 38) mg/L
Reference Toxicant Warning Limits (mean ± 2SD) and CV
28 ± 13 µg/L SDS CV: 23%

TEST CONDITIONS

Dissolved Oxygen Range (mg/L) 8-10.3
Temperature Range (°C) 14.0-16.0
pH Range 6.8-7.7
Conductivity Range (µS/cm) 60-2340
Aeration Provided? (give rate) 6.5 L/min/L
Photoperiod (L:D h) 16:8
No. Organisms/Volume 10/12L
Loading Density (g/L) 0.33
Acclimation Before Testing (days) 31
Mortality In Previous Week of Acclimation (%) 0%
Other -

TEST RESULTS The 96hr LC50 for Sample 2703-2 is >100% (UN)

Data Verified By B. Park

Date Verified 15 Jan 2007

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA**

WHOLE SAMPLE WATER QUALITY

Temp. (°C)	Initial	pH Adjustment ¹	After 30-min Pre-aeration
14			14
pH	6.8		6.8
DO (mg/L)	9.4		9.4
Cond. (µS/cm)	7340		7340

¹. Document pH adjustment procedure (if used) under "Comments".

Client Golder

Lab Project No. 05-144-002
 Lab Work Order No. 0600589
 Trout Batch No. and 7-d Acclimation Mortality 112066106
 No. Fish/Volume 10/12L
 Sample ID 2703-2
 Date/Time Collected Dec 19/06
 Test Initiation Date/Time Dec 19/06 09:16:10

Total Pre-Aeration Time 30 min

Concentration % (0/100)	Number of Survivors (1 to 96 hours)					Dissolved Oxygen (mg/L)					Temperature (°C)					pH					Conductivity (µS/cm)		
	1	2	4	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	24	48	72	96	
control				10	10	10	10	103	9.3	8.8	9.6	10.0	14	16	15	14	14.6	7.5	7.7	7.6	7.7	6.0	6.7
ca 25				10	10	10	10	103	9.4	9.0	9.6	9.7	14	16	15	14	14.0	7.5	7.5	7.6	7.6	3.96	4.02
12.5				10	10	10	10	103	9.2	9.0	9.7	9.9	14	16	15	14	14.0	7.4	7.3	7.5	7.5	5.71	4.14
25				6	0 ⁰			102	8.9	9.0			14	16	15			7.2	7.0	7.2		8.79	5.87
50				10	10	10	10	9.9	9.3	9.1	9.9	9.9	14	16	15	14	14.0	7.0	9.3	7.2	7.4	15.35	15.18
100				10	10	10	10	9.6	9.4	9.2	9.9	9.9	14	16	15	14	14.0	6.8	7.3	7.0	7.4	13.40	13.40
Technician Initials				gm	APF	m		9.0	9.0	9.0	APF	m	9.0	9.0	9.0	APF	9.0	9.0	9.0	APF	9.0	9.0	9.0

WQ Instruments Used: Calibrated H2S Temperature DO II-A-20 Conductivity II-A-03030
 Sample Description light brown pH 11-A-030302 DO II-A-20

Comments Found 6 dead fish in tank. - tanks previously used for Zn testing only

Test Set Up By WU Data Verified By B. P. Smith Date Verified 15-Jan-2007

No 2703

195 Pemberton Avenue
North Vancouver, B.C.
Canada V7P 2R4

Tel: 604-986-4331
Fax: 604-662-8548
www.golder.com



Client Name GOLDER (victoria)
Address 2640 Douglas St.
Victoria, BC
V8T 4M1

Client Contact DWAYNE SMITH
Phone 250-881-7372
Fax 250-881-7470

Shipped to EJS GOLDER
195 Pemberton Ave
North Vancouver V7P2R4

Sampled by N. Goeffer
Attn. Cathy McPherson

Shipping Date 20/12/06[illegible]

- 1 For composite effluent or water samples, the sample collection date/time is the end of the compositing period.
- 2 Receiving Water (RW): Effluent (E); Eutriate (ELU); Sediment (SED); Chemical (CHEM); Stormwater (SW); Other (Please Specify)
- 3 Collapsible Carboy (CC); glass jar (GJ); Jerry Can (JC); Plastic HDPE (P); Other (Please Specify)
- 4 Please note any conditions the lab should be aware of for safety and storage concerns

Distribution of copies:

White, Yellow — accompany the shipment
Pink — kept by consignor (e.g. shipper)
Yellow — kept by consignee (e.g. receiver)
White — returned to consignor by consignee

Please see instructions for completion on back of form



CERTIFICATE OF ANALYSIS

Date: January 16, 2007

ALS File No. Z6280rr

Report On: 06-1414-092-6000 Water Analysis

Report To: **Golder Associates Ltd.**
2640 Douglas Street
Victoria, BC
V8T 4M1

Attention: **Mr. Dwayne Smith**

Received: December 21, 2006

ALS ENVIRONMENTAL

per:

Mahssa Beattie, B.Sc. - Account Manager
Paula Shuto, B.Sc. - Account Manager

File No. Z6280rr

REMARKS



This report, Z6280rr, replaces and supercedes the original file, Z6280r. Certain Total Metals analytes not previously reported have been added. All other data remains unchanged.

This report, Z6280r, replaces and supercedes the original file, Z6280. The Chromium (III) and Chromium (VI) detection limits have been decreased where possible. All other data remains unchanged.

Please note that the Speciated Chromium detection limits have been increased for some of the samples due to colour interference.

Please note that certain Dissolved Anions, Nutrients, Metals, and Sulphide detection limits have been increased for some of the samples due to the matrix interferences encountered during the analysis.

For some of the submitted water samples, the measured concentration of specific dissolved parameters is greater than the corresponding total parameters concentration. The explanation for these findings is one or a combination of the following:

- laboratory method variability;
- field sampling method variability;
- bias introduced during general handling, storage, transportation and/or analysis of the sample;
- field sample grab bias - where separate grab samples are processed to produce total and dissolved samples;
- field sample split bias - where total and dissolved parameters samples are produced from the same grab sample.

For further clarification on any of the above information, please contact your ALS representative.

File No. Z6280rr

RESULTS OF ANALYSIS - Water



Sample ID	0569-01	0569-02	0569-03	0569-04	0569-05
Sample Date	06-12-19	06-12-19	06-12-19	06-12-19	06-12-19
ALS ID	1	2	3	4	5

Physical Tests

Hardness	CaCO ₃	61.5	107	52.9	109	1760
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Dissolved Anions

Bromide	Br	<0.050	<0.050	0.184	0.071	9.8
Chloride	Cl	4.15	7.37	10.2	11.5	5340
Fluoride	F	0.062	0.103	0.229	0.057	1.9
Sulphate	SO ₄	1.83	<0.50	3.71	42.0	747

Nutrients

Nitrate Nitrogen	N	<0.0050	<0.0050	<0.0050	0.0100	<0.25
Nitrite Nitrogen	N	<0.0010	<0.0010	<0.0010	0.0023	<0.050

Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre.
 < = Less than the detection limit indicated.

File No. Z6280rr

RESULTS OF ANALYSIS - Water

Sample ID		0569-01	0569-02	0569-03	0569-04	0569-05
Sample Date		06-12-19	06-12-19	06-12-19	06-12-19	06-12-19
ALS ID		1	2	3	4	5
Total Metals						
Aluminum	T-Al	5.56	24.7	57.3	1.08	23.4
Antimony	T-Sb	<0.0025	<0.0025	<0.0025	<0.0025	<0.025
Arsenic	T-As	<0.0050	0.0111	0.0356	<0.0050	<0.050
Barium	T-Ba	<0.10	<0.10	0.15	0.097	<0.10
Beryllium	T-Be	<0.025	<0.025	<0.025	<0.0050	<0.025
Bismuth	T-Bi	<1.0	<1.0	<1.0	<0.20	<1.0
Boron	T-B	<0.50	<0.50	<0.50	<0.10	1.25
Cadmium	T-Cd	<0.00025	0.00057	0.00230	0.00086	<0.0025
Calcium	T-Ca	17.7	34.2	27.5	24.8	118
Chromium	T-Cr	0.0129	0.0546	0.155	0.0033	0.059
Cobalt	T-Co	<0.0025	0.0134	0.0346	0.0149	<0.025
Copper	T-Cu	0.0151	0.237	0.468	0.0235	0.148
Iron	T-Fe	9.69	43.6	79.7	4.66	35.3
Lead	T-Pb	<0.0050	0.0152	0.0316	<0.0050	<0.050
Lithium	T-Li	<0.25	<0.25	<0.25	<0.050	<0.25
Magnesium	T-Mg	6.95	19.0	30.4	9.62	350
Manganese	T-Mn	0.224	1.61	1.53	1.19	0.787
Mercury	T-Hg	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020
Molybdenum	T-Mo	<0.0050	<0.0050	0.0465	<0.0050	<0.050
Nickel	T-Ni	<0.025	0.036	0.082	<0.025	<0.25
Phosphorus	T-P	<1.5	1.9	3.6	<0.30	2.4
Potassium	T-K	<10	13	15	8.7	112
Selenium	T-Se	<0.0050	<0.0050	<0.0050	<0.0050	<0.10
Silicon	T-Si	13.7	45.5	70.2	15.6	28.8
Silver	T-Ag	<0.00025	<0.00025	0.00041	<0.00025	<0.0025
Sodium	T-Na	<10	17	35	32.9	2890
Strontium	T-Sr	0.100	0.127	0.180	0.0832	2.04
Thallium	T-Tl	<0.0010	<0.0010	<0.0010	<0.0010	<0.010
Tin	T-Sn	<0.15	<0.15	<0.15	<0.030	<0.15
Titanium	T-Ti	0.26	1.02	2.53	<0.050	0.98
Uranium	T-U	<0.0010	0.0060	0.0311	<0.0010	0.020
Vanadium	T-V	<0.15	<0.15	0.27	<0.030	<0.15
Zinc	T-Zn	<0.025	0.087	0.167	0.0352	0.045

Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre.
 < = Less than the detection limit indicated.

RESULTS OF ANALYSIS - Water

Sample ID	0569-01	0569-02	0569-03	0569-04	0569-05
Sample Date	06-12-19	06-12-19	06-12-19	06-12-19	06-12-19
ALS ID	1	2	3	4	5

Dissolved Metals

Aluminum	D-Al	0.239	0.538	1.04	0.985	<0.50
Antimony	D-Sb	<0.0025	<0.0025	<0.0025	<0.0025	<0.025
Arsenic	D-As	<0.0050	<0.0050	<0.0050	<0.0050	<0.050
Barium	D-Ba	<0.10	<0.10	<0.10	0.104	<0.10
Beryllium	D-Be	<0.025	<0.025	<0.025	<0.0050	<0.025
Bismuth	D-Bi	<1.0	<1.0	<1.0	<0.20	<1.0
Boron	D-B	<0.50	<0.50	<0.50	<0.10	1.29
Cadmium	D-Cd	<0.00025	<0.00025	<0.00025	0.00064	<0.0025
Calcium	D-Ca	15.4	25.0	10.4	26.4	119
Chromium	D-Cr	<0.0025	<0.0025	0.0040	<0.0025	<0.025
Cobalt	D-Co	<0.0025	<0.0025	0.0039	0.0183	<0.025
Copper	D-Cu	<0.0050	<0.0050	<0.0050	0.0201	<0.050
Iron	D-Fe	5.60	21.9	13.6	5.04	2.67
Lead	D-Pb	<0.0050	<0.0050	<0.0050	<0.0050	<0.050
Lithium	D-Li	<0.25	<0.25	<0.25	<0.050	<0.25
Magnesium	D-Mg	5.61	10.7	6.52	10.4	355
Manganese	D-Mn	0.188	1.27	0.659	1.29	0.606
Mercury	D-Hg	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020
Molybdenum	D-Mo	<0.0050	<0.0050	<0.0050	<0.0050	<0.050
Nickel	D-Ni	<0.025	<0.025	<0.025	<0.025	<0.25
Phosphorus	D-P	<1.5	<1.5	<1.5	<0.30	<1.5
Potassium	D-K	<10	12	10	9.5	119
Selenium	D-Se	<0.0050	<0.0050	<0.0050	<0.0050	<0.10
Silicon	D-Si	7.70	17.5	15.3	16.8	8.24
Silver	D-Ag	<0.00025	<0.00025	<0.00025	<0.00025	<0.0025
Sodium	D-Na	<10	15	31	34.4	3060
Strontium	D-Sr	0.081	0.083	0.059	0.0883	2.07
Thallium	D-Tl	<0.0010	<0.0010	<0.0010	<0.0010	<0.010
Tin	D-Sn	<0.15	<0.15	<0.15	<0.030	<0.15
Titanium	D-Ti	<0.25	<0.25	<0.25	<0.050	<0.25
Uranium	D-U	<0.0010	<0.0010	0.0012	<0.0010	<0.010
Vanadium	D-V	<0.15	<0.15	<0.15	<0.030	<0.15
Zinc	D-Zn	<0.025	<0.025	<0.025	0.0385	<0.025

Speciated Metals

Chromium (III)	T-Cr	<0.015	<0.060	<0.20	<0.010	<0.060
Chromium (VI)	T-Cr	<0.015	<0.060	<0.20	<0.010	<0.060

Inorganic Parameters

Sulphide	S	0.10	<0.20	0.13	<0.040	0.023
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Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre.
 < = Less than the detection limit indicated.

File No. Z6280rr

RESULTS OF ANALYSIS - Water



Sample ID		0569-01	0569-02	0569-03	0569-04	0569-05
Sample Date		06-12-19	06-12-19	06-12-19	06-12-19	06-12-19
ALS ID		1	2	3	4	5
Organic Parameters						
Chemical Oxygen Demand	COD	95	705	1840	173	570
Phenols		0.0050	0.0320	0.0320	0.0510	0.0220
Tannin and Lignin		4.09	9.3	12.3	7.8	2.13

Remarks regarding the analyses appear at the beginning of this report.
Results are expressed as milligrams per litre.
< = Less than the detection limit indicated.

File No. Z6280rr

RESULTS OF ANALYSIS - Water



Sample ID	0569-06	0569-07	0569-08
Sample Date	06-12-19	06-12-19	06-12-19
ALS ID	6	7	8

Physical Tests

Hardness	CaCO ₃	1720	84.9	205
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Dissolved Anions

Bromide	Br	15.5	0.085	2.25
Chloride	Cl	5320	10.9	660
Fluoride	F	<1.0	0.114	0.072
Sulphate	SO ₄	753	1.79	88.0

Nutrients

Nitrate Nitrogen	N	<0.25	<0.0050	0.342
Nitrite Nitrogen	N	<0.050	<0.0010	0.0015

Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre.
 < = Less than the detection limit indicated.

File No. Z6280rr

RESULTS OF ANALYSIS - Water



Sample ID	0569-06	0569-07	0569-08
Sample Date	06-12-19	06-12-19	06-12-19
ALS ID	6	7	8
Total Metals			
Aluminum T-Al	19.9	0.289	0.269
Antimony T-Sb	<0.025	<0.0025	<0.0025
Arsenic T-As	<0.050	<0.0050	<0.0050
Barium T-Ba	<0.10	<0.10	<0.10
Beryllium T-Be	<0.025	<0.025	<0.025
Bismuth T-Bi	<1.0	<1.0	<1.0
Boron T-B	1.26	<0.50	<0.50
Cadmium T-Cd	<0.0025	<0.00025	<0.00025
Calcium T-Ca	119	17.9	15.2
Chromium T-Cr	0.051	<0.0025	<0.0025
Cobalt T-Co	<0.025	<0.0025	<0.0025
Copper T-Cu	0.110	0.0152	<0.0050
Iron T-Fe	32.7	12.5	0.71
Lead T-Pb	<0.050	<0.0050	<0.0050
Lithium T-Li	<0.25	<0.25	<0.25
Magnesium T-Mg	356	9.76	40.4
Manganese T-Mn	0.770	1.01	<0.050
Mercury T-Hg	<0.00020	<0.00020	<0.00020
Molybdenum T-Mo	<0.050	<0.0050	<0.0050
Nickel T-Ni	<0.25	<0.025	<0.025
Phosphorus T-P	1.9	<1.5	<1.5
Potassium T-K	115	12	15
Selenium T-Se	<0.10	<0.0050	<0.010
Silicon T-Si	27.0	13.7	4.76
Silver T-Ag	<0.0025	<0.00025	<0.00025
Sodium T-Na	2920	33	362
Strontium T-Sr	2.04	0.074	0.231
Thallium T-Tl	<0.010	<0.0010	<0.0010
Tin T-Sn	<0.15	<0.15	<0.15
Titanium T-Ti	0.89	<0.25	<0.25
Uranium T-U	0.016	<0.0010	<0.0010
Vanadium T-V	<0.15	<0.15	<0.15
Zinc T-Zn	0.043	<0.025	<0.025

Remarks regarding the analyses appear at the beginning of this report.
Results are expressed as milligrams per litre.
< = Less than the detection limit indicated.

File No. Z6280rr

RESULTS OF ANALYSIS - Water

Sample ID	0569-06	0569-07	0569-08
Sample Date	06-12-19	06-12-19	06-12-19
ALS ID	6	7	8

Dissolved Metals

Aluminum	D-Al	<0.50	-	-
Antimony	D-Sb	<0.025	-	-
Arsenic	D-As	<0.050	-	-
Barium	D-Ba	<0.10	-	-
Beryllium	D-Be	<0.025	-	-
Bismuth	D-Bi	<1.0	-	-
Boron	D-B	1.26	-	-
Cadmium	D-Cd	<0.0025	-	-
Calcium	D-Ca	116	-	-
Chromium	D-Cr	<0.025	-	-
Cobalt	D-Co	<0.025	-	-
Copper	D-Cu	<0.050	-	-
Iron	D-Fe	2.54	-	-
Lead	D-Pb	<0.050	-	-
Lithium	D-Li	<0.25	-	-
Magnesium	D-Mg	348	-	-
Manganese	D-Mn	0.583	-	-
Mercury	D-Hg	<0.00020	-	-
Molybdenum	D-Mo	<0.050	-	-
Nickel	D-Ni	<0.25	-	-
Phosphorus	D-P	<1.5	-	-
Potassium	D-K	116	-	-
Selenium	D-Se	<0.10	-	-
Silicon	D-Si	8.05	-	-
Silver	D-Ag	<0.0025	-	-
Sodium	D-Na	3000	-	-
Strontium	D-Sr	2.02	-	-
Thallium	D-Tl	<0.010	-	-
Tin	D-Sn	<0.15	-	-
Titanium	D-Ti	<0.25	-	-
Uranium	D-U	<0.010	-	-
Vanadium	D-V	<0.15	-	-
Zinc	D-Zn	<0.025	-	-

Speciated Metals

Chromium (III)	T-Cr	<0.060	<0.010	<0.010
Chromium (VI)	T-Cr	<0.060	<0.010	<0.010

Inorganic Parameters

Sulphide	S	0.026	0.092	<0.020
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Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre.
 < = Less than the detection limit indicated.

File No. Z6280rr

RESULTS OF ANALYSIS - Water



Sample ID	0569-06	0569-07	0569-08
Sample Date	06-12-19	06-12-19	06-12-19
ALS ID	6	7	8

Organic Parameters

Chemical Oxygen Demand	COD	618	115	55
Phenols		0.0190	0.0430	0.0100
Tannin and Lignin		2.14	10.6	1.60

Remarks regarding the analyses appear at the beginning of this report.
Results are expressed as milligrams per litre.
< = Less than the detection limit indicated.

File No. Z6280rr

Appendix 1 - QUALITY CONTROL - Replicates



Water	0569-02	0569-02
	06-12-19	QC # 535374

Physical Tests

Hardness	CaCO3	107	107
Total Metals			
Aluminum	T-Al	24.7	24.4
Antimony	T-Sb	<0.0025	<0.0025
Arsenic	T-As	0.0111	0.0111
Barium	T-Ba	<0.10	<0.10
Beryllium	T-Be	<0.025	<0.025
Boron	T-B	<0.50	<0.50
Cadmium	T-Cd	0.00057	0.00061
Calcium	T-Ca	34.2	34.7
Chromium	T-Cr	0.0546	0.0545
Cobalt	T-Co	0.0134	0.0133
Copper	T-Cu	0.237	0.233
Iron	T-Fe	43.6	44.5
Lead	T-Pb	0.0152	0.0154
Lithium	T-Li	<0.25	<0.25
Magnesium	T-Mg	19.0	19.2
Manganese	T-Mn	1.61	1.63
Mercury	T-Hg	<0.00020	<0.00020
Molybdenum	T-Mo	<0.0050	<0.0050
Nickel	T-Ni	0.036	0.036
Selenium	T-Se	<0.0050	<0.0050
Silver	T-Ag	<0.00025	<0.00025
Sodium	T-Na	17	17
Thallium	T-Tl	<0.0010	<0.0010
Titanium	T-Ti	1.02	1.03
Uranium	T-U	0.0060	0.0063
Vanadium	T-V	<0.15	<0.15
Zinc	T-Zn	0.087	0.083

Remarks regarding the analyses appear at the beginning of this report.
Results are expressed as milligrams per litre.
< = Less than the detection limit indicated.

Appendix 1 - QUALITY CONTROL - Replicates

Water	0569-02	0569-02
	06-12-19	QC # 535374

Dissolved Metals

Aluminum	D-Al	0.538	0.528
Antimony	D-Sb	<0.0025	<0.0025
Arsenic	D-As	<0.0050	<0.0050
Barium	D-Ba	<0.10	<0.10
Beryllium	D-Be	<0.025	<0.025
Bismuth	D-Bi	<1.0	<1.0
Boron	D-B	<0.50	<0.50
Cadmium	D-Cd	<0.00025	<0.00025
Calcium	D-Ca	25.0	25.2
Chromium	D-Cr	<0.0025	<0.0025
Cobalt	D-Co	<0.0025	<0.0025
Copper	D-Cu	<0.0050	<0.0050
Iron	D-Fe	21.9	22.1
Lead	D-Pb	<0.0050	<0.0050
Lithium	D-Li	<0.25	<0.25
Magnesium	D-Mg	10.7	10.8
Manganese	D-Mn	1.27	1.29
Mercury	D-Hg	<0.00020	<0.00020
Molybdenum	D-Mo	<0.0050	<0.0050
Nickel	D-Ni	<0.025	<0.025
Phosphorus	D-P	<1.5	<1.5
Potassium	D-K	12	12
Selenium	D-Se	<0.0050	<0.0050
Silicon	D-Si	17.5	17.5
Silver	D-Ag	<0.00025	<0.00025
Sodium	D-Na	15	15
Strontium	D-Sr	0.083	0.085
Thallium	D-Tl	<0.0010	<0.0010
Tin	D-Sn	<0.15	<0.15
Titanium	D-Ti	<0.25	<0.25
Uranium	D-U	<0.0010	<0.0010
Vanadium	D-V	<0.15	<0.15
Zinc	D-Zn	<0.025	<0.025

Speciated Metals

Chromium (VI)	T-Cr	<0.060	<0.060
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Inorganic Parameters

Sulphide	S	<0.20	0.22
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Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre.
 < = Less than the detection limit indicated.

File No. Z6280rr

Appendix 1 - QUALITY CONTROL - Replicates



Water	0569-02	0569-02
	06-12-19	QC # 535374

Organic Parameters

Chemical Oxygen Demand	COD	705	731
Tannin and Lignin		9.3	9.3

Remarks regarding the analyses appear at the beginning of this report.
Results are expressed as milligrams per litre.
< = Less than the detection limit indicated.

Appendix 1 - QUALITY CONTROL - Reference Materials

Water

		Result	Target	Units	DL	ALSQC#	Method
ALSEV Metals in Water - High, 4GK116							
Total Metals							
Aluminum	T-Al	1.96	2.00	mg/L	0.2	535841	a
Aluminum	T-Al	2.03	2.00	mg/L	0.2	535917	a
Antimony	T-Sb	0.96	1.00	mg/L	0.2	535841	a
Antimony	T-Sb	0.99	1.00	mg/L	0.2	535917	a
Arsenic	T-As	0.99	1.00	mg/L	0.2	535841	a
Arsenic	T-As	0.99	1.00	mg/L	0.2	535917	a
Barium	T-Ba	0.244	0.250	mg/L	0.01	535841	a
Barium	T-Ba	0.238	0.250	mg/L	0.01	535917	a
Beryllium	T-Be	0.0944	0.100	mg/L	0.005	535841	a
Beryllium	T-Be	0.0981	0.100	mg/L	0.005	535917	a
Bismuth	T-Bi	0.94	1.00	mg/L	0.2	535841	a
Bismuth	T-Bi	1.01	1.00	mg/L	0.2	535917	a
Boron	T-B	0.97	1.00	mg/L	0.1	535841	a
Boron	T-B	0.98	1.00	mg/L	0.1	535917	a
Cadmium	T-Cd	0.100	0.100	mg/L	0.01	535841	a
Cadmium	T-Cd	0.097	0.100	mg/L	0.01	535917	a
Calcium	T-Ca	49.4	50.0	mg/L	0.05	535841	a
Calcium	T-Ca	50.8	50.0	mg/L	0.05	535917	a
Chromium	T-Cr	0.241	0.250	mg/L	0.01	535841	a
Chromium	T-Cr	0.242	0.250	mg/L	0.01	535917	a
Cobalt	T-Co	0.242	0.250	mg/L	0.01	535841	a
Cobalt	T-Co	0.246	0.250	mg/L	0.01	535917	a
Copper	T-Cu	0.249	0.250	mg/L	0.01	535841	a
Copper	T-Cu	0.243	0.250	mg/L	0.01	535917	a
Iron	T-Fe	1.02	1.00	mg/L	0.03	535841	a
Iron	T-Fe	0.974	1.00	mg/L	0.03	535917	a
Lead	T-Pb	0.496	0.500	mg/L	0.05	535841	a
Lead	T-Pb	0.490	0.500	mg/L	0.05	535917	a
Lithium	T-Li	0.245	0.250	mg/L	0.01	535841	a
Lithium	T-Li	0.245	0.250	mg/L	0.01	535917	a
Magnesium	T-Mg	50.0	50.0	mg/L	0.1	535841	a
Magnesium	T-Mg	49.4	50.0	mg/L	0.1	535917	a
Manganese	T-Mn	0.251	0.250	mg/L	0.005	535841	a
Manganese	T-Mn	0.242	0.250	mg/L	0.005	535917	a

Methods:

a = ICPOES

Appendix 1 - QUALITY CONTROL - Reference Materials

Water

		Result	Target	Units	DL	ALSQC#	Method
ALSEV Metals in Water - High, 4GK116							
Total Metals							
Molybdenum	T-Mo	0.240	0.250	mg/L	0.03	535841	a
Molybdenum	T-Mo	0.256	0.250	mg/L	0.03	535917	a
Nickel	T-Ni	0.508	0.500	mg/L	0.05	535841	a
Nickel	T-Ni	0.480	0.500	mg/L	0.05	535917	a
Phosphorus	T-P	2.50	2.50	mg/L	0.3	535841	a
Phosphorus	T-P	2.57	2.50	mg/L	0.3	535917	a
Potassium	T-K	50.1	50.0	mg/L	2	535841	a
Potassium	T-K	50.7	50.0	mg/L	2	535917	a
Selenium	T-Se	0.99	1.00	mg/L	0.2	535841	a
Selenium	T-Se	0.97	1.00	mg/L	0.2	535917	a
Silicon	T-Si	1.00	1.00	mg/L	0.05	535841	a
Silicon	T-Si	1.02	1.00	mg/L	0.05	535917	a
Silver	T-Ag	0.101	0.100	mg/L	0.01	535841	a
Silver	T-Ag	0.099	0.100	mg/L	0.01	535917	a
Sodium	T-Na	48.2	50.0	mg/L	2	535841	a
Sodium	T-Na	47.7	50.0	mg/L	2	535917	a
Strontium	T-Sr	0.237	0.250	mg/L	0.005	535841	a
Strontium	T-Sr	0.244	0.250	mg/L	0.005	535917	a
Thallium	T-Tl	0.95	1.00	mg/L	0.2	535841	a
Thallium	T-Tl	0.97	1.00	mg/L	0.2	535917	a
Tin	T-Sn	0.538	0.500	mg/L	0.03	535841	a
Tin	T-Sn	0.506	0.500	mg/L	0.03	535917	a
Titanium	T-Ti	0.260	0.250	mg/L	0.01	535841	a
Titanium	T-Ti	0.254	0.250	mg/L	0.01	535917	a
Vanadium	T-V	0.498	0.500	mg/L	0.03	535841	a
Vanadium	T-V	0.491	0.500	mg/L	0.03	535917	a
Zinc	T-Zn	0.486	0.500	mg/L	0.005	535841	a
Zinc	T-Zn	0.507	0.500	mg/L	0.005	535917	a

ALSEV Metals in Water - Mid, 4GK116**Total Metals**

Aluminum	T-Al	0.101	0.100	mg/L	0.001	535823	b
Aluminum	T-Al	0.101	0.100	mg/L	0.001	535916	b

Methods:

a = ICPOES

b = ICPMS

Appendix 1 - QUALITY CONTROL - Reference Materials

Water

		Result	Target	Units	DL	ALSQC#	Method
ALSEV Metals in Water - Mid, 4GK116							
Total Metals							
Antimony	T-Sb	0.0516	0.0500	mg/L	0.0001	535823	a
Antimony	T-Sb	0.0484	0.0500	mg/L	0.0001	535916	a
Arsenic	T-As	0.0515	0.0500	mg/L	0.0001	535823	a
Arsenic	T-As	0.0493	0.0500	mg/L	0.0001	535916	a
Barium	T-Ba	0.0128	0.0125	mg/L	0.00005	535823	a
Barium	T-Ba	0.0125	0.0125	mg/L	0.00005	535916	a
Beryllium	T-Be	0.00506	0.00500	mg/L	0.0005	535823	a
Beryllium	T-Be	0.00502	0.00500	mg/L	0.0005	535916	a
Bismuth	T-Bi	0.0494	0.0500	mg/L	0.0005	535823	a
Bismuth	T-Bi	0.0469	0.0500	mg/L	0.0005	535916	a
Boron	T-B	0.048	0.050	mg/L	0.01	535823	a
Boron	T-B	0.047	0.050	mg/L	0.01	535916	a
Cadmium	T-Cd	0.00504	0.00500	mg/L	0.00005	535823	a
Cadmium	T-Cd	0.00493	0.00500	mg/L	0.00005	535916	a
Calcium	T-Ca	2.54	2.50	mg/L	0.02	535823	a
Calcium	T-Ca	2.49	2.50	mg/L	0.02	535916	a
Chromium	T-Cr	0.0124	0.0125	mg/L	0.0005	535823	a
Chromium	T-Cr	0.0123	0.0125	mg/L	0.0005	535916	a
Cobalt	T-Co	0.0124	0.0125	mg/L	0.0001	535823	a
Cobalt	T-Co	0.0125	0.0125	mg/L	0.0001	535916	a
Copper	T-Cu	0.0123	0.0125	mg/L	0.0001	535823	a
Copper	T-Cu	0.0126	0.0125	mg/L	0.0001	535916	a
Lead	T-Pb	0.0245	0.0250	mg/L	0.00005	535823	a
Lead	T-Pb	0.0245	0.0250	mg/L	0.00005	535916	a
Lithium	T-Li	0.0120	0.0125	mg/L	0.005	535823	a
Lithium	T-Li	0.0123	0.0125	mg/L	0.005	535916	a
Magnesium	T-Mg	2.45	2.50	mg/L	0.005	535823	a
Magnesium	T-Mg	2.48	2.50	mg/L	0.005	535916	a
Manganese	T-Mn	0.0125	0.0125	mg/L	0.00005	535823	a
Manganese	T-Mn	0.0126	0.0125	mg/L	0.00005	535916	a
Molybdenum	T-Mo	0.0131	0.0125	mg/L	0.00005	535823	a
Molybdenum	T-Mo	0.0125	0.0125	mg/L	0.00005	535916	a
Nickel	T-Ni	0.0251	0.0250	mg/L	0.0005	535823	a
Nickel	T-Ni	0.0246	0.0250	mg/L	0.0005	535916	a

Methods:

a = ICPMS

Appendix 1 - QUALITY CONTROL - Reference Materials

Water

		Result	Target	Units	DL	ALSQC#	Method
ALSEV Metals in Water - Mid, 4GK116							
Total Metals							
Potassium	T-K	2.53	2.50	mg/L	0.05	535823	a
Potassium	T-K	2.49	2.50	mg/L	0.05	535916	a
Selenium	T-Se	0.0523	0.0500	mg/L	0.0005	535823	a
Selenium	T-Se	0.0489	0.0500	mg/L	0.0005	535916	a
Silver	T-Ag	0.00517	0.00500	mg/L	0.00001	535823	a
Silver	T-Ag	0.00500	0.00500	mg/L	0.00001	535916	a
Sodium	T-Na	2.55	2.50	mg/L	0.01	535823	a
Sodium	T-Na	2.54	2.50	mg/L	0.01	535916	a
Strontium	T-Sr	0.0132	0.0125	mg/L	0.0001	535823	a
Strontium	T-Sr	0.0125	0.0125	mg/L	0.0001	535916	a
Thallium	T-Tl	0.0489	0.0500	mg/L	0.0001	535823	a
Thallium	T-Tl	0.0467	0.0500	mg/L	0.0001	535916	a
Tin	T-Sn	0.0254	0.0250	mg/L	0.0001	535823	a
Tin	T-Sn	0.0245	0.0250	mg/L	0.0001	535916	a
Uranium	T-U	0.000245	0.000250	mg/L	0.00001	535823	a
Uranium	T-U	0.000259	0.000250	mg/L	0.00001	535916	a
Vanadium	T-V	0.0252	0.0250	mg/L	0.001	535823	a
Vanadium	T-V	0.0246	0.0250	mg/L	0.001	535916	a
Zinc	T-Zn	0.0247	0.0250	mg/L	0.001	535823	a
Zinc	T-Zn	0.0244	0.0250	mg/L	0.001	535916	a
Alltech Custom Anions Mix (10x), AIT170088							
Dissolved Anions							
Bromide	Br	0.984	1.00	mg/L	0.05	535565	b
Chloride	Cl	51.9	50.1	mg/L	0.5	535565	b
Fluoride	F	1.06	1.00	mg/L	0.02	535565	b
Sulphate	SO4	52.9	50.1	mg/L	0.5	535565	b
Nutrients							
Nitrate Nitrogen	N	0.236	0.225	mg/L	0.005	535565	b
Nitrite Nitrogen	N	0.161	0.152	mg/L	0.001	535565	b
SPEX Demand (2xC), Q2-94DEM							
Organic Parameters							
Chemical Oxygen Demand	COD	221	224	mg/L	20	535779	c

Methods:

a = ICPMS

b = Ion Chromatography

c = Colourimetry

File No. Z6280rr

Appendix 1 - QUALITY CONTROL - Spikes



Water

	Result	Target	Units	DL	ALSQC#	Method
<u>Inorganic Parameters</u>						
Sulphide S	1.03	1.00	mg/L	0.02	535686	a
Sulphide S	1.14	1.00	mg/L	0.02	535781	a
<u>Organic Parameters</u>						
Tannin and Lignin	9.19	10.0	mg/L	0.1	535840	a

Methods:

a = Colourimetry

Appendix 1 - QUALITY CONTROL - Blanks



Water

			Result	Target	Units	DL	ALSQC#	Method
Dissolved Anions								
Bromide	Br		<0.050	<0.050	mg/L	0.05	535557	a
Chloride	Cl		<0.50	<0.50	mg/L	0.5	535557	a
Fluoride	F		<0.020	<0.020	mg/L	0.02	535557	a
Sulphate	SO ₄		<0.50	<0.50	mg/L	0.5	535557	a
Nutrients								
Nitrate Nitrogen		N	<0.0050	<0.0050	mg/L	0.005	535557	a
Nitrite Nitrogen		N	<0.0010	<0.0010	mg/L	0.001	535557	a
Total Metals								
Aluminum	T-Al		<0.0050	<0.0050	mg/L	0.005	535590	b
Antimony	T-Sb		<0.00020	<0.00020	mg/L	0.0002	535590	b
Arsenic	T-As		<0.00020	<0.00020	mg/L	0.0002	535590	b
Barium	T-Ba		<0.0010	<0.0010	mg/L	0.001	535590	b
Beryllium	T-Be		<0.0010	<0.0010	mg/L	0.001	535590	b
Bismuth	T-Bi		<0.0010	<0.0010	mg/L	0.001	535590	b
Boron	T-B		<0.050	<0.050	mg/L	0.05	535590	b
Cadmium	T-Cd		<0.00020	<0.00020	mg/L	0.0002	535590	b
Calcium	T-Ca		<0.050	<0.050	mg/L	0.05	535590	b
Chromium	T-Cr		<0.0010	<0.0010	mg/L	0.001	535590	b
Cobalt	T-Co		<0.0010	<0.0010	mg/L	0.001	535590	b
Copper	T-Cu		<0.0010	<0.0010	mg/L	0.001	535590	b
Lead	T-Pb		<0.0010	<0.0010	mg/L	0.001	535590	b
Lithium	T-Li		<0.010	<0.010	mg/L	0.01	535590	b
Magnesium	T-Mg		<0.050	<0.050	mg/L	0.05	535590	b
Manganese	T-Mn		<0.0010	<0.0010	mg/L	0.001	535590	b
Mercury	T-Hg		<0.000010	<0.000010	mg/L	0.00001	535876	c
Molybdenum	T-Mo		<0.0010	<0.0010	mg/L	0.001	535590	b
Nickel	T-Ni		0.0020	<0.0010	mg/L	0.001	535590	b
Potassium	T-K		<0.050	<0.050	mg/L	0.05	535590	b
Selenium	T-Se		<0.0010	<0.0010	mg/L	0.001	535590	b
Silver	T-Ag		<0.00010	<0.00010	mg/L	0.0001	535590	b
Sodium	T-Na		<0.050	<0.050	mg/L	0.05	535590	b
Strontium	T-Sr		<0.0010	<0.0010	mg/L	0.001	535590	b
Thallium	T-Tl		<0.0010	<0.0010	mg/L	0.001	535590	b
Tin	T-Sn		<0.0010	<0.0010	mg/L	0.001	535590	b
Uranium	T-U		<0.000010	<0.000010	mg/L	0.00001	535590	b

Methods:

a = Ion Chromatography

b = ICPMS

c = CVAFS

File No. Z6280rr

Appendix 1 - QUALITY CONTROL - Blanks



Water

		Result	Target	Units	DL	ALSQC#	Method
Total Metals							
Vanadium	T-V	<0.0050	<0.0050	mg/L	0.005	535590	a
Zinc	T-Zn	<0.0050	<0.0050	mg/L	0.005	535590	a
Inorganic Parameters							
Sulphide	S	<0.02	<0.02	mg/L	0.02	535685	b
Sulphide	S	<0.02	<0.02	mg/L	0.02	535780	b
Organic Parameters							
Chemical Oxygen Demand	COD	<20	<20	mg/L	20	535778	b
Tannin and Lignin		<0.1	<0.1	mg/L	0.1	535839	b

Methods:

a = ICPMS

b = Colourimetry

Appendix 2 - METHODOLOGY



Outlines of the methodologies utilized for the analysis of the samples submitted are as follows

Dissolved Anions in Water by Ion Chromatography

This analysis is carried out using procedures adapted from APHA Method 4110 "Determination of Anions by Ion Chromatography" and EPA Method 300.0 "Determination of Inorganic Anions by Ion Chromatography". Anions are determined by filtering the sample through a 0.45 micron membrane filter and injecting the filtrate onto a Dionex IonPac AG17 anion exchange column with a hydroxide eluent stream. Anions routinely determined by this method include: bromide, chloride, fluoride, nitrate, nitrite and sulphate.

Recommended Holding Time:

Sample: 28 days (bromide, chloride, fluoride, sulphate)

Sample: 2 days (nitrate, nitrite)

Reference: APHA and EPA

Laboratory Location: ALS Environmental, Vancouver

Metals in Water

This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" 21th Edition 2005 published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedures may involve preliminary sample treatment by acid digestion, using either hotplate or microwave oven, or filtration (EPA Method 3005A). Instrumental analysis is by atomic absorption/emission spectrophotometry (EPA Method 7000 series), inductively coupled plasma - optical emission spectrophotometry (EPA Method 6010B), and/or inductively coupled plasma - mass spectrometry (EPA Method 6020).

Recommended Holding Time:

Sample: 6 months

Reference: EPA

Laboratory Location: ALS Environmental, Vancouver

Mercury in Water

This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" 20th Edition 1998 published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedure involves a cold-oxidation of the acidified sample using bromine monochloride prior to reduction of the sample with stannous chloride. Instrumental analysis is by cold vapour atomic fluorescence spectrophotometry (EPA Method 245.7).

File No. Z6280rr

Appendix 2 - METHODOLOGY - Continued



Recommended Holding Time:

Sample: 28 days

Reference: EPA

Laboratory Location: ALS Environmental, Vancouver

Chromium (VI) and Chromium (III) in Water and Seawater

This analysis is carried out using procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846, Method 7196A, published by the United States Environmental Protection Agency (EPA). The procedure involves a preliminary analysis of the sample for total chromium by atomic absorption spectrophotometry (EPA Method 7000 series), inductively coupled plasma - optical emission spectrophotometry (EPA Method 6010B), and/or inductively coupled plasma - mass spectrometry (EPA Method 6020). The sample is then analysed for chromium (VI) by colourimetry using diphenylcarbazide in a sulphuric acid solution. Chromium (III) is calculated as the difference between the total chromium and the chromium (VI) results.

Recommended Holding Time:

Sample: 24 hours

Reference: EPA

Laboratory Location: ALS Environmental, Vancouver

Sulphide in Water

This analysis is carried out using procedures adapted from APHA Method 4500-S2 "Sulphide". Sulphide is determined using the methylene blue colourimetric method.

Recommended Holding Time:

Sample: 7 days

Reference: APHA

Laboratory Location: ALS Environmental, Vancouver

Chemical Oxygen Demand in Water

This analysis is carried out using procedures adapted from APHA Method 5220 "Chemical Oxygen Demand (COD)". Chemical oxygen demand is determined using the closed reflux colourimetric method.

Recommended Holding Time:

Sample: 28 days

Reference: APHA

Laboratory Location: ALS Environmental, Vancouver



Phenols in Water

This analysis is carried out using procedures adapted from APHA Method 5530 "Phenols". Phenols, defined by APHA as "hydroxy derivatives of benzene and its condensed nuclei", is determined by sample distillation and analysis using the antipyrine colourimetric method. The recovery of phenolics is non-uniform; substitution usually reduces recovery and the reported value represents the minimum concentration of phenolic compounds present in the sample.

Recommended Holding Time:

Sample: 28 days

Reference: APHA

For more detail see ALS Environmental "Collection & Sampling Guide"

Laboratory Location: ALS Environmental, Calgary

Conventional Parameters in Water

These analyses are carried out in accordance with procedures described in "Methods for Chemical Analysis of Water and Wastes" (USEPA), "Manual for the Chemical Analysis of Water, Wastewaters, Sediments and Biological Tissues" (BCMOE), and/or "Standard Methods for the Examination of Water and Wastewater" (APHA). Further details are available on request.

Laboratory Location: ALS Environmental, Vancouver

Chloride in Water

This analysis is carried out using procedures adapted from APHA Method 4500 "Chloride". Chloride is determined using the ferricyanide colourimetric method.

Recommended Holding Time:

Sample: 28 days

Reference: APHA

Laboratory Location: ALS Environmental, Vancouver

Results contained within this certificate relate only to the samples as submitted.

This Certificate Of Analysis shall only be reproduced in full, except with the written approval of ALS Environmental.

End of Report

Golder Associates Ltd.

195 Pemberton Avenue
North Vancouver, British Columbia, Canada V7P 2R4
Telephone 604-986-4331
Fax 604-662-8548



April 9, 2007

E/07/1498
07-1414-0039

Golder Associates Ltd.
2640 Douglas Street
Victoria, BC V8T 4M1

Attention: Dwayne Smith

**RE: TOXICITY TESTING ON THE SAMPLES IDENTIFIED AS 2849-01
(AS06-O/A) & 2849-02 (GWS06-01) (COLLECTED ON
MARCH 22, 2007) WORK ORDER: 0700115**

Dear Mr. Smith:

We have conducted two 96-h LC50 toxicity tests using rainbow trout on the above samples, received at the North Vancouver Laboratory on March 23, 2007. The tests were performed according to the Environment Canada protocol for conducting acute toxicity tests using rainbow trout (EPS 1/RM/13, Second Edition, 2000). The results of these tests are based on the appended data and are presented in Table 1 on the following page.

Should you have any questions or comments regarding this report, please do not hesitate to contact the undersigned at 604-986-4331.

Yours very truly,
GOLDER ASSOCIATES LTD.

Robert Harrison, B.Sc. Hons.
Laboratory Biologist – Fish Team

Verified By:

QA/QC Committee:
Julianna Kalocai, M.Sc. R.P.Bio.
Barri Rudolph, M.E.T.

Attachment: Table 1
REH/JGK/pdk

N:\Data\Final\2007\1414\07-1414-0039\LET 0409_07 Golder Tox Test WO0700115.doc



TABLE 1: 96-h Toxicity Test Results

Sample Identification	Sample Collection Date (Time)	96-h LC50 (95% Confidence Limits) [% vol/vol]
2849-01 (AS06-O/A)	March 22, 2007 (0900h)	>100
2849-02 (GWS06-01)	March 22, 2007 (1100h)	>100

Toxicity testing was carried out in accordance with applicable test methodologies and/or standards of practice. Our liability is limited solely to the cost of re-testing in the event of non-compliance with such test specifications or standards of practice. Golder accepts no responsibility or liability for the interpretation or use of these testing results by others, nor for any delay, loss, damage or interruptions of testing, collection, preparation, and delivery of samples or test results resulting from events or circumstances beyond our control.

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA SUMMARY**

Client Golder
Lab Project No. 07-1414-0039
Lab Work Order No. 0700115

Lab Analysts JMS / RGH / AKY
Test Type 96 hr LC50
Test Initiation Date March 23, 2007

SAMPLE

Identification 2849-01
Amount Received 2x20L
Date Collected 22 March 2007
Date Received 23 March 2007
Other -

DILUTION/CONTROL WATER (initial water quality)

Fresh Water (dechlorinated) ✓
Temperature (°C) 15
pH 7.3
Dissolved Oxygen (mg/L) 10.1
Conductivity (µS/cm) 43
Hardness (mg/L as CaCO₃) 14
Alkalinity (mg/L as CaCO₃) 10
Other -

TEST SPECIES INFORMATION

Source Sun Valley
Collection Date/Batch 030707
Control Fish Size (mean, SD and range measured at end of test)
Date Measured March 27, 2007
Fork Length (mm) 36 ± 3 (32-40)
Wet Weight (g) 0.47 ± 0.11 (0.28-0.60)
Reference Toxicant SDS
Current Reference Toxicant Result
Reference Toxicant Test Date Mar 19/07
Duration of Acclimation (days) 12
96-h LC50 (and 95% CL) 24 (18 and 32) µg/L
Reference Toxicant Warning Limits (mean ± 2SD) and CV
29 ± 14 µg/L SDS CV: 24%

TEST CONDITIONS

Dissolved Oxygen Range (mg/L) 8.3 - 10.1
Temperature Range (°C) 15 pm
pH Range 6.8 - 7.98
Conductivity Range (µS/cm) 43 - 345
Aeration Provided? (give rate) 0.5 ± 1 mL/min/L
Photoperiod (L:D h) 16:8
No. Organisms/Volume 10 / 12L
Loading Density (g/L) 0.39
Acclimation Before Testing (days) 16
Mortality In Previous Week of Acclimation (%) 0
Other -

TEST RESULTS The 96 hr LC50 for sample 2849-01 is
estimated to be >100% (v/v)

Data Verified By Qalif Date Verified April 5/07

GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA

WHOLE SAMPLE WATER QUALITY

Temp. (°C)	Initial	pH Adjustment ¹	After 30-min Pre-aeration
15		/	15
pH	6.7	/	7.3 6.8
DO (mg/L)	10.0	/	10.1
Cond. (µS/cm)	337	/	336

1. Document pH adjustment procedure (if used) under "Comments".

Client Colder
Lab Project No. 07-144-0039
Lab Work Order No. 070015
Trout Batch No. and 7-d Acclimation Mortality 030701/0%
No. Fish/Volume 10/12L
Sample ID 2849-01
Date/Time Collected Mar-22-07
Test Initiation Date/Time Mar-23-07 @ 16:00

Total Pre-Aeration Time 30 min

Concentration % (v/v)	Number of Survivors (1 to 96 hours)								Dissolved Oxygen (mg/L)				Temperature (°C)				pH				Conductivity (µS/cm)
	1	2	4	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	24	96	
control				10	10	10	10	10.1	8.4	9.1	9.6	9.5	15	15	15	15	15	7.3	7.8	7.8	4380
6.25				10	10	10	10	10.1	8.3	9.2	9.6	9.2	15	15	15	15	15	7.2	7.8	7.8	7984
12.5				10	10	10	10	10.1	8.6	9.1	9.6	9.0	15	15	15	15	15	7.2	7.7	7.7	99105
25				10	10	10	10	10.1	8.6	9.4	9.6	9.0	15	15	15	15	15	7.2	7.7	7.7	12884
50				10	10	10	10	10.1	8.6	9.4	9.6	9.3	15	15	15	15	15	7.1	7.6	7.6	201207
100				10	10	10	10	10.1	8.6	9.3	9.6	9.4	15	15	15	15	15	6.8	7.6	7.6	336845
Technician Initials				mi	18	ven	ven	10.1	8.6	9.4	9.6	9.0	15	15	15	15	15	7.1	7.6	7.6	336845

WQ Instruments Used: Temperature
Sample Description: light brown, cloudy
Comments: CEP#46 100107
pH II-A-020501 DO II-A-20 Conductivity II-A-030006

Test Set Up By ven Date Verified By Opelidh Date Verified April 307

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA SUMMARY**

Client Golder
Lab Project No. 07-1414-0039
Lab Work Order No. 0700115

Lab Analysts JMS/MJG/REH
Test Type 96hr LC50
Test Initiation Date March 23, 2007

SAMPLE

Identification 2849-02
Amount Received 2 x 20L
Date Collected 22 March 2007
Date Received 23 March 2007
Other -

DILUTION/CONTROL WATER (initial water quality)

Fresh Water (dechlorinated) ✓
Temperature (°C) 15
pH 7.4
Dissolved Oxygen (mg/L) 10.3
Conductivity (µS/cm) 44
Hardness (mg/L as CaCO₃) 14
Alkalinity (mg/L as CaCO₃) 10
Other -

TEST SPECIES INFORMATION

Source Sun Valley
Collection Date/Batch 030707
Control Fish Size (mean, SD and range measured at end of test)
Date Measured March 27, 2007
Fork Length (mm) 36 ± 3.2 (31-38)
Wet Weight (g) 0.44 ± 0.08 (0.27-0.55)
Reference Toxicant SDS
Current Reference Toxicant Result
Reference Toxicant Test Date Mar 19/07
Duration of Acclimation (days) 12
96-h LC50 (and 95% CL) 24 (18 and 32) µg/L
Reference Toxicant Warning Limits (mean ± 2SD) and CV
29 ± 14 µg/L SDS CV: 24%

TEST CONDITIONS

Dissolved Oxygen Range (mg/L) 7.4-10.3
Temperature Range (°C) 15
pH Range 6.5-7.8
Conductivity Range (µS/cm) 44-203
Aeration Provided? (give rate) 0.5 L/min/L
Photoperiod (L:D h) 16:8
No. Organisms/Volume 10/12L
Loading Density (g/L) 0.37
Acclimation Before Testing (days) 16
Mortality In Previous Week of Acclimation (%) 0
Other -

TEST RESULTS The 96hr LC50 for sample 2849-02 is estimated to be >100% (V/V)

Data Verified By Galt

Date Verified April 3/07

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA**

WHOLE SAMPLE WATER QUALITY

Temp. (°C)	Initial	pH Adjustment ¹	After 30-min Pre-aeration
15			15
pH	6.2		6.2
DO (mg/L)	5.0		6.0
Conc. (µS/cm)	202		205

¹ Document pH adjustment procedure (if used) under "Comments".

Client Golder

Lab Project No. 07-1414-0039
 Lab Work Order No. 0700115
 Trout Batch No. and 7-d Acclimation Mortality 030707/07
 No. Fish/Volume 10/12L
 Sample ID 2849-02
 Date/Time Collected Mar 22/07
 Test Initiation Date/Time Mar 23/07 01:35

Total Pre-Aeration Time 90 min

Concentration % (v/v)	Number of Survivors (1 to 96 hours)							Dissolved Oxygen (mg/L)							Temperature (°C)							pH							Conductivity (µS/cm)		
	1	2	4	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	96		
control				10	10	10	10	10.3	9.2	9.3	9.4	9.1	15.0	15	15	15	15	7.4	7.8	7.6	7.6	7.5	44	49							
6.25				10	10	10	10	10.1	9.2	9.1	9.4	9.3	15.0	15	15	15	15	7.2	7.7	7.7	7.6	7.6	62	65							
12.5				10	10	10	10	9.8	9.2	9.3	9.3	9.4	15	15	15	15	15	7.1	7.6	7.7	7.6	7.6	69	72							
25				10	10	10	10	9.5	9.4	9.4	9.4	9.7	15	15	15	15	15	6.9	7.6	7.7	7.7	7.7	89	91							
50				10	10	10	10	9.2	9.0	9.4	9.4	9.3	15	15	15	15	15	6.7	7.7	7.7	7.7	7.8	130	128							
100				00	10	10	10	7.4	9.5	9.5	9.4	9.5	15	15	15	15	15	6.5	7.8	7.8	7.8	7.8	203	190							
Technician Initials				ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms	ms

WQ Instruments Used: CE-2 #14 Temperature 10.9/07
 Sample Description dark brown
 Comments

Conductivity II-A-030306

DO II-A-11C

pH II-A-020501

Test Set Up By Wm Date Verified By Qalif Date Verified April 3/07

**195 Pemberton Avenue
North Vancouver, B.C.
Canada V7P 2R4**

**Tel: 604-986-4331
Fax: 604-662-8548
www.golder.com**

Shipping Date 22/3/07[illegible]

- 1 For composite effluent or water samples, the sample collection date/time is the end of the compositing period.
- 2 Receiving Water (RW): Effluent (E); Elutriate (ELU); Sediment (SED); Chemical (CHEM); Stormwater (SW); Other (Please Specify)
- 3 Collapsible Carboy (CC); glass jar (GJ); Jerry Can (JC); Plastic HDPE (P); Other (Please Specify)

Distribution of copies:

- White, Yellow — accompany the shipment
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Please see instructions for completion on back of form



Environmental Division

ANALYTICAL REPORT

GOLDER ASSOCIATES LTD.

ATTN: DWAYNE SMITH

2640 DOUGLAS STREET

VICTORIA BC V8T 4M1

Reported On: 04-APR-07 03:13 PM

Lab Work Order #: L489415

Date Received: 23-MAR-07

Project P.O. #:

Job Reference: 07-1414-0039-1000

Legal Site Desc:

CofC Numbers: 0687

Other Information:

Comments:

TIMOTHY GUY CROWTHER
General Manager, Vancouver

For any questions about this report please contact your Account Manager:

AMBER SPRINGER

THIS REPORT SHALL NOT BE REPRODUCED EXCEPT IN FULL WITHOUT THE WRITTEN AUTHORITY OF THE LABORATORY.
ALL SAMPLES WILL BE DISPOSED OF AFTER 30 DAYS FOLLOWING ANALYSIS. PLEASE CONTACT THE LAB IF YOU
REQUIRE ADDITIONAL SAMPLE STORAGE TIME.

ALS LABORATORY GROUP ANALYTICAL REPORT

		Sample ID	L489415-1	L489415-2	L489415-3	L489415-4	L489415-5
		Description					
		Sampled Date	22-MAR-07	22-MAR-07	22-MAR-07	22-MAR-07	22-MAR-07
		Sampled Time					
		Client ID	0687-01	0687-02	0687-03	0687-04	0687-05
Grouping	Analyte						
WATER							
Physical Tests	Hardness (as CaCO3) (mg/L)		49.3	49.7	33.3	460	63.2
Anions and Nutrients	Bromide (Br) (mg/L)		<0.050	<0.050	<0.25	5.16	<0.50
	Chloride (Cl) (mg/L)		5.21	5.18	73.8	1880	7.4
	Fluoride (F) (mg/L)		0.040	0.043	0.12	<0.20	0.24
	Sulfate (SO4) (mg/L)		1.25	1.19	9.5	268	<5.0
	Sulphide as S (mg/L)		0.025	0.025	<0.020	0.033	0.191
	Nitrate (as N) (mg/L)		<0.0050	<0.0050	0.398	<0.050	<0.050
	Nitrite (as N) (mg/L)		<0.0010	<0.0010	0.0051	0.017	0.015
Total Metals	Aluminum (Al)-Total (mg/L)		31.0	16.5	0.357	120	0.288
	Antimony (Sb)-Total (mg/L)		<0.0025	<0.0025	<0.0025	<0.010	<0.0025
	Arsenic (As)-Total (mg/L)		0.0070	<0.0050	<0.0050	0.116	<0.0050
	Barium (Ba)-Total (mg/L)		0.109	0.062	<0.020	0.173	<0.020
	Beryllium (Be)-Total (mg/L)		<0.0050	<0.0050	<0.0050	<0.010	<0.0050
	Bismuth (Bi)-Total (mg/L)		<0.20	<0.20	<0.20	<0.40	<0.20
	Boron (B)-Total (mg/L)		<0.10	<0.10	<0.10	0.84	<0.10
	Cadmium (Cd)-Total (mg/L)		0.00032	<0.00025	<0.00025	0.0036	<0.00025
	Calcium (Ca)-Total (mg/L)		22.8	18.9	5.14	58.4	14.5
	Chromium (Cr)-Total (mg/L)		0.0703	0.0385	<0.0025	0.261	<0.0025
	Cobalt (Co)-Total (mg/L)		0.0094	0.0054	<0.0025	0.082	<0.0025
	Copper (Cu)-Total (mg/L)		0.0850	0.0476	0.0065	0.702	0.0112
	Iron (Fe)-Total (mg/L)		24.2	15.5	0.721	158	11.6
	Lead (Pb)-Total (mg/L)		0.0065	<0.0050	<0.0050	0.024	<0.0050
	Lithium (Li)-Total (mg/L)		<0.050	<0.050	<0.050	0.29	<0.050
	Magnesium (Mg)-Total (mg/L)		11.2	7.86	4.97	130	6.59
	Manganese (Mn)-Total (mg/L)		0.426	0.297	0.025	1.17	0.797
	Mercury (Hg)-Total (mg/L)		<0.00020	<0.00020	<0.00020	<0.00020	<0.00020
	Molybdenum (Mo)-Total (mg/L)		<0.0050	<0.0050	<0.0050	0.117	<0.0050
	Nickel (Ni)-Total (mg/L)		0.027	<0.025	<0.025	0.14	<0.025
	Phosphorus (P)-Total (mg/L)		1.32	0.73	<0.30	10.8	0.43
	Potassium (K)-Total (mg/L)		2.1	<2.0	<2.0	56.5	7.3
	Selenium (Se)-Total (mg/L)		<0.0050	<0.0050	<0.0050	0.025	<0.0050
	Silicon (Si)-Total (mg/L)		29.3	20.3	4.20	78.7	11.0
	Silver (Ag)-Total (mg/L)		<0.00025	<0.00025	<0.00025	<0.0010	<0.00025
	Sodium (Na)-Total (mg/L)		6.3	5.2	46.0	1170	11.7
	Strontium (Sr)-Total (mg/L)		0.125	0.102	0.0408	0.863	0.0519
	Thallium (Tl)-Total (mg/L)		<0.0010	<0.0010	<0.0010	<0.0040	<0.0010
	Tin (Sn)-Total (mg/L)		<0.030	<0.030	<0.030	<0.060	<0.030
	Titanium (Ti)-Total (mg/L)		1.18	0.656	<0.050	4.01	<0.050
	Uranium (U)-Total (mg/L)		0.0016	<0.0010	<0.0010	0.0846	<0.0010

ALS LABORATORY GROUP ANALYTICAL REPORT

		Sample ID			
		Description			
		Sampled Date			
		Sampled Time			
		Client ID			
Grouping	Analyte				
WATER					
Physical Tests	Hardness (as CaCO ₃) (mg/L)	46.1	91.9	96.1	
Anions and Nutrients	Bromide (Br) (mg/L)	<0.25	<0.50	<0.50	
	Chloride (Cl) (mg/L)	10.0	6.5	6.6	
	Fluoride (F) (mg/L)	0.19	<0.20	0.36	
	Sulfate (SO ₄) (mg/L)	3.0	<5.0	10.6	
	Sulphide as S (mg/L)	0.025	0.032	0.034	
	Nitrate (as N) (mg/L)	<0.025	<0.050	<0.050	
	Nitrite (as N) (mg/L)	<0.0050	<0.010	0.010	
Total Metals	Aluminum (Al)-Total (mg/L)	115	17.2	0.901	
	Antimony (Sb)-Total (mg/L)	<0.0050	<0.0025	<0.0025	
	Arsenic (As)-Total (mg/L)	0.066	0.0081	0.0057	
	Barium (Ba)-Total (mg/L)	0.231	0.057	0.067	
	Beryllium (Be)-Total (mg/L)	<0.0050	<0.0050	<0.0050	
	Bismuth (Bi)-Total (mg/L)	<0.20	<0.20	<0.20	
	Boron (B)-Total (mg/L)	0.11	<0.10	<0.10	
	Cadmium (Cd)-Total (mg/L)	0.00424	0.00053	<0.00025	
	Calcium (Ca)-Total (mg/L)	37.8	28.3	22.4	
	Chromium (Cr)-Total (mg/L)	0.295	0.0387	0.0029	
	Cobalt (Co)-Total (mg/L)	0.0744	0.0100	0.0069	
	Copper (Cu)-Total (mg/L)	0.900	0.160	0.0098	
	Iron (Fe)-Total (mg/L)	142	35.9	13.1	
	Lead (Pb)-Total (mg/L)	0.043	0.0096	<0.0050	
	Lithium (Li)-Total (mg/L)	0.228	<0.050	<0.050	
	Magnesium (Mg)-Total (mg/L)	43.4	13.4	8.71	
	Manganese (Mn)-Total (mg/L)	1.77	1.35	1.41	
	Mercury (Hg)-Total (mg/L)	<0.00020	<0.00020	<0.00020	
	Molybdenum (Mo)-Total (mg/L)	0.081	<0.0050	<0.0050	
	Nickel (Ni)-Total (mg/L)	0.155	0.028	<0.025	
	Phosphorus (P)-Total (mg/L)	7.36	1.43	<0.30	
	Potassium (K)-Total (mg/L)	15.0	9.5	7.1	
	Selenium (Se)-Total (mg/L)	<0.010	<0.0050	<0.0050	
	Silicon (Si)-Total (mg/L)	85.8	31.7	12.0	
	Silver (Ag)-Total (mg/L)	0.00070	<0.00025	<0.00025	
	Sodium (Na)-Total (mg/L)	34.2	12.0	24.1	
	Strontium (Sr)-Total (mg/L)	0.251	0.0996	0.0788	
	Thallium (Tl)-Total (mg/L)	<0.0020	<0.0010	<0.0010	
	Tin (Sn)-Total (mg/L)	<0.030	<0.030	<0.030	
	Titanium (Ti)-Total (mg/L)	4.25	0.673	<0.050	
	Uranium (U)-Total (mg/L)	0.0527	0.0038	<0.0010	

ALS LABORATORY GROUP ANALYTICAL REPORT

		Sample ID Description Sampled Date Sampled Time Client ID	L489415-1	L489415-2	L489415-3	L489415-4	L489415-5
			22-MAR-07	22-MAR-07	22-MAR-07	22-MAR-07	22-MAR-07
			0687-01	0687-02	0687-03	0687-04	0687-05
Grouping	Analyte						
WATER							
Total Metals	Vanadium (V)-Total (mg/L)	0.130	0.073	<0.030	0.738	<0.030	
	Zinc (Zn)-Total (mg/L)	0.0564	0.0319	<0.0050	0.227	<0.0050	
Dissolved Metals	Aluminum (Al)-Dissolved (mg/L)	0.311	0.301		0.79		
	Antimony (Sb)-Dissolved (mg/L)	<0.0025	<0.0025		<0.010		
	Arsenic (As)-Dissolved (mg/L)	<0.0050	<0.0050		<0.020		
	Barium (Ba)-Dissolved (mg/L)	<0.020	<0.020		<0.040		
	Beryllium (Be)-Dissolved (mg/L)	<0.0050	<0.0050		<0.010		
	Bismuth (Bi)-Dissolved (mg/L)	<0.20	<0.20		<0.40		
	Boron (B)-Dissolved (mg/L)	<0.10	<0.10		0.77		
	Cadmium (Cd)-Dissolved (mg/L)	<0.00025	<0.00025		<0.0010		
	Calcium (Ca)-Dissolved (mg/L)	13.5	13.7		32.5		
	Chromium (Cr)-Dissolved (mg/L)	<0.0025	<0.0025		<0.010		
	Cobalt (Co)-Dissolved (mg/L)	<0.0025	<0.0025		<0.010		
	Copper (Cu)-Dissolved (mg/L)	<0.0050	<0.0050		<0.020		
	Iron (Fe)-Dissolved (mg/L)	5.33	5.30		2.44		
	Lead (Pb)-Dissolved (mg/L)	<0.0050	<0.0050		<0.020		
	Lithium (Li)-Dissolved (mg/L)	<0.050	<0.050		<0.10		
	Magnesium (Mg)-Dissolved (mg/L)	3.76	3.79		92.1		
	Manganese (Mn)-Dissolved (mg/L)	0.169	0.166		0.193		
	Mercury (Hg)-Dissolved (mg/L)	<0.00020	<0.00020		<0.00020		
	Molybdenum (Mo)-Dissolved (mg/L)	<0.0050	<0.0050		<0.020		
	Nickel (Ni)-Dissolved (mg/L)	<0.025	<0.025		<0.10		
	Phosphorus (P)-Dissolved (mg/L)	<0.30	<0.30		<0.60		
	Potassium (K)-Dissolved (mg/L)	<2.0	<2.0		48.4		
	Selenium (Se)-Dissolved (mg/L)	<0.0050	<0.0050		<0.020		
	Silicon (Si)-Dissolved (mg/L)	6.74	6.86		7.99		
	Silver (Ag)-Dissolved (mg/L)	<0.00025	<0.00025		<0.0010		
	Sodium (Na)-Dissolved (mg/L)	4.2	4.2		1170		
	Strontium (Sr)-Dissolved (mg/L)	0.0684	0.0692		0.528		
	Thallium (Tl)-Dissolved (mg/L)	<0.0010	<0.0010		<0.0040		
	Tin (Sn)-Dissolved (mg/L)	<0.030	<0.030		<0.060		
	Titanium (Ti)-Dissolved (mg/L)	<0.050	<0.050		<0.10		
	Uranium (U)-Dissolved (mg/L)	<0.0010	<0.0010		<0.0040		
	Vanadium (V)-Dissolved (mg/L)	<0.030	<0.030		<0.060		
	Zinc (Zn)-Dissolved (mg/L)	<0.0050	<0.0050		<0.010		
Speciated Metals	Chromium (III)-Total (mg/L)	<0.080	<0.040	<0.010	<0.27	<0.010	
	Chromium (VI)-Total (mg/L)	<0.50	<0.50	<1.0	<1.0	<1.0	
Organic Parameters	COD (mg/L)	170	330	30	220	90	
	Phenols (4AAP) (mg/L)	<0.001	<0.001	<0.001	<0.001	<0.001	

ALS LABORATORY GROUP ANALYTICAL REPORT

		Sample ID Description Sampled Date Sampled Time Client ID	L489415-6 22-MAR-07 0687-06	L489415-7 22-MAR-07 0687-07	L489415-8 22-MAR-07 0687-08		
Grouping	Analyte						
WATER							
Total Metals	Vanadium (V)-Total (mg/L)	0.551	0.082	<0.030			
	Zinc (Zn)-Total (mg/L)	0.295	0.0532	0.0226			
Dissolved Metals	Aluminum (Al)-Dissolved (mg/L)	1.19	0.474	0.735			
	Antimony (Sb)-Dissolved (mg/L)	<0.0050	<0.0025	<0.0025			
	Arsenic (As)-Dissolved (mg/L)	<0.010	<0.0050	<0.0050			
	Barium (Ba)-Dissolved (mg/L)	<0.020	<0.020	0.063			
	Beryllium (Be)-Dissolved (mg/L)	<0.0050	<0.0050	<0.0050			
	Bismuth (Bi)-Dissolved (mg/L)	<0.20	<0.20	<0.20			
	Boron (B)-Dissolved (mg/L)	0.12	<0.10	<0.10			
	Cadmium (Cd)-Dissolved (mg/L)	<0.00050	<0.00025	<0.00025			
	Calcium (Ca)-Dissolved (mg/L)	7.84	23.0	23.4			
	Chromium (Cr)-Dissolved (mg/L)	<0.0050	0.0048	0.0038			
	Cobalt (Co)-Dissolved (mg/L)	<0.0050	<0.0025	0.0103			
	Copper (Cu)-Dissolved (mg/L)	<0.010	<0.0050	0.0065			
	Iron (Fe)-Dissolved (mg/L)	11.6	19.3	12.4			
	Lead (Pb)-Dissolved (mg/L)	<0.010	<0.0050	<0.0050			
	Lithium (Li)-Dissolved (mg/L)	<0.050	<0.050	<0.050			
	Magnesium (Mg)-Dissolved (mg/L)	6.43	8.37	9.15			
	Manganese (Mn)-Dissolved (mg/L)	0.398	1.15	1.46			
	Mercury (Hg)-Dissolved (mg/L)	<0.00020	<0.00020	<0.00020			
	Molybdenum (Mo)-Dissolved (mg/L)	<0.010	<0.0050	<0.0050			
	Nickel (Ni)-Dissolved (mg/L)	<0.050	<0.025	<0.025			
	Phosphorus (P)-Dissolved (mg/L)	<0.30	0.54	<0.30			
	Potassium (K)-Dissolved (mg/L)	7.6	8.9	7.6			
	Selenium (Se)-Dissolved (mg/L)	<0.010	<0.0050	<0.0050			
	Silicon (Si)-Dissolved (mg/L)	13.8	14.9	12.8			
	Silver (Ag)-Dissolved (mg/L)	<0.00050	<0.00025	<0.00025			
	Sodium (Na)-Dissolved (mg/L)	30.2	10.8	24.0			
	Strontium (Sr)-Dissolved (mg/L)	0.0454	0.0712	0.0815			
	Thallium (Tl)-Dissolved (mg/L)	<0.0020	<0.0010	<0.0010			
	Tin (Sn)-Dissolved (mg/L)	<0.030	<0.030	<0.030			
	Titanium (Ti)-Dissolved (mg/L)	<0.050	<0.050	<0.050			
	Uranium (U)-Dissolved (mg/L)	<0.0020	<0.0010	<0.0010			
	Vanadium (V)-Dissolved (mg/L)	<0.030	<0.030	<0.030			
	Zinc (Zn)-Dissolved (mg/L)	<0.0050	<0.0050	0.0166			
Speciated Metals	Chromium (III)-Total (mg/L)	<0.30	<0.040	<0.010			
	Chromium (VI)-Total (mg/L)	<1.0	<1.0	<0.50			
Organic Parameters	COD (mg/L)	500	310	150			
	Phenols (4AAP) (mg/L)	<0.001	<0.001	<0.001			

ALS LABORATORY GROUP ANALYTICAL REPORT

		Sample ID	L489415-1	L489415-2	L489415-3	L489415-4	L489415-5
		Description					
		Sampled Date	22-MAR-07	22-MAR-07	22-MAR-07	22-MAR-07	22-MAR-07
		Sampled Time					
		Client ID	0687-01	0687-02	0687-03	0687-04	0687-05
Grouping	Analyte						
WATER							
Organic Parameters	Tannin & lignin (mg/L)		1	1	2	6	9

ALS LABORATORY GROUP ANALYTICAL REPORT

		Sample ID Description Sampled Date Sampled Time Client ID	L489415-6 22-MAR-07 0687-06	L489415-7 22-MAR-07 0687-07	L489415-8 22-MAR-07 0687-08		
Grouping	Analyte						
WATER							
Organic Parameters	Tannin & lignin (mg/L)		7	9	8		

Reference Information

Methods Listed (if applicable):

ALS Test Code	Matrix	Test Description	Analytical Method Reference(Based On)
ANIONS-BR-IC-VA	Water	Bromide by Ion Chromatography	APHA 4110 "Determination of Anions by IC
This analysis is carried out using procedures adapted from APHA Method 4110 "Determination of Anions by Ion Chromatography" and EPA Method 300.0 "Determination of Inorganic Anions by Ion Chromatography". Anions routinely determined by this method include: bromide, chloride, fluoride, nitrate, nitrite and sulphate.			
ANIONS-CL-IC-VA	Water	Chloride by Ion Chromatography	APHA 4110 "Determination of Anions by IC
This analysis is carried out using procedures adapted from APHA Method 4110 "Determination of Anions by Ion Chromatography" and EPA Method 300.0 "Determination of Inorganic Anions by Ion Chromatography". Anions routinely determined by this method include: bromide, chloride, fluoride, nitrate, nitrite and sulphate.			
ANIONS-F-IC-VA	Water	Fluoride by Ion Chromatography	APHA 4110 "Determination of Anions by IC
This analysis is carried out using procedures adapted from APHA Method 4110 "Determination of Anions by Ion Chromatography" and EPA Method 300.0 "Determination of Inorganic Anions by Ion Chromatography". Anions routinely determined by this method include: bromide, chloride, fluoride, nitrate, nitrite and sulphate.			
ANIONS-NO2-IC-VA	Water	Nitrite by Ion Chromatography	APHA 4110 "Determination of Anions by IC
This analysis is carried out using procedures adapted from APHA Method 4110 "Determination of Anions by Ion Chromatography" and EPA Method 300.0 "Determination of Inorganic Anions by Ion Chromatography". Anions routinely determined by this method include: bromide, chloride, fluoride, nitrate, nitrite and sulphate.			
ANIONS-NO3-IC-VA	Water	Nitrate by Ion Chromatography	APHA 4110 "Determination of Anions by IC
This analysis is carried out using procedures adapted from APHA Method 4110 "Determination of Anions by Ion Chromatography" and EPA Method 300.0 "Determination of Inorganic Anions by Ion Chromatography". Anions routinely determined by this method include: bromide, chloride, fluoride, nitrate, nitrite and sulphate.			
ANIONS-SO4-IC-VA	Water	Sulfate by Ion Chromatography	APHA 4110 "Determination of Anions by IC
This analysis is carried out using procedures adapted from APHA Method 4110 "Determination of Anions by Ion Chromatography" and EPA Method 300.0 "Determination of Inorganic Anions by Ion Chromatography". Anions routinely determined by this method include: bromide, chloride, fluoride, nitrate, nitrite and sulphate.			
COD-COL-VA	Water	Chemical Oxygen Demand by Colorimetric	APHA 5220 "Chemical Oxygen Demand"
This analysis is carried out using procedures adapted from APHA Method 5220 "Chemical Oxygen Demand (COD)". Chemical oxygen demand is determined using the closed reflux colourimetric method.			
CR-CR3-TOT-VA	Water	Total Trivalent Chromium in Water	EPA SW-846 7196A
This analysis is carried out using procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846, Method 7196A, published by the United States Environmental Protection Agency (EPA). The procedure involves analysis for chromium (VI) by colourimetry using diphenylcarbazide in a sulphuric acid solution. Total Chromium (III) is determined by subtraction of chromium (VI) from total chromium.			
CR-CR6-TOT-VA	Water	Total Hexavalent Chromium in Water	EPA SW-846 7196A
This analysis is carried out using procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846, Method 7196A, published by the United States Environmental Protection Agency (EPA). The procedure involves analysis for chromium (VI) by colourimetry using diphenylcarbazide in a sulphuric acid solution. Chromium (III) is calculated as the difference between the total chromium and the chromium (VI) results.			
HARDNESS-CALC-VA	Water	Hardness	APHA 2340B
Hardness is calculated from Calcium and Magnesium concentrations, and is expressed as calcium carbonate equivalents.			
HG-DIS-CSR-CVAFS-VA	Water	Diss. Mercury in Water by CVAFS (CSR)	EPA 3005A/245.7
This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedures may involve preliminary sample treatment by filtration (EPA Method 3005A) and involves a cold-oxidation of the acidified sample using bromine monochloride prior to reduction of the sample with stannous chloride. Instrumental analysis is by cold vapour atomic fluorescence spectrophotometry (EPA Method 245.7).			
HG-TOT-CSR-CVAFS-VA	Water	Total Mercury in Water by CVAFS (CSR)	EPA 245.7
This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedure involves a cold-oxidation of the acidified sample using bromine monochloride prior to reduction of the sample with stannous chloride. Instrumental analysis is by cold vapour atomic fluorescence spectrophotometry (EPA Method 245.7).			

Reference Information

Methods Listed (if applicable):

ALS Test Code	Matrix	Test Description	Analytical Method Reference(Based On)
MET-DIS-CSR-ICP-VA	Water	Diss. Metals in Water by ICPOES (CSR)	EPA SW-846 3005A/6010B
This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedures may involve preliminary sample treatment by acid digestion, using either hotblock or microwave oven, or filtration (EPA Method 3005A). Instrumental analysis is by inductively coupled plasma - optical emission spectrophotometry (EPA Method 6010B).			
MET-DIS-CSR-MS-VA	Water	Diss. Metals in Water by ICPMS (CSR)	EPA SW-846 3005A/6020A
This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedures may involve preliminary sample treatment by acid digestion, using either hotblock or microwave oven, or filtration (EPA Method 3005A). Instrumental analysis is by inductively coupled plasma - mass spectrometry (EPA Method 6020A).			
MET-DIS-ICP-VA	Water	Dissolved Metals in Water by ICPOES	EPA SW-846 3005A/6010B
This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedure involves filtration (EPA Method 3005A) and analysis by inductively coupled plasma - optical emission spectrophotometry (EPA Method 6010B).			
MET-TOT-CSR-ICP-VA	Water	Total Metals in Water by ICPOES (CSR)	EPA SW-846 3005A/6010B
This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedures may involve preliminary sample treatment by acid digestion, using either hotblock or microwave oven, or filtration (EPA Method 3005A). Instrumental analysis is by inductively coupled plasma - optical emission spectrophotometry (EPA Method 6010B).			
MET-TOT-CSR-MS-VA	Water	Total Metals in Water by ICPMS (CSR)	EPA SW-846 3005A/6020A
This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedures may involve preliminary sample treatment by acid digestion, using either hotblock or microwave oven, or filtration (EPA Method 3005A). Instrumental analysis is by inductively coupled plasma - mass spectrometry (EPA Method 6020A).			
MET-TOT-ICP-VA	Water	Total Metals in Water by ICPOES	EPA SW-846 3005A/6010B
This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedures may involve preliminary sample treatment by acid digestion, using either hotblock or microwave oven (EPA Method 3005A). Instrumental analysis is by inductively coupled plasma - optical emission spectrophotometry (EPA Method 6010B).			
PHENOLS-4AAP-WT	Water	Phenols (4AAP)	APHA 5530
S2-T-COL-VA	Water	Total Sulphide by Colorimetric	APHA 4500-S2 "Sulphide"
This analysis is carried out using procedures adapted from APHA Method 4500-S2 "Sulphide". Sulphide is determined using the methylene blue colourimetric method.			
T+L-COL-VA	Water	Tannin & Lignins by Extraction & Colour	APHA 5550 'TANNIN & LIGNIN'
This analysis is carried out using procedures adapted from APHA Method 5550 "Tannin & Lignin".			

** Laboratory Methods employed follow in-house procedures, which are generally based on nationally or internationally accepted methodologies. The last two letters of the above ALS Test Code column indicate the laboratory that performed analytical analysis for that test. Refer to the list below:

Laboratory Definition Code	Laboratory Location	Laboratory Definition Code	Laboratory Location
VA	ALS LABORATORY GROUP - VANCOUVER, BC, CANADA	WT	ALS LABORATORY GROUP - WATERLOO (SENTINEL), ONTARIO, CAN

Reference Information

Methods Listed (If applicable):

ALS Test Code	Matrix	Test Description	Analytical Method Reference(Based On)
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GLOSSARY OF REPORT TERMS

Surr - A surrogate is an organic compound that is similar to the target analyte(s) in chemical composition and behavior but not normally detected in environmental samples. Prior to sample processing, samples are fortified with one or more surrogate compounds.

The reported surrogate recovery value provides a measure of method efficiency.

mg/kg (units) - unit of concentration based on mass, parts per million

mg/L (units) - unit of concentration based on volume, parts per million

N/A - Result not available. Refer to qualifier code and definition for explanation

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Although test results are generated under strict QA/QC protocols, any unsigned test reports, faxes, or emails are considered preliminary.

ALS Laboratory Group has an extensive QA/QC program where all analytical data reported is analyzed using approved referenced procedures followed by checks and reviews by senior managers and quality assurance personnel. However, since the results are obtained from chemical measurements and thus cannot be guaranteed, ALS Laboratory Group assumes no liability for the use or interpretation of the results.



Environmental Division

ALS Laboratory Group Quality Control Report

Workorder: L489415

Report Date: 04-APR-07

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Client: GOLDER ASSOCIATES LTD.
2640 DOUGLAS STREET
VICTORIA BC V8T 4M1

Contact: DWAYNE SMITH

Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
ANIONS-BR-IC-VA		Water						
Batch	R506831							
WG575746-2	CRM	VA-ALLT-170088						
Bromide (Br)			98		%		90-110	27-MAR-07
WG575746-8	CRM	VA-ALLT-170088						
Bromide (Br)			100		%		90-110	27-MAR-07
WG575746-1	MB							
Bromide (Br)			<0.050		mg/L		0.05	27-MAR-07
WG575746-7	MB							
Bromide (Br)			<0.050		mg/L		0.05	27-MAR-07
ANIONS-CL-IC-VA		Water						
Batch	R506831							
WG575746-2	CRM	VA-ALLT-170088						
Chloride (Cl)			104		%		94-106	27-MAR-07
WG575746-8	CRM	VA-ALLT-170088						
Chloride (Cl)			101		%		94-106	27-MAR-07
WG575746-1	MB							
Chloride (Cl)			<0.50		mg/L		0.5	27-MAR-07
WG575746-7	MB							
Chloride (Cl)			<0.50		mg/L		0.5	27-MAR-07
ANIONS-F-IC-VA		Water						
Batch	R506831							
WG575746-2	CRM	VA-ALLT-170088						
Fluoride (F)			98		%		93-107	27-MAR-07
WG575746-8	CRM	VA-ALLT-170088						
Fluoride (F)			100		%		93-107	27-MAR-07
WG575746-1	MB							
Fluoride (F)			<0.020		mg/L		0.02	27-MAR-07
WG575746-7	MB							
Fluoride (F)			<0.020		mg/L		0.02	27-MAR-07
ANIONS-NO2-IC-VA		Water						
Batch	R506831							
WG575746-2	CRM	VA-ALLT-170088						
Nitrite (as N)			104		%		91-109	27-MAR-07
WG575746-8	CRM	VA-ALLT-170088						
Nitrite (as N)			102		%		91-109	27-MAR-07
WG575746-1	MB							
Nitrite (as N)			<0.0010		mg/L		0.001	27-MAR-07
WG575746-7	MB							

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Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
ANIONS-NO2-IC-VA								
Batch	R506831							
WG575746-7	MB							
Nitrite (as N)			<0.0010		mg/L		0.001	27-MAR-07
ANIONS-NO3-IC-VA								
Batch	R506831							
WG575746-2	CRM	VA-ALLT-170088						
Nitrate (as N)			103		%		91-109	27-MAR-07
WG575746-8	CRM	VA-ALLT-170088						
Nitrate (as N)			102		%		91-109	27-MAR-07
WG575746-1	MB							
Nitrate (as N)			<0.0050		mg/L		0.005	27-MAR-07
WG575746-7	MB							
Nitrate (as N)			<0.0050		mg/L		0.005	27-MAR-07
ANIONS-SO4-IC-VA								
Batch	R506831							
WG575746-2	CRM	VA-ALLT-170088						
Sulfate (SO4)			104		%		93-107	27-MAR-07
WG575746-8	CRM	VA-ALLT-170088						
Sulfate (SO4)			103		%		93-107	27-MAR-07
WG575746-1	MB							
Sulfate (SO4)			<0.50		mg/L		0.5	27-MAR-07
WG575746-7	MB							
Sulfate (SO4)			<0.50		mg/L		0.5	27-MAR-07
COD-COL-VA								
Batch	R508388							
WG577986-2	DUP	L489415-1						
COD		170	170	J	mg/L	0	80	30-MAR-07
WG577986-1	MB							
COD			<20		mg/L		20	30-MAR-07
CR-CR3-TOT-VA								
Batch	R509319							
CR-CR6-TOT-VA								
	Water							

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Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
MET-DIS-CSR-ICP-VA Water								
Batch R506783								
WG575331-1 MB								
Beryllium (Be)-Dissolved			<0.0050		mg/L		0.005	27-MAR-07
Boron (B)-Dissolved			<0.10		mg/L		0.1	27-MAR-07
Calcium (Ca)-Dissolved			<0.10		mg/L		0.1	27-MAR-07
Iron (Fe)-Dissolved			<0.030		mg/L		0.03	27-MAR-07
Lithium (Li)-Dissolved			<0.050		mg/L		0.05	27-MAR-07
Magnesium (Mg)-Dissolved			<0.10		mg/L		0.1	27-MAR-07
Manganese (Mn)-Dissolved			<0.010		mg/L		0.01	27-MAR-07
Sodium (Na)-Dissolved			<2.0		mg/L		2	27-MAR-07
Titanium (Ti)-Dissolved			<0.050		mg/L		0.05	27-MAR-07
Vanadium (V)-Dissolved			<0.030		mg/L		0.03	27-MAR-07
Zinc (Zn)-Dissolved			<0.0050		mg/L		0.005	27-MAR-07
Batch R508031								
WG575331-4 DUP L489415-1								
Barium (Ba)-Dissolved		<0.020	<0.020	RPD-NA	mg/L	N/A	20	30-MAR-07
Beryllium (Be)-Dissolved		<0.0050	<0.0050	RPD-NA	mg/L	N/A	20	30-MAR-07
Boron (B)-Dissolved		<0.10	<0.10	RPD-NA	mg/L	N/A	20	30-MAR-07
Calcium (Ca)-Dissolved		13.5	13.6		mg/L	0.19	20	30-MAR-07
Iron (Fe)-Dissolved		5.33	5.37		mg/L	0.84	20	30-MAR-07
Lithium (Li)-Dissolved		<0.050	<0.050	RPD-NA	mg/L	N/A	20	30-MAR-07
Magnesium (Mg)-Dissolved		3.76	3.78		mg/L	0.38	20	30-MAR-07
Manganese (Mn)-Dissolved		0.169	0.169		mg/L	0.13	20	30-MAR-07
Sodium (Na)-Dissolved		4.2	4.2	J	mg/L	0.0	8	30-MAR-07
Titanium (Ti)-Dissolved		<0.050	<0.050	RPD-NA	mg/L	N/A	20	30-MAR-07
Vanadium (V)-Dissolved		<0.030	<0.030	RPD-NA	mg/L	N/A	20	30-MAR-07
Zinc (Zn)-Dissolved		<0.0050	<0.0050	RPD-NA	mg/L	N/A	20	30-MAR-07
MET-DIS-CSR-MS-VA Water								
Batch R506721								
WG575331-1 MB								
Aluminum (Al)-Dissolved			<0.010		mg/L		0.01	27-MAR-07
Antimony (Sb)-Dissolved			<0.00050		mg/L		0.0005	27-MAR-07
Arsenic (As)-Dissolved			<0.0010		mg/L		0.001	27-MAR-07
Cadmium (Cd)-Dissolved			<0.000050		mg/L		0.00005	27-MAR-07
Cobalt (Co)-Dissolved			<0.00050		mg/L		0.0005	27-MAR-07

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Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
MET-DIS-CSR-MS-VA								
Batch R506721								
WG575331-1 MB								
Copper (Cu)-Dissolved			<0.0010		mg/L		0.001	27-MAR-07
Chromium (Cr)-Dissolved			<0.00050		mg/L		0.0005	27-MAR-07
Lead (Pb)-Dissolved			<0.0010		mg/L		0.001	27-MAR-07
Molybdenum (Mo)-Dissolved			<0.0010		mg/L		0.001	27-MAR-07
Nickel (Ni)-Dissolved			<0.0050		mg/L		0.005	27-MAR-07
Selenium (Se)-Dissolved			<0.0010		mg/L		0.001	27-MAR-07
Silver (Ag)-Dissolved			<0.000050		mg/L		0.00005	27-MAR-07
Thallium (Tl)-Dissolved			<0.00020		mg/L		0.0002	27-MAR-07
Uranium (U)-Dissolved			<0.00020		mg/L		0.0002	27-MAR-07
Batch R506977								
WG575331-5 CRM								
VA-HIGH-WATRM								
Aluminum (Al)-Dissolved			111	G	%		90-110	27-MAR-07
Antimony (Sb)-Dissolved			106		%		90-110	27-MAR-07
Arsenic (As)-Dissolved			105		%		90-110	27-MAR-07
Cadmium (Cd)-Dissolved			101		%		90-110	27-MAR-07
Cobalt (Co)-Dissolved			104		%		90-110	27-MAR-07
Copper (Cu)-Dissolved			102		%		90-110	27-MAR-07
Chromium (Cr)-Dissolved			104		%		90-110	27-MAR-07
Lead (Pb)-Dissolved			104		%		90-110	27-MAR-07
Molybdenum (Mo)-Dissolved			109		%		90-110	27-MAR-07
Nickel (Ni)-Dissolved			105		%		90-110	27-MAR-07
Selenium (Se)-Dissolved			106		%		90-110	27-MAR-07
Silver (Ag)-Dissolved			107		%		90-110	27-MAR-07
Thallium (Tl)-Dissolved			98		%		85-115	27-MAR-07
Uranium (U)-Dissolved			106		%		90-110	27-MAR-07
Batch R508275								
WG576196-3 CRM								
VA-HIGH-WATRM								
Arsenic (As)-Dissolved			109		%		90-110	30-MAR-07
Cadmium (Cd)-Dissolved			106		%		90-110	30-MAR-07
Cobalt (Co)-Dissolved			110		%		90-110	30-MAR-07
Copper (Cu)-Dissolved			107		%		90-110	30-MAR-07
Lead (Pb)-Dissolved			107		%		90-110	30-MAR-07
Nickel (Ni)-Dissolved			108		%		90-110	30-MAR-07

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Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
MET-DIS-CSR-MS-VA		Water						
Batch	R508275							
WG576196-3	CRM	VA-HIGH-WATRM						
Silver (Ag)-Dissolved			108		%		90-110	30-MAR-07
Thallium (Tl)-Dissolved			106		%		85-115	30-MAR-07
Uranium (U)-Dissolved			108		%		90-110	30-MAR-07
WG575331-4	DUP	L489415-1						
Aluminum (Al)-Dissolved		0.311	0.268	J	mg/L	0.042	0.2	30-MAR-07
Antimony (Sb)-Dissolved		<0.0025	<0.0025	RPD-NA	mg/L	N/A	20	30-MAR-07
Arsenic (As)-Dissolved		<0.0050	<0.0050	RPD-NA	mg/L	N/A	20	30-MAR-07
Cadmium (Cd)-Dissolved		<0.00025	<0.00025	RPD-NA	mg/L	N/A	20	30-MAR-07
Cobalt (Co)-Dissolved		<0.0025	<0.0025	RPD-NA	mg/L	N/A	20	30-MAR-07
Copper (Cu)-Dissolved		<0.0050	<0.0050	RPD-NA	mg/L	N/A	20	30-MAR-07
Chromium (Cr)-Dissolved		<0.0025	<0.0025	RPD-NA	mg/L	N/A	20	30-MAR-07
Lead (Pb)-Dissolved		<0.0050	<0.0050	RPD-NA	mg/L	N/A	20	30-MAR-07
Molybdenum (Mo)-Dissolved		<0.0050	<0.0050	RPD-NA	mg/L	N/A	20	30-MAR-07
Nickel (Ni)-Dissolved		<0.025	<0.025	RPD-NA	mg/L	N/A	20	30-MAR-07
Selenium (Se)-Dissolved		<0.0050	<0.0050	RPD-NA	mg/L	N/A	20	30-MAR-07
Silver (Ag)-Dissolved		<0.00025	<0.00025	RPD-NA	mg/L	N/A	20	30-MAR-07
Thallium (Tl)-Dissolved		<0.0010	<0.0010	RPD-NA	mg/L	N/A	20	30-MAR-07
Uranium (U)-Dissolved		<0.0010	<0.0010	RPD-NA	mg/L	N/A	20	30-MAR-07
Batch	R508747							
WG576196-3	CRM	VA-HIGH-WATRM						
Aluminum (Al)-Dissolved			108		%		90-110	02-APR-07
Antimony (Sb)-Dissolved			110		%		90-110	02-APR-07
Chromium (Cr)-Dissolved			105		%		90-110	02-APR-07
Molybdenum (Mo)-Dissolved			108		%		90-110	02-APR-07
Selenium (Se)-Dissolved			106		%		90-110	02-APR-07
MET-DIS-ICP-VA		Water						
Batch	R506783							
WG575331-5	CRM	VA-HIGH-WATRM						
Bismuth (Bi)-Dissolved			103		%		90-110	27-MAR-07
Phosphorus (P)-Dissolved			109		%		90-110	27-MAR-07
Potassium (K)-Dissolved			103		%		85-115	27-MAR-07
Silicon (Si)-Dissolved			101		%		90-110	27-MAR-07
Strontium (Sr)-Dissolved			104		%		90-110	27-MAR-07

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Test	Matrix	Reference	Result	Qualflier	Units	RPD	Limit	Analyzed
MET-DIS-ICP-VA								
Batch R506783								
WG575331-5 CRM VA-HIGH-WATRM								
Tin (Sn)-Dissolved			92		%		90-110	27-MAR-07
WG575331-1 MB								
Bismuth (Bi)-Dissolved			<0.20		mg/L		0.2	27-MAR-07
Phosphorus (P)-Dissolved			<0.30		mg/L		0.3	27-MAR-07
Potassium (K)-Dissolved			<2.0		mg/L		2	27-MAR-07
Silicon (Si)-Dissolved			<0.050		mg/L		0.05	27-MAR-07
Strontium (Sr)-Dissolved			<0.0050		mg/L		0.005	27-MAR-07
Tin (Sn)-Dissolved			<0.030		mg/L		0.03	27-MAR-07
Batch R508031								
WG575331-4 DUP L489415-1								
Bismuth (Bi)-Dissolved		<0.20	<0.20	RPD-NA	mg/L	N/A	20	30-MAR-07
Phosphorus (P)-Dissolved		<0.30	<0.30	RPD-NA	mg/L	N/A	20	30-MAR-07
Potassium (K)-Dissolved		<2.0	<2.0	RPD-NA	mg/L	N/A	20	30-MAR-07
Silicon (Si)-Dissolved		6.74	6.76		mg/L	0.34	20	30-MAR-07
Strontium (Sr)-Dissolved		0.0684	0.0693		mg/L	1.2	20	30-MAR-07
Tin (Sn)-Dissolved		<0.030	<0.030	RPD-NA	mg/L	N/A	20	30-MAR-07
MET-TOT-CSR-ICP-VA								
Batch R507089								
WG575819-4 CRM VA-HIGH-WATRM								
Barium (Ba)-Total			101		%		90-110	28-MAR-07
Beryllium (Be)-Total			97		%		90-110	28-MAR-07
Boron (B)-Total			95		%		85-115	28-MAR-07
Calcium (Ca)-Total			90		%		85-115	28-MAR-07
Iron (Fe)-Total			97		%		90-110	28-MAR-07
Lithium (Li)-Total			94		%		90-110	28-MAR-07
Magnesium (Mg)-Total			91		%		85-115	28-MAR-07
Manganese (Mn)-Total			94		%		90-110	28-MAR-07
Sodium (Na)-Total			100		%		85-115	28-MAR-07
Titanium (Ti)-Total			103		%		90-110	28-MAR-07
Vanadium (V)-Total			98		%		90-110	28-MAR-07
Zinc (Zn)-Total			96		%		85-115	28-MAR-07
WG575819-1 MB								
Barium (Ba)-Total			<0.020		mg/L		0.02	28-MAR-07
Beryllium (Be)-Total			<0.0050				0.005	

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Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
MET-TOT-CSR-ICP-VA								
Water								
Batch R507089								
WG575819-1 MB								
Beryllium (Be)-Total			<0.0050		mg/L		0.005	28-MAR-07
Boron (B)-Total			<0.10		mg/L		0.1	28-MAR-07
Calcium (Ca)-Total			<0.10		mg/L		0.1	28-MAR-07
Iron (Fe)-Total			<0.030		mg/L		0.03	28-MAR-07
Lithium (Li)-Total			<0.050		mg/L		0.05	28-MAR-07
Magnesium (Mg)-Total			<0.10		mg/L		0.1	28-MAR-07
Manganese (Mn)-Total			<0.010		mg/L		0.01	28-MAR-07
Sodium (Na)-Total			<2.0		mg/L		2	28-MAR-07
Titanium (Ti)-Total			<0.050		mg/L		0.05	28-MAR-07
Vanadium (V)-Total			<0.030		mg/L		0.03	28-MAR-07
Zinc (Zn)-Total			<0.0050		mg/L		0.005	28-MAR-07
MET-TOT-CSR-MS-VA								
Water								
Batch R506721								
WG575819-1 MB								
Aluminum (Al)-Total			<0.010		mg/L		0.01	27-MAR-07
Antimony (Sb)-Total			<0.00050		mg/L		0.0005	27-MAR-07
Arsenic (As)-Total			<0.0010		mg/L		0.001	27-MAR-07
Cadmium (Cd)-Total			<0.000050		mg/L		0.00005	27-MAR-07
Chromium (Cr)-Total			<0.00050		mg/L		0.0005	27-MAR-07
Cobalt (Co)-Total			<0.00050		mg/L		0.0005	27-MAR-07
Copper (Cu)-Total			<0.0010		mg/L		0.001	27-MAR-07
Lead (Pb)-Total			<0.0010		mg/L		0.001	27-MAR-07
Molybdenum (Mo)-Total			<0.0010		mg/L		0.001	27-MAR-07
Nickel (Ni)-Total			<0.0050		mg/L		0.005	27-MAR-07
Selenium (Se)-Total			<0.0010		mg/L		0.001	27-MAR-07
Silver (Ag)-Total			<0.000050		mg/L		0.00005	27-MAR-07
Thallium (Tl)-Total			<0.00020		mg/L		0.0002	27-MAR-07
Uranium (U)-Total			<0.00020		mg/L		0.0002	27-MAR-07
Batch R507415								
WG575819-4 CRM								
VA-HIGH-WATRM								
Aluminum (Al)-Total			109		%		90-110	28-MAR-07
Antimony (Sb)-Total			108		%		90-110	28-MAR-07
Arsenic (As)-Total			107		%		90-110	28-MAR-07

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Test	Matrix	Reference	Result	Qualifier	Units	RPD	Lmlt	Analyzed
MET-TOT-CSR-MS-VA		Water						
Batch	R507415							
WG575819-4	CRM	VA-HIGH-WATRM						
Cadmium (Cd)-Total			103		%		90-110	28-MAR-07
Chromium (Cr)-Total			106		%		90-110	28-MAR-07
Cobalt (Co)-Total			107		%		90-110	28-MAR-07
Copper (Cu)-Total			106		%		90-110	28-MAR-07
Lead (Pb)-Total			104		%		90-110	28-MAR-07
Molybdenum (Mo)-Total			109		%		90-110	28-MAR-07
Nickel (Ni)-Total			106		%		90-110	28-MAR-07
Selenium (Se)-Total			109		%		90-110	28-MAR-07
Silver (Ag)-Total			106		%		90-110	28-MAR-07
Thallium (Tl)-Total			101		%		85-115	28-MAR-07
Uranium (U)-Total			104		%		90-110	28-MAR-07
MET-TOT-ICP-VA		Water						
Batch	R507089							
WG575819-4	CRM	VA-HIGH-WATRM						
Bismuth (Bi)-Total			99		%		90-110	28-MAR-07
Phosphorus (P)-Total			93		%		90-110	28-MAR-07
Potassium (K)-Total			94		%		85-115	28-MAR-07
Silicon (Si)-Total			96		%		90-110	28-MAR-07
Strontium (Sr)-Total			94		%		90-110	28-MAR-07
Tin (Sn)-Total			104		%		90-110	28-MAR-07
WG575819-1	MB							
Bismuth (Bi)-Total			<0.20		mg/L		0.2	28-MAR-07
Phosphorus (P)-Total			<0.30		mg/L		0.3	28-MAR-07
Potassium (K)-Total			<2.0		mg/L		2	28-MAR-07
Silicon (Si)-Total			<0.050		mg/L		0.05	28-MAR-07
Strontium (Sr)-Total			<0.0050		mg/L		0.005	28-MAR-07
Tin (Sn)-Total			<0.030		mg/L		0.03	28-MAR-07
PHENOLS-4AAP-WT		Water						
Batch	R507883							
WG577354-2	CVS							
Phenols (4AAP)			95		%		85-115	29-MAR-07
WG577354-1	MB							
Phenols (4AAP)			<0.001		mg/L		0.001	29-MAR-07
	Water							

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Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
S2-T-COL-VA	Water							
Batch	R507000							
WG576372-1	MB							
Sulphide as S			<0.020		mg/L		0.02	27-MAR-07
T+L-COL-VA	Water							
Batch	R507446							
WG576537-3	DUP	L489415-3						
Tannin & lignin		2	2	J	mg/L	0	4	30-MAR-07
WG576537-1	MB							
Tannin & lignin			<0.1		mg/L		0.1	29-MAR-07
WG576537-2	MB							
Tannin & lignin			<0.1		mg/L		0.1	30-MAR-07

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Legend:

Limit	99% Confidence Interval (Laboratory Control Limits)
DUP	Duplicate
RPD	Relative Percent Difference
N/A	Not Available
LCS	Laboratory Control Sample
SRM	Standard Reference Material
MS	Matrix Spike
MSD	Matrix Spike Duplicate
ADE	Average Desorption Efficiency
MB	Method Blank
IRM	Internal Reference Material
CRM	Certified Reference Material
CCV	Continuing Calibration Verification
CVS	Calibration Verification Standard
LCSD	Laboratory Control Sample Duplicate

Qualifier:

RPD-NA	Relative Percent Difference Not Available due to result(s) being less than detection limit.
A	Method blank exceeds acceptance limit. Blank correction not applied, unless the qualifier "RAMB" (result adjusted for method blank) appears in the Analytical Report.
B	Method blank result exceeds acceptance limit, however, it is less than 5% of sample concentration. Blank correction not applied.
E	Matrix spike recovery may fall outside the acceptance limits due to high sample background.
F	Silver recovery low, likely due to elevated chloride levels in sample.
G	Outlier - No assignable cause for nonconformity has been determined.
J	Duplicate results and limit(s) are expressed in terms of absolute difference.
K	The sample referenced above is of a non-standard matrix type; standard QC acceptance criteria may not be achievable.
L	Low matrix spike recovery due to instability of spiked analyte in the sample matrix.



**Golder
Associates**

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CHAIN OF CUSTODY RECORD/ANALYSIS REQUEST

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Project Number: 02-16114-2039-1000		Laboratory Name: ALS ENVIRONMENTAL	
Golder Contact: DANN SMITH		Address: 1988 TRUMPH ST. VAN, BC	
Golder E-mail Address: DSMITH		Telephone/Fax: 180065 0242	
@golder.com		Contact: MARIKA REATHIE	

Analyses Required

Sample Control Number (SCN)	Sample Matrix (over)	Date Sampled (D / M / Y)	Number of Containers	COD	TANNIN BLIGNIN	PHENOLICS	ANIONIC	SULPHIDE	TOTAL METALS	DISSOLVED METAL	HARDNESS	SPECIATED CHROMIUM	RUSH	Remarks (over)
0682		22/3/07	6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	contamin
-01			6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	contamin
-02			6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	contamin
-03			6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	contamin
-04			6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	contamin
-05			6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	contamin
-06			6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	contamin
-07			6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	contamin
-08			6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	contamin
-09														
-10														
-11														
-12														

Sampler's Signature: [Signature]	Relinquished by: Signature [Signature]	Company: [Signature]	Date: 22/3/07	Time:	Received by: Signature [Signature]	Company:
Sample Storage (C): 15	Relinquished by: Signature [Signature]	Company:	Date:	Time:	Received by: Signature [Signature]	Company:
Comments:	Method of Shipment: [Signature]	Waybill No.: 1-2410082889	Received for Lab by: [Signature]	Date: 07/05/23	Time: 09:08	
	Shipped by: [Signature]	Shipment Condition: Seal Intact:	Temp (°C): 2	Cooler opened by:	Date:	Time:

WHITE: Golder Copy YELLOW: Lab Copy PINK: Lab Returns with Final Report

Golder Associates Ltd.

500 - 4260 Still Creek Drive
Burnaby, British Columbia, Canada V5C 6C6
Telephone: 604-296-4200
Fax: 604-298-5253



October 27, 2006

E/06/1182
06-1414-092

Golder Associates
195 Pemberton Avenue
North Vancouver, BC V7P 2R4

Attention: Mr. Dwayne Smith

**RE: TOXICITY TESTING ON THE SAMPLES IDENTIFIED AS GWS06-1
(2535-01) & AS06-1(2535-02) (COLLECTED ON SEPTEMBER 27, 2006)
WORK ORDER: 0600437**

Dear Mr. Smith:

We have conducted two 96-h LC50 toxicity tests using rainbow trout on the above samples, received at Golder's North Vancouver Laboratory on September 29, 2006. The tests were performed according to the Environment Canada protocol for conducting acute toxicity tests using rainbow trout (EPS 1/RM/13, Second Edition, 2000). The results of these tests are based on the appended data and are presented in Table 1.

If you wish to schedule additional testing or have any questions regarding the data presented in this report, please do not hesitate to contact me by e-mail (rharrison@golder.com) or by telephone at 604-986-4331.

Yours very truly,
GOLDER ASSOCIATES LTD.

Robert Harrison, B.Sc. Hons.
Laboratory Biologist – Fish Team

Verified By:

QA/QC Committee:
Julianna Kalocai, M.Sc.
Barri Rudolph, B.Sc.

Attachment: Table 1
REH/JGK/pdk

O:\Data\Final\2006\1414\06-1414-092\LET 1027_06 Golder Tox Test W0600437.doc



TABLE 1: 96-h LC50 Toxicity Test Results

Sample Identification	Sample Collection Date (Time)	96-h LC50 (95% Confidence Limits) [% vol/vol]
GWS06-1 (2535-01)	September 27, 2006 (1100h)	>100
AS06-1 (2535-02)	September 27, 2006 (1000h)	100

Toxicity testing was carried out in accordance with applicable test methodologies and/or standards of practice. Our liability is limited solely to the cost of re-testing in the event of non-compliance with such test specifications or standards of practice. Golder accepts no responsibility or liability for the interpretation or use of these testing results by others, nor for any delay, loss, damage or interruptions of testing, collection, preparation, and delivery of samples or test results resulting from events or circumstances beyond our control.

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA SUMMARY**

Client Golder
Lab Project No. 06-1414-092
Lab Work Order No. 0600437

Lab Analysts Deft
Test Type 96-h LC50
Test Initiation Date Oct 2/06 @ 1645

SAMPLE

Identification GW506-1 (2535-011)
Amount Received 2420L
Date Collected Sept 27/06
Date Received Sept 29/06
Other —

DILUTION/CONTROL WATER (initial water quality)

Fresh Water (dechlorinated) ✓
Temperature (°C) 14
pH 6.9
Dissolved Oxygen (mg/L) 10.3
Conductivity (µS/cm) 43
Hardness (mg/L as CaCO₃) 12
Alkalinity (mg/L as CaCO₃) 10
Other —

TEST SPECIES INFORMATION

Source San Valley
Collection Date/Batch 090606
Control Fish Size (mean, SD and range measured at end of test)
Date Measured Oct 6/06
Fork Length (mm) 39±2 (36 and 44)
Wet Weight (g) 0.56±0.07 (0.45 and 0.66)
Reference Toxicant SDS

Current Reference Toxicant Result

Reference Toxicant Test Date Sept 25/06
Duration of Acclimation (days) 19
96-h LC50 (and 95% CL) 19 (15 and 22) mg/L
Reference Toxicant Warning Limits (mean ± 2SD) and CV
28±13 mg/L SDS CV: 22%

TEST CONDITIONS

Dissolved Oxygen Range (mg/L) 6.7-10.3
Temperature Range (°C) 14
pH Range 5.9-8.2
Conductivity Range (µS/cm) 43-400
Aeration Provided? (give rate) 0.5 L/min/L
Photoperiod (L:D h) 16:8
No. Organisms/Volume 10/12L
Loading Density (g/L) 0.47
Acclimation Before Testing (days) 26
Mortality In Previous Week of Acclimation (%) 1.5
Other —

TEST RESULTS

The 96-h LC50 is estimated to be > 100% (0/10)

Data Verified By

Galpith

Date Verified

Oct 25/06

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA**

WHOLE SAMPLE WATER QUALITY

Temp. (°C)	Initial	pH Adjustment ¹	After 30-min Pre-aeration
	14	/	14
pH	5.6	/	5.8
DO (mg/L)	3.0	/	4.0
Cond. (µS/cm)	401	/	400

¹ Document pH adjustment procedure (if used) under "Comments".

Client Golder
 Lab Project No. 06-1414-092
 Lab Work Order No. 0600437
 Trout Batch No. and 7-d Acclimation Mortality 0910606/156
 No. Fish/Volume 10/12L
 Sample ID GW506-1 (2535-01)
 Date/Time Collected Sept 27/06 @ 1100
 Test Initiation Date/Time Oct 2/06 @ 1645

Total Pre-Aeration Time 120 min

Concentration % (v/v)	Number of Survivors (1 to 96 hours)							Dissolved Oxygen (mg/L)					Temperature (°C)					pH					Conductivity (µS/cm)	
	1	2	4	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	96
control				10	10	10	10	10.3	9.5	9.6	9.5	9.6	14	14	14	14	14	6.9	6.7	6.7	6.6	6.8	43	50
6.25				10	9	9	9	9.6	9.3	9.7	9.5	9.4	14	14	14	14	14	6.8	6.9	7.0	6.9	7.3	89	92
12.5				10	10	10	10	9.2	9.3	9.7	9.5	9.5	14	14	14	14	14	6.2	7.1	7.2	7.1	7.7	173	179
25				9	9	8	8	8.8	9.2	9.5	9.5	9.7	14	14	14	14	14	6.1	7.2	7.4	7.3	7.9	166	170
50				10	9	9	9	8.3	9.3	9.4	9.5	9.8	14	14	14	14	14	6.0	7.3	7.5	7.5	8.1	156	160
100				9	6	6	6	6.7	9.2	9.4	9.6	9.8	14	14	14	14	14	5.9	7.4	7.5	7.5	8.2	400	392
Technician Initials				901A	Deed	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A	999A

WQ Instruments Used: Temperature calibrated by Freymeyer DO IA-A-20 Conductivity IA-A-030306
 Sample Description cloudy dark brown/black
 Comments

Test Set Up By RUA Date Verified By Gatth Date Verified Oct -15/06

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA SUMMARY**

Client Golder
Lab Project No. 06-1414-092
Lab Work Order No. 0600437

Lab Analysts Rut
Test Type 96-h LC50
Test Initiation Date Oct 2/06 @ 1530

SAMPLE

Identification AS06-1 (2535-02)
Amount Received 2x20L
Date Collected Sept 27/06
Date Received Sept 29/06
Other _____

DILUTION/CONTROL WATER (initial water quality)

Fresh Water (dechlorinated) ☒
Temperature (°C) 14
pH 6.9
Dissolved Oxygen (mg/L) 10.2
Conductivity (µS/cm) 44
Hardness (mg/L as CaCO₃) 12
Alkalinity (mg/L as CaCO₃) 10
Other _____

TEST SPECIES INFORMATION

Source San Valley
Collection Date/Batch 090606
Control Fish Size (mean, SD and range measured at end of test)
Date Measured Oct 6/06
Fork Length (mm) 39±2 (35 and 42)
Wet Weight (g) 0.56±0.11 (0.37 and 0.71)
Reference Toxicant SDS

Current Reference Toxicant Result

Reference Toxicant Test Date Sept 25/06
Duration of Acclimation (days) 19
96-h LC50 (and 95% CL) 19 (15 and 22) mg/L
Reference Toxicant Warning Limits (mean ± 2SD) and CV
28±13 mg/L SDS CV: 22%

TEST CONDITIONS

Dissolved Oxygen Range (mg/L) 8.6 - 10.2
Temperature Range (°C) 14
pH Range 6.6 - 7.7
Conductivity Range (µS/cm) 44 - 24700
Aeration Provided? (give rate) 0.5 L/min/L
Photoperiod (L:D h) 16:8
No. Organisms/Volume 10/12L
Loading Density (g/L) 0.47
Acclimation Before Testing (days) 26
Mortality In Previous Week of Acclimation (%) 1.5
Other _____

TEST RESULTS

The 96-h LC50 is estimated to be 100% (0/10)

Data Verified By

Qualifit

Date Verified

Oct. 26/06

**GOLDER ASSOCIATES-NORTH VANCOUVER LABORATORY
RAINBOW TROUT ACUTE TOXICITY TEST DATA**

WHOLE SAMPLE WATER QUALITY

Temp. (°C)	pH	After 30-min Pre-aeration
14		14
7.0		7.1
8.3		8.6
Cond. (µS/cm)	24800	24700

1. Document pH adjustment procedure (if used) under "Comments".

Client AlS Golder
 Lab Project No. W884-474-038 06-144-092
 Lab Work Order No. 0600437
 Trout Batch No. and 7-d Acclimation Mortality 090606/1.5%
 No. Fish/Volume 10/12L
 Sample ID A506-1 (2535-02)
 Date/Time Collected Sept 27/06 @ 1000
 Test Initiation Date/Time Oct 2/06 @ 1530

Total Pre-Aeration Time 30 min

Concentration % (v/v)	Number of Survivors (1 to 96 hours)						Dissolved Oxygen (mg/L)					Temperature (°C)					pH					Conductivity (µS/cm)		
	1	2	4	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	24	48	72	96	0	96
control				10	10	10	10	10.2	9.8	9.7	9.5	9.5	14	14	14	14	14	6.9	6.7	6.6	6.6	6.7	44	51
6.25				10	10	10	9	9.7	9.8	9.6	9.6	9.6	14	14	14	14	14	7.1	7.0	7.1	7.2	6.9	36303660	
12.5				9	8	8	8	9.6	9.7	9.7	9.6	9.9	14	14	14	14	14	7.0	7.2	7.3	7.3	7.0	53305550	
25				9	7	6	5	9.3	9.6	9.7	9.5	9.7	14	14	14	14	14	7.1	7.3	7.3	7.5	7.3	86308780	
50				7	6	5	5	8.8	9.5	9.6	9.4	9.7	14	14	14	14	14	7.1	7.5	7.6	7.6	7.5	192408280	
100				9	7	6	5	8.6	9.3	9.6	9.5	9.4	14	14	14	14	14	7.1	7.6	7.6	7.7	7.6	247002400	
Technician Initials																								

WQ Instruments Used: Temperature Calibrated by pH TD-A-03507 DO TD-A-20 Conductivity TD-A-030306
 Sample Description slightly cloudy, light yellow
 Comments

Test Set Up By DLA Data Verified By Gaff Date Verified Oct-25/06

No 2535



195 Pemberton Avenue
North Vancouver, B.C.
Canada V7P 2R4
Tel: 604-986-4331
Fax: 604-662-8548
www.golder.com

Shipping Date_

Client Name GOLDER Client Contact DWAYNE SMITH Ship to EVS GOLDER
Address 2640 Douglas St Phone 250-881-7372 195 Pemberton Ave
VICTORIA, BC Fax 250-881-7470 North Vancouver V7P 2R4
VST 4m1 Sampled by P. Goeller Attn. Cathy McDunnon

[illegible]

- 1 For composite effluent or water samples, the sample collection date/time is the end of the compositing period.
- 2 Receiving Water (RW); Effluent (E); Eutriate (ELU); Sediment (SED); Chemical (CHEM); Stormwater (SW); Other (Please Specify)
- 3 Collapsible Carboy (CC); glass jar (GJ); Jerry Can (JC); Plastic HDPE (P); Other (Please Specify)
- 4 Please note any conditions the lab should be aware of for safety and storage concerns

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Please see instructions for completion on back of form



CERTIFICATE OF ANALYSIS

Date: October 23, 2006

ALS File No. Z3248r

Report On: 06-1414-092-4000 Water Analysis

Report To: **Golder Associates Ltd.**
2640 Douglas Street
Victoria, BC
V8T 4M1

Attention: **Mr. Dwayne Smith**

Received: September 29, 2006

ALS ENVIRONMENTAL

per:

Mahssa Rostambake

Mahssa Rostambake, B.Sc. - Account Manager
Paula Shuto, B.Sc. - Account Manager

File No. Z3248r

REMARKS



This report, Z3248r, replaces and supercedes the original file, Z3248. Several samples have been re-analyzed for metals to achieve lower detection limits where possible. All other data remains unchanged.

Please note that the Phenols analysis was done at ALS Environmental in Waterloo, Ontario.

Please note that certain Dissolved Anions, Nutrients, and Metals detection limits have been increased for some of the samples due to the matrix interferences encountered during the analysis.

File No. Z3248r

RESULTS OF ANALYSIS - Water



Sample ID	0557-01	0557-02	0557-03	0557-04	0557-05
Sample Date	06-09-27	06-09-27	06-09-27	06-09-27	06-09-27
ALS ID	1	2	3	4	5

Physical Tests

Hardness	CaCO3	113	108	58.3	71.5	2510
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Dissolved Anions

Bromide	Br	<2.5	<2.5	<1.0	<2.5	16.0
Chloride	Cl	<25	<25	13	27	6540
Fluoride	F	1.6	<1.0	<0.40	<1.0	<0.40
Sulphate	SO4	<25	<25	<10	173	870

Nutrients

Nitrate Nitrogen	N	0.26	0.30	0.16	0.45	0.30
Nitrite Nitrogen	N	0.103	0.113	<0.020	0.067	<0.020

Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre except where noted.
 < = Less than the detection limit indicated.
 Results are expressed as milligrams per dry kilogram except where noted.

File No. Z3248r

RESULTS OF ANALYSIS - Water

Sample ID	0557-01	0557-02	0557-03	0557-04	0557-05
Sample Date	06-09-27	06-09-27	06-09-27	06-09-27	06-09-27
ALS ID	1	2	3	4	5

Total Metals

Aluminum	T-Al	77.7	84.5	48.5	19.4	67.5
Antimony	T-Sb	<0.0025	<0.0025	<0.0025	<0.0025	<0.0050
Arsenic	T-As	0.0325	0.0363	0.0279	0.0231	0.083
Barium	T-Ba	0.267	0.346	0.182	0.089	0.17
Beryllium	T-Be	<0.0050	<0.0050	<0.0050	<0.0050	<0.025
Bismuth	T-Bi	<0.20	<0.20	<0.20	<0.20	<1.0
Boron	T-B	0.11	<0.10	0.11	0.13	1.26
Cadmium	T-Cd	0.00203	0.00247	0.00205	0.00047	0.00121
Calcium	T-Ca	45.0	48.6	22.1	19.9	180
Chromium	T-Cr	0.179	0.202	0.143	0.0519	0.187
Cobalt	T-Co	0.0487	0.0570	0.0349	0.0154	0.0575
Copper	T-Cu	0.749	0.817	0.441	0.0716	0.400
Iron	T-Fe	97.4	105	67.2	20.7	111
Lead	T-Pb	0.0493	0.0555	0.0163	0.0426	0.016
Lithium	T-Li	0.069	0.075	0.095	<0.050	<0.25
Magnesium	T-Mg	33.2	35.4	24.2	5.54	516
Manganese	T-Mn	2.39	2.46	1.20	0.941	2.66
Mercury	T-Hg	<0.00020	<0.00020	<0.00020	0.00038	<0.00020
Molybdenum	T-Mo	0.0120	0.0133	0.0435	0.0051	0.087
Nickel	T-Ni	0.123	0.144	0.088	0.028	0.089
Phosphorus	T-P	4.67	5.22	2.81	0.93	7.0
Potassium	T-K	16.7	15.9	8.6	3.8	139
Selenium	T-Se	<0.0050	<0.0050	<0.0050	0.0115	<0.010
Silicon	T-Si	70.1	73.3	55.8	23.8	55.1
Silver	T-Ag	0.00046	0.00047	0.00035	<0.00025	<0.00050
Sodium	T-Na	26.7	27.2	37.4	137	3750
Strontium	T-Sr	0.224	0.244	0.156	0.117	3.03
Thallium	T-Tl	<0.0010	<0.0010	<0.0010	<0.0010	<0.0020
Tin	T-Sn	<0.030	<0.030	<0.030	<0.030	0.19
Titanium	T-Ti	2.83	3.20	1.98	0.941	2.89
Uranium	T-U	0.0201	0.0226	0.0268	0.0025	0.0690
Vanadium	T-V	0.340	0.384	0.269	0.153	0.45
Zinc	T-Zn	0.237	0.273	0.145	0.0519	0.139

Remarks regarding the analyses appear at the beginning of this report.

Results are expressed as milligrams per litre except where noted.

< = Less than the detection limit indicated.

Results are expressed as milligrams per dry kilogram except where noted.

RESULTS OF ANALYSIS - Water

Sample ID	0557-01	0557-02	0557-03	0557-04	0557-05
Sample Date	06-09-27	06-09-27	06-09-27	06-09-27	06-09-27
ALS ID	1	2	3	4	5

Dissolved Metals

Aluminum	D-Al	0.386	0.415	0.936	14.0	0.20
Antimony	D-Sb	<0.0025	<0.0025	<0.0025	<0.0025	<0.0050
Arsenic	D-As	<0.0050	<0.0050	<0.0050	0.0183	0.014
Barium	D-Ba	0.047	0.048	<0.020	0.076	<0.10
Beryllium	D-Be	<0.0050	<0.0050	<0.0050	<0.0050	<0.025
Bismuth	D-Bi	<0.20	<0.20	<0.20	<0.20	<1.0
Boron	D-B	0.10	<0.10	<0.10	0.13	1.17
Cadmium	D-Cd	<0.00025	<0.00025	<0.00025	0.00033	<0.00050
Calcium	D-Ca	23.8	23.6	10.3	20.0	165
Chromium	D-Cr	0.0030	0.0032	0.0049	0.0327	<0.0050
Cobalt	D-Co	0.0077	0.0091	0.0102	0.0184	0.0068
Copper	D-Cu	<0.0050	0.0056	<0.0050	0.0525	<0.010
Iron	D-Fe	20.5	20.9	6.38	16.2	4.96
Lead	D-Pb	<0.0050	<0.0050	<0.0050	0.0333	<0.010
Lithium	D-Li	<0.050	<0.050	<0.050	<0.050	<0.25
Magnesium	D-Mg	11.7	11.5	7.19	5.53	492
Manganese	D-Mn	1.30	1.29	0.513	0.952	1.39
Mercury	D-Hg	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020
Molybdenum	D-Mo	<0.0050	<0.0050	<0.0050	<0.0050	<0.010
Nickel	D-Ni	<0.025	<0.025	<0.025	<0.025	<0.050
Phosphorus	D-P	<0.30	0.34	<0.30	0.52	<1.5
Potassium	D-K	12.9	12.6	5.1	3.7	134
Selenium	D-Se	<0.0050	<0.0050	<0.0050	0.0105	<0.010
Silicon	D-Si	17.8	17.8	14.4	20.8	9.73
Silver	D-Ag	<0.00025	<0.00025	<0.00025	<0.00025	<0.00050
Sodium	D-Na	22.2	22.3	38.5	135	3220
Strontium	D-Sr	0.116	0.115	0.0838	0.113	2.83
Thallium	D-Tl	<0.0010	<0.0010	<0.0010	<0.0010	<0.0020
Tin	D-Sn	<0.030	<0.030	<0.030	<0.030	0.21
Titanium	D-Ti	<0.050	<0.050	<0.050	0.312	<0.25
Uranium	D-U	<0.0010	<0.0010	0.0016	0.0019	<0.0020
Vanadium	D-V	<0.030	<0.030	0.043	0.119	<0.15
Zinc	D-Zn	0.0120	0.0141	0.0131	0.0482	<0.025

Inorganic Parameters

Sulphide	S	<0.20	<0.20	<0.20	<0.20	0.046
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Remarks regarding the analyses appear at the beginning of this report.

Results are expressed as milligrams per litre except where noted.

< = Less than the detection limit indicated.

Results are expressed as milligrams per dry kilogram except where noted.

File No. Z3248r

RESULTS OF ANALYSIS - Water



Sample ID	0557-01	0557-02	0557-03	0557-04	0557-05
Sample Date	06-09-27	06-09-27	06-09-27	06-09-27	06-09-27
ALS ID	1	2	3	4	5

Organic Parameters

Chemical Oxygen Demand	COD	312	341	177	661	328
Phenols		<0.0010	<0.0010	<0.0010	0.0130	<0.0010
Tannin and Lignin		22.0	24.5	13.0	29.4	2.87

Remarks regarding the analyses appear at the beginning of this report.
Results are expressed as milligrams per litre except where noted.
< = Less than the detection limit indicated.
Results are expressed as milligrams per dry kilogram except where noted.

File No. Z3248r

RESULTS OF ANALYSIS - Water



Sample ID	0557-06	0557-07	0557-08
Sample Date	06-09-27	06-09-27	06-09-27
ALS ID	6	7	8

Physical Tests

Hardness	CaCO3	46.4	119	3320
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Dissolved Anions

Bromide	Br	<0.050	<2.5	31.0
Chloride	Cl	6.11	38	11500
Fluoride	F	0.056	<1.0	<2.0
Sulphate	SO4	<0.50	<25	1600

Nutrients

Nitrate Nitrogen	N	<0.0050	0.55	1.47
Nitrite Nitrogen	N	<0.0010	<0.050	<0.10

Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre except where noted.
 < = Less than the detection limit indicated.
 Results are expressed as milligrams per dry kilogram except where noted.

File No. Z3248r

RESULTS OF ANALYSIS - Water



Sample ID	0557-06	0557-07	0557-08
Sample Date	06-09-27	06-09-27	06-09-27
ALS ID	6	7	8

Total Metals

Aluminum	T-Al	17.1	0.251	<1.0
Antimony	T-Sb	<0.0025	<0.0025	<0.050
Arsenic	T-As	<0.0050	<0.0050	<0.10
Barium	T-Ba	0.083	<0.020	<0.10
Beryllium	T-Be	<0.0050	<0.0050	<0.025
Bismuth	T-Bi	<0.20	<0.20	<2.0
Boron	T-B	<0.10	<0.10	2.19
Cadmium	T-Cd	0.00032	<0.00010	<0.0050
Calcium	T-Ca	15.7	20.8	202
Chromium	T-Cr	0.0355	<0.0025	<0.050
Cobalt	T-Co	0.0114	<0.0025	<0.050
Copper	T-Cu	0.0580	0.0105	<0.10
Iron	T-Fe	12.0	12.7	0.80
Lead	T-Pb	<0.0050	<0.0050	<0.10
Lithium	T-Li	<0.050	<0.050	<0.25
Magnesium	T-Mg	7.26	13.9	683
Manganese	T-Mn	0.392	1.12	0.060
Mercury	T-Hg	<0.00020	<0.00020	<0.00020
Molybdenum	T-Mo	<0.0050	<0.0050	<0.10
Nickel	T-Ni	<0.025	<0.025	<0.50
Phosphorus	T-P	0.59	0.46	<1.5
Potassium	T-K	<2.0	14.6	202
Selenium	T-Se	<0.0050	<0.0050	<0.10
Silicon	T-Si	18.9	17.0	3.77
Silver	T-Ag	<0.00025	<0.00025	<0.0050
Sodium	T-Na	5.2	39.6	5090
Strontium	T-Sr	0.0883	0.108	3.62
Thallium	T-Tl	<0.0010	<0.0010	<0.020
Tin	T-Sn	<0.030	<0.030	0.29
Titanium	T-Ti	0.589	<0.050	<0.25
Uranium	T-U	<0.0010	<0.0010	<0.020
Vanadium	T-V	0.072	<0.030	<0.15
Zinc	T-Zn	0.0350	<0.0050	<0.025

Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre except where noted.
 < = Less than the detection limit indicated.
 Results are expressed as milligrams per dry kilogram except where noted.

File No. Z3248r

RESULTS OF ANALYSIS - Water

Sample ID	0557-06	0557-07	0557-08
Sample Date	06-09-27	06-09-27	06-09-27
ALS ID	6	7	8

Dissolved Metals

Aluminum	D-Al	0.346	-	-
Antimony	D-Sb	<0.0025	-	-
Arsenic	D-As	<0.0050	-	-
Barium	D-Ba	0.031	-	-
Beryllium	D-Be	<0.0050	-	-
Bismuth	D-Bi	<0.20	-	-
Boron	D-B	<0.10	-	-
Cadmium	D-Cd	<0.00025	-	-
Calcium	D-Ca	11.8	-	-
Chromium	D-Cr	<0.0025	-	-
Cobalt	D-Co	0.0063	-	-
Copper	D-Cu	0.0071	-	-
Iron	D-Fe	2.63	-	-
Lead	D-Pb	<0.0050	-	-
Lithium	D-Li	<0.050	-	-
Magnesium	D-Mg	3.69	-	-
Manganese	D-Mn	0.236	-	-
Mercury	D-Hg	<0.00020	-	-
Molybdenum	D-Mo	<0.0050	-	-
Nickel	D-Ni	<0.025	-	-
Phosphorus	D-P	<0.30	-	-
Potassium	D-K	<2.0	-	-
Selenium	D-Se	<0.0050	-	-
Silicon	D-Si	6.85	-	-
Silver	D-Ag	<0.00025	-	-
Sodium	D-Na	4.1	-	-
Strontium	D-Sr	0.0629	-	-
Thallium	D-Tl	<0.0010	-	-
Tin	D-Sn	<0.030	-	-
Titanium	D-Ti	<0.050	-	-
Uranium	D-U	<0.0010	-	-
Vanadium	D-V	<0.030	-	-
Zinc	D-Zn	0.0075	-	-
<u>Inorganic Parameters</u>				
Sulphide	S	<0.020	<0.040	<0.020

Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre except where noted.
 < = Less than the detection limit indicated.
 Results are expressed as milligrams per dry kilogram except where noted.

File No. Z3248r

RESULTS OF ANALYSIS - Water



Sample ID	0557-06	0557-07	0557-08
Sample Date	06-09-27	06-09-27	06-09-27
ALS ID	6	7	8

Organic Parameters

Chemical Oxygen Demand	COD	149	127	981
Phenols		<0.0010	<0.0010	<0.0010
Tannin and Lignin		4.05	13.3	0.94

Remarks regarding the analyses appear at the beginning of this report.
Results are expressed as milligrams per litre except where noted.
< = Less than the detection limit indicated.
Results are expressed as milligrams per dry kilogram except where noted.

File No. Z3248r

Appendix 1 - QUALITY CONTROL - Replicates



Water	0557-03	0557-03
	06-09-27	QC # 525177

Physical Tests

Hardness	CaCO3	58.3	54.8
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Dissolved Anions

Bromide	Br	<1.0	<1.0
Chloride	Cl	13	14
Fluoride	F	<0.40	<0.40
Sulphate	SO4	<10	<10

Nutrients

Nitrate Nitrogen	N	0.16	0.18
Nitrite Nitrogen	N	<0.020	<0.020

Inorganic Parameters

Sulphide	S	<0.20	<0.20
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Organic Parameters

Chemical Oxygen Demand	COD	177	183
Tannin and Lignin		13.0	13.0

Remarks regarding the analyses appear at the beginning of this report.
 Results are expressed as milligrams per litre except where noted.
 < = Less than the detection limit indicated.
 Results are expressed as milligrams per dry kilogram except where noted.

Appendix 1 - QUALITY CONTROL - Reference Materials

Water

		Result	Target	Units	DL	ALSQC#	Method
ALSEV Metals in Water - High, 4GK116							
Total Metals							
Aluminum	T-Al	2.03	2.00	mg/L	0.2	526121	a
Aluminum	T-Al	2.00	2.00	mg/L	0.2	526464	a
Aluminum	T-Al	1.97	2.00	mg/L	0.2	527616	a
Antimony	T-Sb	0.92	1.00	mg/L	0.2	526121	a
Antimony	T-Sb	0.96	1.00	mg/L	0.2	526464	a
Antimony	T-Sb	0.96	1.00	mg/L	0.2	527616	a
Arsenic	T-As	0.97	1.00	mg/L	0.2	526121	a
Arsenic	T-As	0.96	1.00	mg/L	0.2	526464	a
Arsenic	T-As	0.95	1.00	mg/L	0.2	527616	a
Barium	T-Ba	0.243	0.250	mg/L	0.01	526121	a
Barium	T-Ba	0.237	0.250	mg/L	0.01	526464	a
Barium	T-Ba	0.269	0.250	mg/L	0.01	527616	a
Beryllium	T-Be	0.0978	0.100	mg/L	0.005	526121	a
Beryllium	T-Be	0.0965	0.100	mg/L	0.005	526464	a
Beryllium	T-Be	0.101	0.100	mg/L	0.005	527616	a
Bismuth	T-Bi	1.00	1.00	mg/L	0.2	526121	a
Bismuth	T-Bi	0.96	1.00	mg/L	0.2	526464	a
Bismuth	T-Bi	0.95	1.00	mg/L	0.2	527616	a
Boron	T-B	0.97	1.00	mg/L	0.1	526121	a
Boron	T-B	0.96	1.00	mg/L	0.1	526464	a
Boron	T-B	0.94	1.00	mg/L	0.1	527616	a
Cadmium	T-Cd	0.097	0.100	mg/L	0.01	526121	a
Cadmium	T-Cd	0.094	0.100	mg/L	0.01	526464	a
Cadmium	T-Cd	0.098	0.100	mg/L	0.01	527616	a
Calcium	T-Ca	51.6	50.0	mg/L	0.05	526121	a
Calcium	T-Ca	48.2	50.0	mg/L	0.05	526464	a
Calcium	T-Ca	45.8	50.0	mg/L	0.05	527616	a
Chromium	T-Cr	0.245	0.250	mg/L	0.01	526121	a
Chromium	T-Cr	0.231	0.250	mg/L	0.01	526464	a
Chromium	T-Cr	0.241	0.250	mg/L	0.01	527616	a
Cobalt	T-Co	0.243	0.250	mg/L	0.01	526121	a
Cobalt	T-Co	0.229	0.250	mg/L	0.01	526464	a
Cobalt	T-Co	0.236	0.250	mg/L	0.01	527616	a
Copper	T-Cu	0.249	0.250	mg/L	0.01	526121	a

Methods:

a = ICPOES

Appendix 1 - QUALITY CONTROL - Reference Materials



Water

		Result	Target	Units	DL	ALSQC#	Method
ALSEV Metals in Water - High, 4GK116							
Total Metals							
Copper	T-Cu	0.246	0.250	mg/L	0.01	526464	a
Copper	T-Cu	0.244	0.250	mg/L	0.01	527616	a
Iron	T-Fe	0.958	1.00	mg/L	0.03	526121	a
Iron	T-Fe	0.984	1.00	mg/L	0.03	526464	a
Iron	T-Fe	1.05	1.00	mg/L	0.03	527616	a
Lead	T-Pb	0.487	0.500	mg/L	0.05	526121	a
Lead	T-Pb	0.461	0.500	mg/L	0.05	526464	a
Lead	T-Pb	0.490	0.500	mg/L	0.05	527616	a
Lithium	T-Li	0.256	0.250	mg/L	0.01	526121	a
Lithium	T-Li	0.237	0.250	mg/L	0.01	526464	a
Lithium	T-Li	0.240	0.250	mg/L	0.01	527616	a
Magnesium	T-Mg	48.3	50.0	mg/L	0.1	526121	a
Magnesium	T-Mg	49.9	50.0	mg/L	0.1	526464	a
Magnesium	T-Mg	45.8	50.0	mg/L	0.1	527616	a
Manganese	T-Mn	0.244	0.250	mg/L	0.005	526121	a
Manganese	T-Mn	0.238	0.250	mg/L	0.005	526464	a
Manganese	T-Mn	0.248	0.250	mg/L	0.005	527616	a
Molybdenum	T-Mo	0.246	0.250	mg/L	0.03	526121	a
Molybdenum	T-Mo	0.245	0.250	mg/L	0.03	526464	a
Molybdenum	T-Mo	0.246	0.250	mg/L	0.03	527616	a
Nickel	T-Ni	0.484	0.500	mg/L	0.05	526121	a
Nickel	T-Ni	0.485	0.500	mg/L	0.05	526464	a
Nickel	T-Ni	0.485	0.500	mg/L	0.05	527616	a
Phosphorus	T-P	2.49	2.50	mg/L	0.3	526121	a
Phosphorus	T-P	2.47	2.50	mg/L	0.3	526464	a
Phosphorus	T-P	2.59	2.50	mg/L	0.3	527616	a
Potassium	T-K	49.9	50.0	mg/L	2	526121	a
Potassium	T-K	49.5	50.0	mg/L	2	526464	a
Potassium	T-K	49.8	50.0	mg/L	2	527616	a
Selenium	T-Se	0.98	1.00	mg/L	0.2	526121	a
Selenium	T-Se	0.98	1.00	mg/L	0.2	526464	a
Selenium	T-Se	0.95	1.00	mg/L	0.2	527616	a
Silver	T-Ag	0.095	0.100	mg/L	0.01	526121	a
Silver	T-Ag	0.102	0.100	mg/L	0.01	526464	a

Methods:

a = ICPOES

Appendix 1 - QUALITY CONTROL - Reference Materials



Water

		Result	Target	Units	DL	ALSQC#	Method
ALSEV Metals in Water - High, 4GK116							
Total Metals							
Silver	T-Ag	0.101	0.100	mg/L	0.01	527616	a
Sodium	T-Na	49.3	50.0	mg/L	2	526121	a
Sodium	T-Na	45.4	50.0	mg/L	2	526464	a
Sodium	T-Na	50.7	50.0	mg/L	2	527616	a
Strontium	T-Sr	0.253	0.250	mg/L	0.005	526121	a
Strontium	T-Sr	0.251	0.250	mg/L	0.005	526464	a
Strontium	T-Sr	0.242	0.250	mg/L	0.005	527616	a
Thallium	T-Tl	0.95	1.00	mg/L	0.2	526121	a
Thallium	T-Tl	0.92	1.00	mg/L	0.2	526464	a
Thallium	T-Tl	0.95	1.00	mg/L	0.2	527616	a
Tin	T-Sn	0.456	0.500	mg/L	0.03	526121	a
Tin	T-Sn	0.476	0.500	mg/L	0.03	526464	a
Tin	T-Sn	0.528	0.500	mg/L	0.03	527616	a
Titanium	T-Ti	0.241	0.250	mg/L	0.01	526121	a
Titanium	T-Ti	0.241	0.250	mg/L	0.01	526464	a
Titanium	T-Ti	0.249	0.250	mg/L	0.01	527616	a
Vanadium	T-V	0.498	0.500	mg/L	0.03	526121	a
Vanadium	T-V	0.515	0.500	mg/L	0.03	526464	a
Vanadium	T-V	0.505	0.500	mg/L	0.03	527616	a
Zinc	T-Zn	0.506	0.500	mg/L	0.005	526121	a
Zinc	T-Zn	0.493	0.500	mg/L	0.005	526464	a
Zinc	T-Zn	0.481	0.500	mg/L	0.005	527616	a
ALSEV Metals in Water - Mid, 4GK116							
Total Metals							
Aluminum	T-Al	0.0944	0.100	mg/L	0.001	526060	b
Aluminum	T-Al	0.100	0.100	mg/L	0.001	526376	b
Antimony	T-Sb	0.0500	0.0500	mg/L	0.0001	526060	b
Antimony	T-Sb	0.0501	0.0500	mg/L	0.0001	526376	b
Arsenic	T-As	0.0487	0.0500	mg/L	0.0001	526060	b
Arsenic	T-As	0.0488	0.0500	mg/L	0.0001	526376	b
Barium	T-Ba	0.0123	0.0125	mg/L	0.00005	526060	b
Barium	T-Ba	0.0123	0.0125	mg/L	0.00005	526376	b

Methods:

a = ICPOES

b = ICPMS

Appendix 1 - QUALITY CONTROL - Reference Materials



Water

		Result	Target	Units	DL	ALSQC#	Method
ALSEV Metals in Water - Mid, 4GK116							
Total Metals							
Beryllium	T-Be	0.00497	0.00500	mg/L	0.0005	526060	a
Beryllium	T-Be	0.00495	0.00500	mg/L	0.0005	526376	a
Bismuth	T-Bi	0.0487	0.0500	mg/L	0.0005	526060	a
Bismuth	T-Bi	0.0495	0.0500	mg/L	0.0005	526376	a
Boron	T-B	0.050	0.050	mg/L	0.01	526060	a
Boron	T-B	0.046	0.050	mg/L	0.01	526376	a
Cadmium	T-Cd	0.00477	0.00500	mg/L	0.00005	526060	a
Cadmium	T-Cd	0.00483	0.00500	mg/L	0.00005	526376	a
Calcium	T-Ca	2.40	2.50	mg/L	0.02	526060	a
Calcium	T-Ca	2.46	2.50	mg/L	0.02	526376	a
Chromium	T-Cr	0.0123	0.0125	mg/L	0.0005	526060	a
Chromium	T-Cr	0.0124	0.0125	mg/L	0.0005	526376	a
Cobalt	T-Co	0.0120	0.0125	mg/L	0.0001	526060	a
Cobalt	T-Co	0.0121	0.0125	mg/L	0.0001	526376	a
Copper	T-Cu	0.0114	0.0125	mg/L	0.0001	526060	a
Copper	T-Cu	0.0122	0.0125	mg/L	0.0001	526376	a
Lead	T-Pb	0.0243	0.0250	mg/L	0.00005	526060	a
Lead	T-Pb	0.0247	0.0250	mg/L	0.00005	526376	a
Lithium	T-Li	0.0116	0.0125	mg/L	0.005	526060	a
Lithium	T-Li	0.0114	0.0125	mg/L	0.005	526376	a
Magnesium	T-Mg	2.34	2.50	mg/L	0.005	526060	a
Magnesium	T-Mg	2.48	2.50	mg/L	0.005	526376	a
Manganese	T-Mn	0.0119	0.0125	mg/L	0.00005	526060	a
Manganese	T-Mn	0.0123	0.0125	mg/L	0.00005	526376	a
Molybdenum	T-Mo	0.0123	0.0125	mg/L	0.00005	526060	a
Molybdenum	T-Mo	0.0124	0.0125	mg/L	0.00005	526376	a
Nickel	T-Ni	0.0242	0.0250	mg/L	0.0005	526060	a
Nickel	T-Ni	0.0245	0.0250	mg/L	0.0005	526376	a
Potassium	T-K	2.35	2.50	mg/L	0.05	526060	a
Potassium	T-K	2.39	2.50	mg/L	0.05	526376	a
Selenium	T-Se	0.0490	0.0500	mg/L	0.0005	526060	a
Selenium	T-Se	0.0488	0.0500	mg/L	0.0005	526376	a
Silver	T-Ag	0.00493	0.00500	mg/L	0.00001	526060	a
Silver	T-Ag	0.00496	0.00500	mg/L	0.00001	526376	a

Methods:

a = ICPMS

Appendix 1 - QUALITY CONTROL - Reference Materials

Water

		Result	Target	Units	DL	ALSQC#	Method
ALSEV Metals in Water - Mid, 4GK116							
Total Metals							
Sodium	T-Na	2.41	2.50	mg/L	0.01	526060	a
Sodium	T-Na	2.50	2.50	mg/L	0.01	526376	a
Strontium	T-Sr	0.0121	0.0125	mg/L	0.0001	526060	a
Strontium	T-Sr	0.0126	0.0125	mg/L	0.0001	526376	a
Thallium	T-Tl	0.0473	0.0500	mg/L	0.0001	526060	a
Thallium	T-Tl	0.0478	0.0500	mg/L	0.0001	526376	a
Tin	T-Sn	0.0242	0.0250	mg/L	0.0001	526060	a
Tin	T-Sn	0.0243	0.0250	mg/L	0.0001	526376	a
Uranium	T-U	0.000250	0.000250	mg/L	0.00001	526060	a
Uranium	T-U	0.000254	0.000250	mg/L	0.00001	526376	a
Vanadium	T-V	0.0239	0.0250	mg/L	0.001	526060	a
Vanadium	T-V	0.0243	0.0250	mg/L	0.001	526376	a
Zinc	T-Zn	0.0242	0.0250	mg/L	0.001	526060	a
Zinc	T-Zn	0.0250	0.0250	mg/L	0.001	526376	a
ALSEV Mercury in Water, 246049A							
Total Metals							
Mercury	T-Hg	0.0109	0.0100	mg/L	0.00001	525525	b
Alltech Custom Anions Mix (10x), AllT170088							
Dissolved Anions							
Bromide	Br	1.01	1.00	mg/L	0.05	525786	c
Chloride	Cl	51.5	50.1	mg/L	0.5	525786	c
Fluoride	F	1.05	1.00	mg/L	0.02	525786	c
Sulphate	SO ₄	52.5	50.1	mg/L	0.5	525786	c
Nutrients							
Nitrate Nitrogen	N	0.215	0.225	mg/L	0.005	525786	c
Nitrite Nitrogen	N	0.156	0.152	mg/L	0.001	525786	c
SPEX Demand (2xC), Q2-94DEM							
Organic Parameters							
Chemical Oxygen Demand	COD	244	224	mg/L	20	525675	d

Methods:

a = ICPMS

b = CVAFS

c = Ion Chromatography

d = Colourimetry

File No. Z3248r

Appendix 1 - QUALITY CONTROL - Spikes



Water

	Result	Target	Units	DL	ALSQC#	Method
<u>Inorganic Parameters</u>						
Sulphide S	1.03	1.00	mg/L	0.02	525737	a
<u>Organic Parameters</u>						
Tannin and Lignin	9.64	10.0	mg/L	0.1	525845	a

Methods:

a = Colourimetry

Appendix 1 - QUALITY CONTROL - Blanks



Water

			Result	Target	Units	DL	ALSQC#	Method
Dissolved Anions								
Bromide	Br		<0.050	<0.050	mg/L	0.05	525782	a
Chloride	Cl		<0.50	<0.50	mg/L	0.5	525782	a
Fluoride	F		<0.020	<0.020	mg/L	0.02	525782	a
Sulphate	SO ₄		<0.50	<0.50	mg/L	0.5	525782	a
Nutrients								
Nitrate Nitrogen		N	<0.0050	<0.0050	mg/L	0.005	525782	a
Nitrite Nitrogen		N	<0.0010	<0.0010	mg/L	0.001	525782	a
Total Metals								
Aluminum	T-Al		0.0055	<0.0050	mg/L	0.005	526038	b
Aluminum	T-Al		<0.0050	<0.0050	mg/L	0.005	526040	b
Aluminum	T-Al		<0.0050	<0.0050	mg/L	0.005	526041	b
Aluminum	T-Al		<0.0050	<0.0050	mg/L	0.005	526042	b
Aluminum	T-Al		<0.20	<0.20	mg/L	0.2	526165	c
Antimony	T-Sb		<0.00020	<0.00020	mg/L	0.0002	526038	b
Antimony	T-Sb		<0.00020	<0.00020	mg/L	0.0002	526040	b
Antimony	T-Sb		<0.00020	<0.00020	mg/L	0.0002	526041	b
Antimony	T-Sb		<0.00020	<0.00020	mg/L	0.0002	526042	b
Antimony	T-Sb		<0.20	<0.20	mg/L	0.2	526165	c
Arsenic	T-As		<0.00020	<0.00020	mg/L	0.0002	526038	b
Arsenic	T-As		<0.00020	<0.00020	mg/L	0.0002	526040	b
Arsenic	T-As		<0.00020	<0.00020	mg/L	0.0002	526041	b
Arsenic	T-As		<0.00020	<0.00020	mg/L	0.0002	526042	b
Arsenic	T-As		<0.20	<0.20	mg/L	0.2	526165	c
Barium	T-Ba		<0.0010	<0.0010	mg/L	0.001	526038	b
Barium	T-Ba		<0.0010	<0.0010	mg/L	0.001	526040	b
Barium	T-Ba		<0.0010	<0.0010	mg/L	0.001	526041	b
Barium	T-Ba		<0.0010	<0.0010	mg/L	0.001	526042	b
Barium	T-Ba		<0.010	<0.010	mg/L	0.01	526165	c
Beryllium	T-Be		<0.0010	<0.0010	mg/L	0.001	526038	b
Beryllium	T-Be		<0.0010	<0.0010	mg/L	0.001	526040	b
Beryllium	T-Be		<0.0010	<0.0010	mg/L	0.001	526041	b
Beryllium	T-Be		<0.0010	<0.0010	mg/L	0.001	526042	b
Beryllium	T-Be		<0.0050	<0.0050	mg/L	0.005	526165	c
Bismuth	T-Bi		<0.0010	<0.0010	mg/L	0.001	526038	b
Bismuth	T-Bi		<0.0010	<0.0010	mg/L	0.001	526040	b

Methods:

a = Ion Chromatography

b = ICPMS

c = ICPOES

Appendix 1 - QUALITY CONTROL - Blanks



Water

		Result	Target	Units	DL	ALSQC#	Method
Total Metals							
Bismuth	T-Bi	<0.0010	<0.0010	mg/L	0.001	526041	a
Bismuth	T-Bi	<0.0010	<0.0010	mg/L	0.001	526042	a
Bismuth	T-Bi	<0.10	<0.10	mg/L	0.1	526165	b
Boron	T-B	<0.050	<0.050	mg/L	0.05	526038	a
Boron	T-B	<0.050	<0.050	mg/L	0.05	526040	a
Boron	T-B	<0.050	<0.050	mg/L	0.05	526041	a
Boron	T-B	<0.050	<0.050	mg/L	0.05	526042	a
Boron	T-B	<0.10	<0.10	mg/L	0.1	526165	b
Cadmium	T-Cd	<0.00020	<0.00020	mg/L	0.0002	526038	a
Cadmium	T-Cd	<0.00020	<0.00020	mg/L	0.0002	526040	a
Cadmium	T-Cd	<0.00020	<0.00020	mg/L	0.0002	526041	a
Cadmium	T-Cd	<0.00020	<0.00020	mg/L	0.0002	526042	a
Cadmium	T-Cd	<0.010	<0.010	mg/L	0.01	526165	b
Calcium	T-Ca	<0.050	<0.050	mg/L	0.05	526038	a
Calcium	T-Ca	<0.050	<0.050	mg/L	0.05	526040	a
Calcium	T-Ca	<0.050	<0.050	mg/L	0.05	526041	a
Calcium	T-Ca	<0.050	<0.050	mg/L	0.05	526042	a
Calcium	T-Ca	<0.050	<0.050	mg/L	0.05	526165	b
Chromium	T-Cr	<0.0010	<0.0010	mg/L	0.001	526038	a
Chromium	T-Cr	<0.0010	<0.0010	mg/L	0.001	526040	a
Chromium	T-Cr	<0.0010	<0.0010	mg/L	0.001	526041	a
Chromium	T-Cr	<0.0010	<0.0010	mg/L	0.001	526042	a
Chromium	T-Cr	<0.010	<0.010	mg/L	0.01	526165	b
Cobalt	T-Co	<0.0010	<0.0010	mg/L	0.001	526038	a
Cobalt	T-Co	<0.0010	<0.0010	mg/L	0.001	526040	a
Cobalt	T-Co	<0.0010	<0.0010	mg/L	0.001	526041	a
Cobalt	T-Co	<0.0010	<0.0010	mg/L	0.001	526042	a
Cobalt	T-Co	<0.010	<0.010	mg/L	0.01	526165	b
Copper	T-Cu	<0.0010	<0.0010	mg/L	0.001	526038	a
Copper	T-Cu	<0.0010	<0.0010	mg/L	0.001	526040	a
Copper	T-Cu	<0.0010	<0.0010	mg/L	0.001	526041	a
Copper	T-Cu	<0.0010	<0.0010	mg/L	0.001	526042	a
Copper	T-Cu	<0.010	<0.010	mg/L	0.01	526165	b
Iron	T-Fe	<0.030	<0.030	mg/L	0.03	526165	b
Lead	T-Pb	<0.0010	<0.0010	mg/L	0.001	526038	a
Lead	T-Pb	<0.0010	<0.0010	mg/L	0.001	526040	a
Lead	T-Pb	<0.0010	<0.0010	mg/L	0.001	526041	a
Lead	T-Pb	<0.0010	<0.0010	mg/L	0.001	526042	a
Lead	T-Pb	<0.010	<0.010	mg/L	0.01	526165	b

Methods:

a = ICPMS

b = ICPOES

Appendix 1 - QUALITY CONTROL - Blanks



Water

		Result	Target	Units	DL	ALSQC#	Method
Total Metals							
Lead	T-Pb	<0.0010	<0.0010	mg/L	0.001	526041	a
Lead	T-Pb	<0.0010	<0.0010	mg/L	0.001	526042	a
Lead	T-Pb	<0.050	<0.050	mg/L	0.05	526165	b
Lithium	T-Li	<0.010	<0.010	mg/L	0.01	526038	a
Lithium	T-Li	<0.010	<0.010	mg/L	0.01	526040	a
Lithium	T-Li	<0.010	<0.010	mg/L	0.01	526041	a
Lithium	T-Li	<0.010	<0.010	mg/L	0.01	526042	a
Lithium	T-Li	<0.010	<0.010	mg/L	0.01	526165	b
Magnesium	T-Mg	<0.050	<0.050	mg/L	0.05	526038	a
Magnesium	T-Mg	<0.050	<0.050	mg/L	0.05	526040	a
Magnesium	T-Mg	<0.050	<0.050	mg/L	0.05	526041	a
Magnesium	T-Mg	<0.050	<0.050	mg/L	0.05	526042	a
Magnesium	T-Mg	<0.10	<0.10	mg/L	0.1	526165	b
Manganese	T-Mn	<0.0010	<0.0010	mg/L	0.001	526038	a
Manganese	T-Mn	<0.0010	<0.0010	mg/L	0.001	526040	a
Manganese	T-Mn	<0.0010	<0.0010	mg/L	0.001	526041	a
Manganese	T-Mn	<0.0010	<0.0010	mg/L	0.001	526042	a
Manganese	T-Mn	<0.0050	<0.0050	mg/L	0.005	526165	b
Mercury	T-Hg	<0.000010	<0.000010	mg/L	0.00001	525524	c
Mercury	T-Hg	<0.000010	<0.000010	mg/L	0.00001	526138	c
Molybdenum	T-Mo	<0.0010	<0.0010	mg/L	0.001	526038	a
Molybdenum	T-Mo	<0.0010	<0.0010	mg/L	0.001	526040	a
Molybdenum	T-Mo	<0.0010	<0.0010	mg/L	0.001	526041	a
Molybdenum	T-Mo	<0.0010	<0.0010	mg/L	0.001	526042	a
Molybdenum	T-Mo	<0.030	<0.030	mg/L	0.03	526165	b
Nickel	T-Ni	<0.0010	<0.0010	mg/L	0.001	526038	a
Nickel	T-Ni	<0.0010	<0.0010	mg/L	0.001	526040	a
Nickel	T-Ni	<0.0010	<0.0010	mg/L	0.001	526041	a
Nickel	T-Ni	<0.0010	<0.0010	mg/L	0.001	526042	a
Nickel	T-Ni	<0.050	<0.050	mg/L	0.05	526165	b
Phosphorus	T-P	<0.30	<0.30	mg/L	0.3	526165	b
Potassium	T-K	<0.050	<0.050	mg/L	0.05	526038	a
Potassium	T-K	<0.050	<0.050	mg/L	0.05	526040	a
Potassium	T-K	<0.050	<0.050	mg/L	0.05	526041	a
Potassium	T-K	<0.050	<0.050	mg/L	0.05	526042	a
Potassium	T-K	<2.0	<2.0	mg/L	2	526165	b

Methods:

a = ICPMS
b = ICPOES
c = CVAFS

Appendix 1 - QUALITY CONTROL - Blanks



Water

		Result	Target	Units	DL	ALSQC#	Method
Total Metals							
Selenium	T-Se	<0.0010	<0.0010	mg/L	0.001	526038	a
Selenium	T-Se	<0.0010	<0.0010	mg/L	0.001	526040	a
Selenium	T-Se	<0.0010	<0.0010	mg/L	0.001	526041	a
Selenium	T-Se	<0.0010	<0.0010	mg/L	0.001	526042	a
Selenium	T-Se	<0.20	<0.20	mg/L	0.2	526165	b
Silicon	T-Si	<0.050	<0.050	mg/L	0.05	526165	b
Silver	T-Ag	<0.00010	<0.00010	mg/L	0.0001	526038	a
Silver	T-Ag	<0.00010	<0.00010	mg/L	0.0001	526040	a
Silver	T-Ag	<0.00010	<0.00010	mg/L	0.0001	526041	a
Silver	T-Ag	<0.00010	<0.00010	mg/L	0.0001	526042	a
Silver	T-Ag	<0.010	<0.010	mg/L	0.01	526165	b
Sodium	T-Na	<0.050	<0.050	mg/L	0.05	526038	a
Sodium	T-Na	<0.050	<0.050	mg/L	0.05	526040	a
Sodium	T-Na	<0.050	<0.050	mg/L	0.05	526041	a
Sodium	T-Na	<0.050	<0.050	mg/L	0.05	526042	a
Sodium	T-Na	<2.0	<2.0	mg/L	2	526165	b
Strontium	T-Sr	<0.0010	<0.0010	mg/L	0.001	526038	a
Strontium	T-Sr	<0.0010	<0.0010	mg/L	0.001	526040	a
Strontium	T-Sr	<0.0010	<0.0010	mg/L	0.001	526041	a
Strontium	T-Sr	<0.0010	<0.0010	mg/L	0.001	526042	a
Strontium	T-Sr	<0.0050	<0.0050	mg/L	0.005	526165	b
Thallium	T-Tl	<0.0010	<0.0010	mg/L	0.001	526038	a
Thallium	T-Tl	<0.0010	<0.0010	mg/L	0.001	526040	a
Thallium	T-Tl	<0.0010	<0.0010	mg/L	0.001	526041	a
Thallium	T-Tl	<0.0010	<0.0010	mg/L	0.001	526042	a
Thallium	T-Tl	<0.20	<0.20	mg/L	0.2	526165	b
Tin	T-Sn	<0.0010	<0.0010	mg/L	0.001	526038	a
Tin	T-Sn	<0.0010	<0.0010	mg/L	0.001	526040	a
Tin	T-Sn	<0.0010	<0.0010	mg/L	0.001	526041	a
Tin	T-Sn	<0.0010	<0.0010	mg/L	0.001	526042	a
Tin	T-Sn	<0.030	<0.030	mg/L	0.03	526165	b
Titanium	T-Ti	<0.010	<0.010	mg/L	0.01	526165	b
Uranium	T-U	<0.000010	<0.000010	mg/L	0.00001	526038	a
Uranium	T-U	<0.000010	<0.000010	mg/L	0.00001	526040	a
Uranium	T-U	<0.000010	<0.000010	mg/L	0.00001	526041	a
Uranium	T-U	<0.000010	<0.000010	mg/L	0.00001	526042	a

Methods:

a = ICPMS

b = ICPOES

File No. Z3248r

Appendix 1 - QUALITY CONTROL - Blanks



Water

		Result	Target	Units	DL	ALSQC#	Method
Total Metals							
Vanadium	T-V	<0.0050	<0.0050	mg/L	0.005	526038	a
Vanadium	T-V	<0.0050	<0.0050	mg/L	0.005	526040	a
Vanadium	T-V	<0.0050	<0.0050	mg/L	0.005	526041	a
Vanadium	T-V	<0.0050	<0.0050	mg/L	0.005	526042	a
Vanadium	T-V	<0.030	<0.030	mg/L	0.03	526165	b
Zinc	T-Zn	<0.0050	<0.0050	mg/L	0.005	526038	a
Zinc	T-Zn	<0.0050	<0.0050	mg/L	0.005	526040	a
Zinc	T-Zn	<0.0050	<0.0050	mg/L	0.005	526041	a
Zinc	T-Zn	<0.0050	<0.0050	mg/L	0.005	526042	a
Zinc	T-Zn	<0.0050	<0.0050	mg/L	0.005	526165	b
Inorganic Parameters							
Sulphide	S	<0.02	<0.02	mg/L	0.02	525738	c
Organic Parameters							
Chemical Oxygen Demand	COD	<20	<20	mg/L	20	525674	c
Tannin and Lignin		<0.1	<0.1	mg/L	0.1	525844	c

Methods:

a = ICPMS
b = ICPOES
c = Colourimetry

Appendix 2 - METHODOLOGY



Outlines of the methodologies utilized for the analysis of the samples submitted are as follows

Dissolved Anions in Water by Ion Chromatography

This analysis is carried out using procedures adapted from APHA Method 4110 "Determination of Anions by Ion Chromatography" and EPA Method 300.0 "Determination of Inorganic Anions by Ion Chromatography". Anions are determined by filtering the sample through a 0.45 micron membrane filter and injecting the filtrate onto a Dionex IonPac AG17 anion exchange column with a hydroxide eluent stream. Anions routinely determined by this method include: bromide, chloride, fluoride, nitrate, nitrite and sulphate.

Recommended Holding Time:

Sample: 28 days (bromide, chloride, fluoride, sulphate)

Sample: 2 days (nitrate, nitrite)

Reference: APHA and EPA

Laboratory Location: ALS Environmental, Vancouver

Metals in Water

This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" 21th Edition 2005 published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedures may involve preliminary sample treatment by acid digestion, using either hotplate or microwave oven, or filtration (EPA Method 3005A). Instrumental analysis is by atomic absorption/emission spectrophotometry (EPA Method 7000 series), inductively coupled plasma - optical emission spectrophotometry (EPA Method 6010B), and/or inductively coupled plasma - mass spectrometry (EPA Method 6020).

Recommended Holding Time:

Sample: 6 months

Reference: EPA

Laboratory Location: ALS Environmental, Vancouver

Mercury in Water

This analysis is carried out using procedures adapted from "Standard Methods for the Examination of Water and Wastewater" 20th Edition 1998 published by the American Public Health Association, and with procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846 published by the United States Environmental Protection Agency (EPA). The procedure involves a cold-oxidation of the acidified sample using bromine monochloride prior to reduction of the sample with stannous chloride. Instrumental analysis is by cold vapour atomic fluorescence spectrophotometry (EPA Method 245.7).

File No. Z3248r

Appendix 2 - METHODOLOGY - Continued



Recommended Holding Time:

Sample: 28 days

Reference: EPA

Laboratory Location: ALS Environmental, Vancouver

Sulphide in Water

This analysis is carried out using procedures adapted from APHA Method 4500-S2 "Sulphide". Sulphide is determined using the methylene blue colourimetric method.

Recommended Holding Time:

Sample: 7 days

Reference: APHA

Laboratory Location: ALS Environmental, Vancouver

Chemical Oxygen Demand in Water

This analysis is carried out using procedures adapted from APHA Method 5220 "Chemical Oxygen Demand (COD)". Chemical oxygen demand is determined using the closed reflux colourimetric method.

Recommended Holding Time:

Sample: 28 days

Reference: APHA

Laboratory Location: ALS Environmental, Vancouver

Phenols in Water

This analysis is carried out using procedures adapted from APHA Method 5530 "Phenols". Phenols, defined by APHA as "hydroxy derivatives of benzene and its condensed nuclei", is determined by sample distillation and analysis using the antipyrine colourimetric method. The recovery of phenolics is non-uniform; substitution usually reduces recovery and the reported value represents the minimum concentration of phenolic compounds present in the sample.

Recommended Holding Time:

Sample: 28 days

Reference: APHA

For more detail see ALS Environmental "Collection & Sampling Guide"

Laboratory Location: ALS Environmental, Calgary

File No. Z3248r

Appendix 2 - METHODOLOGY - Continued



Conventional Parameters in Water

These analyses are carried out in accordance with procedures described in "Methods for Chemical Analysis of Water and Wastes" (USEPA), "Manual for the Chemical Analysis of Water, Wastewaters, Sediments and Biological Tissues" (BCMOE), and/or "Standard Methods for the Examination of Water and Wastewater" (APHA). Further details are available on request.

Laboratory Location: ALS Environmental, Vancouver

Results contained within this certificate relate only to the samples as submitted.

This Certificate Of Analysis shall only be reproduced in full, except with the written approval of ALS Environmental.

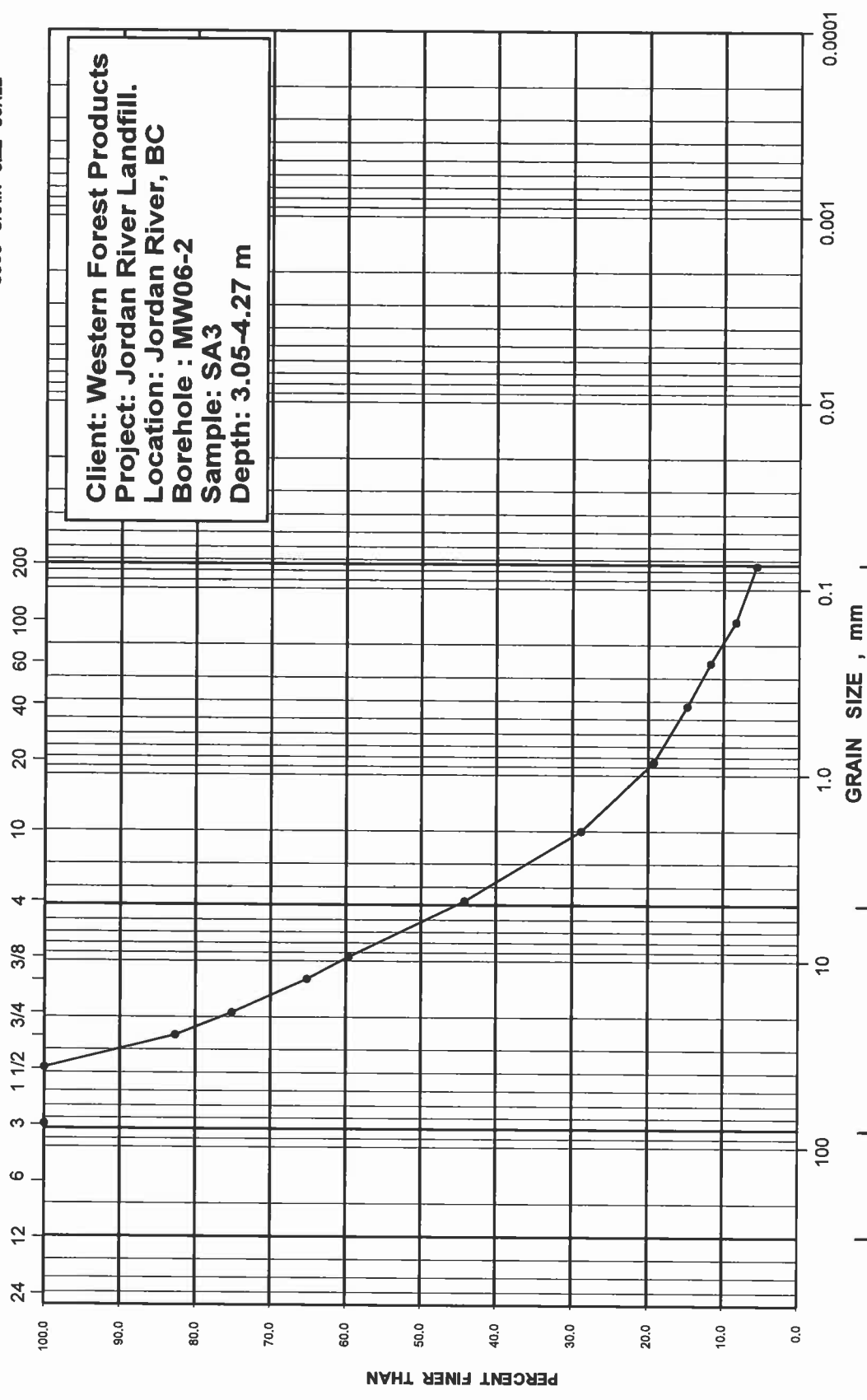
End of Report

USCS GRAIN SIZE SCALE

U. S. S. sieve size, meshes / inch

Size of opening, inches

Client: Western Forest Products
 Project: Jordan River Landfill.
 Location: Jordan River, BC
 Borehole : MW06-2
 Sample: SA3
 Depth: 3.05-4.27 m



BOULDER SIZE		COBBLE SIZE		GRAVEL SIZE		SAND SIZE		FINE GRAINED	

Project No. 06-1414-092
 Drawn DGM
 Reviewed PS
 Date 29 Sept. 06



GRAIN SIZE DISTRIBUTION

GRAPH V-1



SIEVE ANALYSIS

Project No:	06-1414-092	Client :	Western Forest Products		Auger Hole	MW06-02	
Date :	28-Sep-06	Project :	Jordan River Landfill		Sample :	SA3	
Lab Work :	jt	Location :	Jordan River		Depth :	3.05-4.27	
	1st SIEVING		2nd SIEVING		Wash Sieving		
	Weight before sieving		Weight before sieving		Weight after wash		795.1
	Total weight	828.8	1/4 Pass #4		Residual #200		13.0
	Pass #4	0.0			Minus #200		46.7
Sieve (USS)	Weight Retained	% Retained	Weight Retained	% Retained	% Retained of Total	Diameter (mm)	% Passing
3"		0.0			0.0	75.0	100.0
1 1/2"		0.0			0.0	37.5	100.0
1"	143.9	17.4			17.4	25.0	82.6
3/4"	62.2	7.5			7.5	19.0	75.1
1/2"	82.8	10.0			10.0	12.5	65.1
3/8"	45.9	5.5			5.5	9.5	59.6
#4	127.7	15.4			15.4	4.75	44.2
#10	127.4	15.4			15.4	2.00	28.8
#20	79.3	9.6			9.6	0.85	19.3
#40	36.5	4.4			4.4	0.425	14.9
#60	25.8	3.1			3.1	0.25	11.7
#100	27.8	3.4			3.4	0.15	8.4
#200	22.8	2.8			2.8	0.075	5.6
Pan	46.7	5.6			5.6		

REMARKS :



SIEVE ANALYSIS

Project No:	06-1414-092	Client :	Western Forest Products		Auger Hole	MW06-03	
Date :	28-Sep-06	Project :	Jordan River Landfill		Sample :	SA1	
Lab Work :	jt	Location :	Jordan River		Depth :	1.8-2.7 m	
	1st SIEVING		2nd SIEVING		Wash Sieving		
	Weight before sieving		Weight before sieving		Weight after wash		882.3
	Total weight	909.3	1/4 Pass #4		Residual #200		2.0
	Pass #4	0.0			Minus #200		29.0
Sieve (USS)	Weight Retained	% Retained	Weight Retained	% Retained	% Retained of Total	Diameter (mm)	% Passing
3"		0.0			0.0	75.0	100.0
1 1/2"		0.0			0.0	37.5	100.0
1"	0.0	0.0			0.0	25.0	100.0
3/4"	0.0	0.0			0.0	19.0	100.0
1/2"	0.0	0.0			0.0	12.5	100.0
3/8"	1.1	0.1			0.1	9.5	99.9
#4	2.2	0.2			0.2	4.75	99.6
#10	2.2	0.2			0.2	2.00	99.4
#20	3.8	0.4			0.4	0.85	99.0
#40	75.6	8.3			8.3	0.425	90.7
#60	327.2	36.0			36.0	0.25	54.7
#100	409.9	45.1			45.1	0.15	9.6
#200	58.3	6.4			6.4	0.075	3.2
Pan	29.0	3.2			3.2		

REMARKS :



Date: July 12, 2000

Sampled on June 2000

Client: WESTERN FOREST PRODUCTS
JORDAN RIVER OPERATION
GENERAL DELIVERY
JORDAN RIVER BC VOS 1LO

Attn: Mr. D. C. Copp

Sample #01 - 2000-06-14 : Water sample

Sample #01

COD	mg/L	150
pH	pH	5.5
96h LC50 (% v/v)	%	>100
Phenols	mg/L	0.002
Chloride	mg/L	240
Tannin & Lignin	mg/L Tan. Acid	16.1

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jblabs@islandnet.com

Job No.: 1439B955

LR No.: 30124

LR Sub No.: 1

Analyses performed according to "A Laboratory Manual for the Chemical Analysis of Water, Wastewaters and Biological Tissues", Chemistry Laboratory, Water Resource Service and/or "Standard Methods / Water and Wastewater", American Public Health Association.

Sampling Agent: Client

The sample(s) submitted by the agent have been tested as requested and we report as follows:

Report prepared by:

Barbara M. Klassen, B.Sc.

JB Laboratories Ltd.

water/wastewaters



